

Powell-Style Model-Based Derivative-Free Optimization with Complexity Guarantees

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Abstract. We propose variants of model-based trust region derivative free algorithms that are closest to methods initially proposed and implemented by Powell in [22, 21]. These methods rely on low degree polynomial interpolation and carefully maintain geometry of the interpolation sets. We are able to derive complexity bounds for these methods that make them theoretically competitive to other derivative free methods. Applying these methods in randomly generated subspaces recovers what we believe to be nearly tight complexity. This paper builds on recent results in [11] where complexity of a much simplified version of Powell’s methods was derived. Here we extend the analysis to fully incorporate Powell’s geometry handling approach, and conduct extensive numerical comparison of the model-based trust region methods connecting practical and theoretical performance. We also extend the analysis of subspace model-based trust region methods initially developed in [11] to the case of noisy function evaluations.

MSC Classification: 90C30, 90C56.

1 Introduction. In this paper, we focus on the complexity of model-based derivative free algorithms that aim to solve unconstrained nonlinear optimization problems of the form

$$(1.1) \quad \min_{x \in \mathbb{R}^n} \phi(x),$$

where $\phi : \mathbb{R}^n \rightarrow \mathbb{R}$ is a smooth objective function that may not be convex. The key premise of these methods is to economize on function values, possibly at the expense of additional linear algebra, since in most applications the function evaluations cost dominates all else. The complexity will be measured in terms of the number of function evaluations needed to achieve an ϵ -stationary point, that is a point x for which $\|\nabla \phi(x)\| \leq \epsilon$.

The main goal of this paper is to narrow the gap between theory and practice in model based derivative free optimization (DFO). Derivative free optimization, also known under names *zeroth-order* and *gradient-free* optimization, is an area of optimization concerned with developing optimization methods based purely on function value computation without applying any direct differentiation. The area has experienced significant growth both in theory and in a variety of applications in the past several decades. There is a rich literature of DFO starting from around mid 90s which is rapidly growing with new interest spurred by new applications in engineering, machine learning and artificial intelligence. Aside from an increasing number of papers, there are two books [13] and [1], and a survey on the topic [19]. Examples of applications can be found in [1] and [25].

Model-based DFO methods approach the optimization problem by constructing and maintaining (usually local) models of the objective function from function value samples. A particular class of such methods, model-based trust region methods pioneered in the 90s by M.J.D. Powell [22, 21], proved to be very effective in practice for many applications [20]. These methods use polynomial interpolation models and maintain sample sets using carefully engineered techniques, supported by mathematical properties of Lagrange interpolation polynomials. The complex structure of the algorithms, especially as presented in Powell’s papers, made them difficult to implement, let alone to analyze rigorously. Some underlying theory of asymptotic convergence was developed in [13] and later complexity bounds were provided for those algorithms in [15], however, methods analyzed there are much simplified and include elements that significantly depart from Powell’s ideas. More specifically, Powell’s methods only use one or two function evaluations per iteration, attempting to select the sample points in an optimal way, to improve the objective value while also maintaining or improving geometry of sample points. In other words, they carefully balance exploration and exploitation. On the other hand, the algorithms which

具有复杂度保证的Powell风格基于模型的无导数优化

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摘要。 我们提出了与Powell在 [22, 21]中最初提出并实现的方法最为接近的基于模型的信赖域无导数算法的若干变体。这些方法依赖于低次多项式插值，并谨慎地维护插值集的几何。我们能够为这些方法推导出复杂度界，使其在理论上与其他无导数方法具有竞争力。将这些方法应用于随机生成子空间可恢复我们认为近乎紧的复杂度。本文建立在 [11] 中近期结果的基础上，其中推导了Powell方法的一个大幅简化版本的复杂度。在此，我们扩展分析，以充分纳入Powell的几何处理策略，并对基于模型的信赖域方法进行了广泛的数值比较，将实际与理论性能联系起来。我们还将最初在 [11]中开发的子空间基于模型的信赖域方法分析扩展到带噪函数评估的情形。

MSC 分类: 90C30, 90C56。

1 引言。 本文关注基于模型的无导数算法的复杂度，这些算法旨在求解如下形式的无约束非线性优化问题

$$(1.1) \quad \min_{x \in \mathbb{R}^n} \phi(x),$$

其中 $\phi : \mathbb{R}^n \rightarrow \mathbb{R}$ 是一个可能非凸的光滑目标函数。这些方法的关键前提是节约函数值，可能以增加线性代数运算为代价，因为在大多数应用中，函数求值的成本起主导作用。复杂度将按达到一个 ϵ -驻点所需的函数求值次数来衡量，即一个点 x ，使得 $\|\nabla \phi(x)\| \leq \epsilon$ 。

本文的主要目标是缩小基于模型的无导数优化（DFO）中理论与实践之间的差距。无导数优化也被称为 零阶 和 无梯度优化，它是优化领域的一个方向，关注在不应用任何直接求导的情况下，仅基于函数值计算来开发优化方法。过去几十年，该领域在理论和各种应用方面都经历了显著增长。大约从90年代中期开始，DFO 已有丰富的文献，并且随着工程领域新应用激发的兴趣而迅速增长，

机器学习和人工智能。除了越来越多的论文之外，还有两本书 [13]和 [1]，以及一篇关于该主题的综述 [19]。应用示例可见于 [1]和 [25]。

基于模型的无导数优化方法通过从函数值样本中构建并维护目标函数的（通常是局部的）模型来求解优化问题。这类方法中的一种特殊类别——由 M.J.D. Powell 在90年代开创的基于模型的信赖域方法 [22, 21]，在许多应用中被证明在实践中非常有效 [20]。这些方法使用多项式插值模型，并利用精心设计的技术维护样本集，这些技术得到拉格朗日插值多项式数学性质的支持。算法的复杂结构，尤其是如 Powell 论文中所呈现的那样，使其难以实现，更不用说进行严格分析了。一些关于渐近收敛的基础理论在[13]中被提出，后来在 [15]，中为这些算法给出了复杂度界；然而，其中分析的方法被大幅简化，并包含一些显著偏离 Powell 思想的元素。更具体地说，Powell方法每次迭代仅使用一到两次函数求值，试图以最优方式选择样本点，以改进目标值，同时保持或改进样本点几何。换句话说，它们仔细平衡探索与利用。另一方面，那些

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have enjoyed complexity bounds so far require occasional complete model reconstruction, which requires far more function evaluations and essentially abandon Powell’s careful geometry correcting approach.

For the past several decades there was a general lack of understanding of how Powell’s methods work and most importantly if they enjoy favorable complexity bounds. Recently, in a considerable implementational effort Powell’s software has been reincarnated in modern platforms by Z. Zhang and his colleagues [24, 27]. Their software packages have seen considerable success and are now being widely used by practitioners. This raises the interest in providing solid theory for these methods. In [11] a *geometry correcting* method inspired by Powell’s algorithms was proposed and its complexity bounds derived. There were several important questions addressed in that paper.

- It was shown that a model-based TR method which performs only one or two function evaluations per iteration has good complexity guarantees in the case of general polynomials.
- In the case of linear polynomial interpolation the worst case complexity bound is $\mathcal{O}(n^2\epsilon^{-2})$ which matches those of other DFO methods, such as direct search or methods based on finite difference gradient approximation.
- It was shown that model-based trust region methods (Powell’s or other) can be applied within a random subspace algorithm which results in a further improvement in the worst case complexity bound to $\mathcal{O}(n\epsilon^{-2})$.

In this paper we extend the results of [11] in several important ways.

- The method in [11] still departs from Powell’s method in its geometry handling approach. We will explain the details of this when we describe the algorithm but the key difference is that the method in [11] discards some potentially useful sample points, which Powell’s methods do not. Including these points complicates the theory substantially, but we are able to provide such theory here.
- We derive the theory based on linear interpolation since this is sufficient for the first-order convergence analysis, however, with an easy modification we allow for quadratic models without any additional complexity cost.
- We extend the analysis to functions with deterministic noise, which is essential when addressing real DFO applications. Deterministic noise implies a lower bound on the best reachable optimality criterion and we derive such a lower bound.
- We extend the analysis of the random subspace DFO trust region method to the case of deterministic noise. As we will show, there is an additional complication which requires algorithmic modifications in this case (as opposed to the full-space case or noise-free subspace case).
- Finally, we provide a careful implementation of our ideas and demonstrate that our theoretically supported algorithm can match the performance of other trust region DFO methods, including those in [24, 27].

1.1 Preliminaries The following are the standard assumptions on $\phi(x)$ for our setting.

ASSUMPTION 1.1 (Lower bound on ϕ). *The function ϕ is bounded below by a scalar ϕ^* on $\mathbb{R}^n$.*

ASSUMPTION 1.2 (Lipschitz continuous gradient). *The function ϕ is continuously differentiable, and the gradient of ϕ is L -Lipschitz continuous on $\mathbb{R}^n$, i.e., $\|\nabla\phi(y) - \nabla\phi(x)\| \leq L\|y - x\|$ for all $(y, x) \in \mathbb{R}^n \times \mathbb{R}^n$.*

Throughout the paper, we assume that we do not have access to $\nabla\phi(x)$ or its approximation using any form of differentiation. Instead we have access to an inexact zeroth-order oracle $f(x) \approx \phi(x)$. Specifically, for all x we assume

$$|f(x) - \phi(x)| \leq \epsilon_f$$

for some $\epsilon_f > 0$.

Our main objective is to derive algorithms with the following type of guarantees. For a given $\epsilon > 0$, let C_ϵ denote the total number of calls to oracle $f(x)$ performed by the given algorithm that guarantees reaching a point x_ϵ for which $\|\nabla\phi(x_\epsilon)\| \leq \epsilon$. Then we seek the following guarantees: $\forall \epsilon > \psi(\epsilon_f)$ $C_\epsilon \leq \Psi(n, \epsilon)$, if the algorithm is deterministic, or $\mathbb{E}[C_\epsilon] \leq \Psi(n, \epsilon)$, if the algorithm is stochastic. Here ψ is some function of the oracle noise that

迄今为止享有复杂度界的算法，需要偶尔进行完整的模型重构；这需要多得多的函数求值，并且它们实质上放弃了 Powell 精心设计的几何修正方法。

在过去的几十年里，人们普遍不了解 Powell 方法如何工作，更重要的是，它们是否具有有利的复杂度界。最近，经过大量实现工作，Powell 软件由 Z. Zhang 及其同事 [24, 27] 在现代平台上重获新生。他们的软件包取得了相当大的成功，如今正被实践者广泛使用。这激发了为这些方法提供坚实理论的兴趣。在 [11] 中，提出了一种受 Powell 算法启发的几何校正方法，并推导了其复杂度界。该论文探讨了几个重要问题。

- 研究表明，一种每次迭代仅执行一到两次函数求值的基于模型的信赖域方法，在一般多项式情形下具有良好的复杂度保证。
- 在线性多项式插值情形下，最坏情况复杂度界为 $\mathcal{O}(n^2\epsilon^{-2})$ ，这与其它无导数优化方法（如直接搜索或基于有限差分梯度近似的方法）的复杂度界相匹配。

- 研究表明，基于模型的信赖域方法（Powell 方法或其他方法）可以应用于随机子空间算法中，从而将最坏情况复杂度界进一步改进为 $\mathcal{O}(n\epsilon^{-2})$ 。

在本文中，我们在若干重要方面扩展了 [11] 的结果。

- [11] 中的方法在几何处理策略上仍然偏离 Powell 方法。我们将在描述算法时解释其细节，但关键区别在于，[11] 中的方法会丢弃一些可能有用的样本点，而 Powell 方法不会。纳入这些点会使理论变得相当复杂，但我们能够在此提供这样的理论。

- 我们基于线性插值推导理论，因为这对于一阶收敛分析已经足够；然而，通过一个简单的修改，我们允许使用二次模型，而不会产生任何额外的复杂度代价。

- 我们将分析扩展到具有确定性噪声的函数，这在处理真实的无导数优化应用时至关重要。确定性噪声意味着对最佳可达的最优性准则存在一个下界，我们推导了这样的下界。

- 我们将随机子空间无导数优化信赖域方法的分析扩展到确定性噪声情形。正如我们将展示的，在这种情况下存在一个额外的复杂因素，需要进行算法修改（这与全空间情形或无噪声子空间情形不同）。

- 最后，我们给出了对我们想法的细致实现，并证明我们具有理论支持的算法能够匹敌其他信赖域无导数优化方法的性能，包括 [24, 27] 中的那些方法。

1.1 预备知识 以下是我们设定中关于 $\phi(x)$ 的标准假设。

假设 1.1 (下界: ϕ)。函数 ϕ 在 $\mathbb{R}^n$ 上以标量 ϕ^* 为下界。

假设 1.2 (Lipschitz 连续梯度)。函数 ϕ 连续可微，且 ϕ 的梯度在 $\mathbb{R}^n$ 上是 L -Lipschitz 连续的，即， $\|\nabla\phi(y) - \nabla\phi(x)\| \leq L\|y - x\|$ 对于所有 $(y, x) \in \mathbb{R}^n \times \mathbb{R}^n$ 。

在整篇论文中，我们假设无法访问 $\nabla\phi(x)$ ，也无法通过任何微分形式获得其近似。相反，我们可以访问非精确零阶预言机 $f(x) \approx \phi(x)$ 。具体而言，对于所有 x ，我们假设

$$|f(x) - \phi(x)| \leq \epsilon_f$$

对于某个 $\epsilon_f > 0$ 。

我们的主要目标是推导出具有以下类型保证的算法。对于给定的 $\epsilon > 0$ ，令 C_ϵ 表示给定算法对预言机 $f(x)$ 的总调用次数，该算法保证能达到一个点 x_ϵ ，使得 $\|\nabla\phi(x_\epsilon)\| \leq \epsilon$ 。于是我们寻求如下保证： $\forall \epsilon > \psi(\epsilon_f)$ $C_\epsilon \leq \Psi(n, \epsilon)$ ，如果算法是确定性的；或者 $\mathbb{E}[C_\epsilon] \leq \Psi(n, \epsilon)$ ，如果算法是随机的。这里 ψ 是预言机噪声的某个函数，该函数

gives the best achievable optimality criterion and Ψ is the complexity bound. We will make use of $\mathcal{O}()$ notation to suppress dependence on constants in upper bounds, $\Omega()$ notation to do the same in the case of lower bounds and $\Theta()$ to be used with an equality up to a constant factor. For all methods considered here as for all methods of similar type $\Psi() = \mathcal{O}(\epsilon^{-2})$. Thus the main focus of this work is in the dependence of $\Psi(n, \epsilon)$ on n and the dependence of $\psi()$ on n and ϵ_f . This reflects practical concerns with respect to derivative-free optimization, which in contrast to the usual derivative-based optimization has complexity dependence on n and is known not to scale well for large dimensional problems.

The paper is organized as follows: In Section 2 we present the analysis of the basic model-based trust region framework and present some key results based on what is known as fully-linear models that drive complexity analysis. In Section 3 we establish how these fully linear models arise in polynomial interpolation and quantify their properties in terms of Lagrange polynomials associated with the sample sets. In Section 4 we propose our main algorithm that uses Lagrange polynomials as a tool for model maintenance and analyze its complexity. Section 5 focuses on the subspace trust region method with inexact zeroth-order oracle. Finally, in Section 6 we describe implementational enhancements and present our computational comparisons.

2 Basic algorithm and elements of complexity analysis. We first present and analyze the trust region framework with the focus on its key elements.

As in most trust region algorithms, at every iteration $k \in \{0, 1, \dots\}$, we construct a quadratic model to approximate $\phi(x)$ near the iterate x_k

$$(2.1) \quad m_k(x_k + s) = \phi(x_k) + g_k(x_k)^T s + \frac{1}{2} s^T H_k(x_k) s.$$

The model is then minimized (approximately) over the trust region $B(x_k, \Delta_k)$ - a Euclidean ball around x_k of a radius Δ_k . In the paper we use the abbreviation $g_k := g(x_k) = \nabla m_k(x_k)$ and $H_k := H(x_k) = \nabla^2 m_k(x_k)$.¹

The following definition, also widely used in the literature [13], helps us identify the requirements on models m_k that are critical for convergence.

DEFINITION 2.1 (Fully-linear model). *Given a ball around point x of radius Δ , $B(x, \Delta)$, we say that model $m(x + s)$ is a κ_{ef}, κ_{eg} -fully linear model of $\phi(x + s)$ on $B(x, \Delta)$ if*

$$\|\nabla m(x) - \nabla \phi(x)\| \leq \kappa_{eg} \Delta$$

and

$$|m(x + s) - \phi(x + s)| \leq \kappa_{ef} \Delta^2$$

for all $\|s\| \leq \Delta$.

We now state the algorithmic framework where at each iteration updates are made based on progress and also on whether or not the model is known to be fully linear. We do not need to specify κ_{eg} and κ_{ef} constants in the algorithm but we assume they exist and are fixed throughout the iterations. In the framework below we do not even discuss how to verify if a model is κ_{ef}, κ_{eg} -fully-linear model, we simply assume such mechanism is given as an input. In the later section we will address this aspect in detail and incorporate it into the algorithm.

¹Note that the constant term $\phi(x_k)$ appears in the definition (2.1) but will not be needed in the algorithm, since only changes in the model value $m_k(x_k) - m_k(x_k + s)$ are of interest.

给出可达到的最佳最优性准则，而 Ψ 是复杂度界。我们将使用 $\mathcal{O}()$ 记号来在上界中忽略对常数的依赖，使用 $\Omega()$ 记号在下界中做同样的事，并使用 $\Theta()$ 表示在相差一个常数因子意义下相等。对于这里考虑的所有方法，如同所有类似类型的方法一样， $\Psi() = \mathcal{O}(\epsilon^{-2})$ 。因此，这项工作的主要关注点是 $\Psi(n, \epsilon)$ 对 n 的依赖，以及 $\psi()$ 对 n 和 ϵ_f 的依赖。这反映了对无导数优化的实际关切：与通常的基于导数的优化不同，无导数优化的复杂度依赖于 n ，并且已知在高维问题中不能很好地扩展。

本文组织如下：在第2节中，我们分析基本的基于模型的信赖域框架，并给出一些基于所谓完全线性模型的关键结果，这些结果驱动复杂度分析。在第3节中，我们说明这些完全线性模型如何在多项式插值中产生，并以与样本集相关联的拉格朗日多项式来量化其性质。在第4节中，我们提出使用拉格朗日多项式作为模型维护工具的主算法，并分析其复杂度。第5节关注带有非精确零阶预言机的子空间信赖域方法。最后，在第6节中，我们描述实现增强并给出我们的计算比较。

2 基本算法与复杂度分析要素。 我们首先介绍并分析信赖域框架，重点放在其关键要素上。

与大多数信赖域算法一样，在每次迭代 $k \in \{0, 1, \dots\}$ 时，我们构造一个二次模型，用以在迭代点 x_k 附近近似 $\phi(x)$

$$(2.1) \quad m_k(x_k + s) = \phi(x_k) + g_k(x_k)^T s + \frac{1}{2} s^T H_k(x_k) s.$$

随后，在信赖域 $B(x_k, \Delta_k)$ 上（近似地）最小化模型——该信赖域是以 x_k 为中心、半径为 Δ_k 的欧几里得球。在本文中，我们使用缩写 $g_k := g(x_k) = \nabla m_k(x_k)$ 和 $H_k := H(x_k) = \nabla^2 m_k(x_k)$ 。¹

以下定义同样在文献 [13] 中被广泛使用，它有助于我们确定对模型 m_k 的要求，这些要求对收敛至关重要。

定义 2.1（完全线性模型）。给定以点 x 为中心、半径为 Δ 的球 $B(x, \Delta)$ ，我们说模型 $m(x + s)$ 是 $\phi(x + s)$ 在 $B(x, \Delta)$ 上的一个 κ_{ef}, κ_{eg} -完全线性模型，如果

$$\|\nabla m(x) - \nabla \phi(x)\| \leq \kappa_{eg} \Delta$$

and

$$|m(x + s) - \phi(x + s)| \leq \kappa_{ef} \Delta^2$$

对于所有 $\|s\| \leq \Delta$ 。

我们现在陈述算法框架，其中在每次迭代中，根据进展以及模型是否已知为完全线性来进行更新。我们不需要在算法中具体指定 κ_{eg} 和 κ_{ef} 常数，但我们假设它们存在，并在整个迭代过程中保持固定。在下面的框架中，我们甚至不讨论如何验证一个模型是否为 κ_{ef}, κ_{eg} -完全线性模型，我们只是假设这种机制作为输入给出。在后续章节中，我们将详细讨论这一方面，并将其纳入算法。

¹注意，常数项 $\phi(x_k)$ 出现在定义（2.1）中，但在算法中并不需要，因为只有模型值 $m_k(x_k) - m_k(x_k + s)$ 的变化才是我们关心的。

Algorithm 1: Trust region method based on fully-linear models

Inputs: A zeroth-order oracle $f(x) \approx \phi(x)$, a starting point x_0 , TR radius Δ_0 , and hyperparameters $\eta_1 \in (0, 1)$, $\eta_2 > 0$, and $\gamma \in (0, 1)$. Mechanism for establishing if a model is κ_{ef}, κ_{eg} -fully-linear for some fixed κ_{ef}, κ_{eg} .

for $k = 0, 1, 2, \dots$ **do**

1 Compute model m_k and a trial step $x_k + s_k$ where $s_k \approx \arg \min_s \{m_k(x_k + s) : s \in B(0, \Delta_k)\}$.

2 Compute the ratio ρ_k as

$$\rho_k = \frac{f(x_k) - f(x_k + s_k)}{m_k(x_k) - m_k(x_k + s_k)}.$$

3 Update the iterate and the TR radius as

$$(x_{k+1}, \Delta_{k+1}) \leftarrow \begin{cases} (x_k + s_k, \gamma^{-1} \Delta_k) & \text{if } \rho_k \geq \eta_1 \text{ and } \|g_k\| \geq \eta_2 \Delta_k, \\ (x_k, \Delta_k) & \text{else, if model is not FL in } B(x_k, \Delta_k). \\ (x_k, \gamma \Delta_k) & \text{otherwise.} \end{cases}$$

4 Perform some model improvement steps if appropriate.

We will make the following standard assumption on the models m_k and their minimization.

ASSUMPTION 2.2. 1. *The trust region subproblem is solved sufficiently accurately in each iteration k so that $x_k + s_k$ provides at least a fraction of Cauchy decrease, i.e. for some constant $0 < \kappa_{fcd} < 1$*

$$(2.2) \quad m_k(x_k) - m_k(x_k + s_k) \geq \frac{\kappa_{fcd}}{2} \|g_k\| \min \left\{ \frac{\|g_k\|}{\|H_k\|}, \Delta_k \right\}.$$

2. *There exists a constant $\kappa_{bhm} > 0$ such that, for all x_k generated by Algorithm 1, the spectral norm of the Hessian of the model is bounded as*

$$\|H_k\| \leq \kappa_{bhm}.$$

Condition (2.2) is commonly used in the literature and is satisfied by the Cauchy point with $\kappa_{fcd} = 1$. See [12, Section 6.3.2] for more details.

Algorithm 1 is a variant of a standard trust region method, well studied in the literature [12]. There are two key differences that appear in trust region methods specifically in the DFO context. The first one is the condition $\|g_k\| \geq \eta_2 \Delta_k$. This condition is not used in classical TR methods where $\nabla m_k(x_k) = g_k = \nabla \phi(x_k)$ and was first introduced in [3] for the case of a TR method based on random models. The reason for this condition is tied to the fact that the trust region radii have a dual function in this setting - controlling the step of the algorithm and also controlling the model accuracy. For the same reasons, most prior deterministic DFO literature relies on a much less practical and cumbersome "criticality step". The condition $\|g_k\| \geq \eta_2 \Delta_k$ ensures that the trust region radius and thus the gradient accuracy stays on track as the norm of the gradient reduces which removes the need for a separate criticality step. The second difference for the derivative-free model-based TR (versus the classical TR) method is the necessity to ensure that the model is fully-linear in the trust region when the trust region radius is reduced. One can simply assume that this is guaranteed, for example, by using finite difference gradient approximation to build the model at each iteration. This gives an easy and convenient way to analyze the algorithm, but this does not result in a practical method. In various prior works [22, 13, 11] efficient methods have been proposed to recognize whether a model is fully-linear and if not, to make an improving step. The purpose of this paper is to improve on these methods both in terms of theory and practice.

Step 3 of Algorithm 1 identifies three types of iterations: *successful* iteration, where the trial step is accepted and the trust region radius is increased, *unsuccessful* iteration, where the step is rejected and the trust region radius is decreased and what we will call *model improving* iteration where the step is rejected but the trust region radius is not decreased because the model is not fully-linear.

In this section we will derive the bound on the total number of successful and unsuccessful iterations and in the following section we will provide a mechanism for model improvement and will bound the number of the model improving iterations. For a given $\epsilon > 0$, let K_ϵ be the first iteration of Algorithms 1 for which $\|\nabla \phi(x_k)\| \leq \epsilon$. We

算法1：基于完全线性模型的信赖域方法

输入：一个零阶预言机 $f(x) \approx \phi(x)$ 、初始点 x_0 、信赖域半径 Δ_0 ，以及超参数 $\eta_1 \in (0, 1)$ 、 $\eta_2 > 0$

和 $\gamma \in (0, 1)$ 。用于确定模型对于某个固定的 κ_{ef} 、 κ_{eg} 是否为 κ_{ef} 、 κ_{eg} -完全线性的机制。对于 $k = 0$,

1, 2, $\dots$ **执行**1 计算模型 m_k 和试探步 $x_k + s_k$ ，其中 $s_k \approx \arg \min_s \{m_k(x_k + s) : s \in B(0, \Delta_k)\}$ 。

2 计算比率 ρ_k 如下：3 按如下方式更新迭代点和信赖域半径：4 如果合适，执行一些模型改进步骤。

$$\rho_k = \frac{f(x_k) - f(x_k + s_k)}{m_k(x_k) - m_k(x_k + s_k)}.$$

$$(x_{k+1}, \Delta_{k+1}) \leftarrow \begin{cases} (x_k + s_k, \gamma^{-1} \Delta_k) & \text{if } \rho_k \geq \eta_1 \text{ and } \|g_k\| \geq \eta_2 \Delta_k, \\ (x_k, \Delta_k) & \text{else, if model is not FL in } B(x_k, \Delta_k). \\ (x_k, \gamma \Delta_k) & \text{otherwise.} \end{cases}$$

We 将就模型 m_k 及其最小化作出以下标准假设

假设2.2. 1. 在每个迭代 k 中，信赖域子问题都被足够精确地求解，以使得 $x_k + s_k$ 至少提供一定比例的柯西下降，即对于某个常数 $0 < \kappa_{fcd} < 1$

$$(2.2) \quad m_k(x_k) - m_k(x_k + s_k) \geq \frac{\kappa_{fcd}}{2} \|g_k\| \min \left\{ \frac{\|g_k\|}{\|H_k\|}, \Delta_k \right\}.$$

2. 存在一个常数 $\kappa_{bhm} > 0$ ，使得对于由算法1生成的所有 x_k ，模型的Hessian矩阵的谱范数有界如下

$$\|H_k\| \leq \kappa_{bhm}.$$

条件(2.2)在文献中常用，并且可由柯西点在 $\kappa_{fcd} = 1$ 下满足。更多细节参见[12, 第6.3节2]。

算法1是标准信赖域方法的一种变体，该方法在文献 [12] 中已有充分研究。在无导数优化背景下，信赖域方法中会出现两个关键差异。第一个差异是条件 $\|g_k\| \geq \eta_2 \Delta_k$ 。该条件不用于经典信赖域方法，其中 $\nabla m_k(x_k) = g_k = \nabla \phi(x_k)$ ；它首次在 [3] 中针对基于随机模型的信赖域方法情形被引入。该条件的原因在于：在此情形下，信赖域半径具有双重功能——控制算法的步长，同时也控制模型精度。出于同样原因，大多数此前的确定性无导数优化文献都依赖于一个远不那么实用且繁琐的“临界步”。条件 $\|g_k\| \geq \eta_2 \Delta_k$ 确保随着梯度范数降低，信赖域半径以及因此的梯度精度保持在正轨上，从而无需单独的临界步。对于基于模型的无导数信赖域方法（相较于经典信赖域方法），第二个差异是必须确保当信赖域半径减小时，模型在信赖域内是完全线性的。例如，通过使用有限差分梯度近似在每次迭代中构建模型，就可以简单地假设这一点能够得到保证。这提供了一种简单方便的分析算法的方式，但这并不能产生实用的方法。在此前的多项工作 [22, 13, 11] 中，已经提出了高效方法，用于识别模型是否完全线性，并在不是完全线性时进行改进步。本文的目的在于从理论和实践两方面改进这些方法。

算法1的第3步识别出三种类型的迭代：成功迭代，其中试探步被接受且信赖域半径增加，不成功迭代，其中该步被拒绝且信赖域半径减小，以及我们称之为 模型改进迭代，其中该步被拒绝但信赖域半径未减小，因为模型不是完全线性的。

在本节中，我们将推导成功迭代和不成功迭代总数的界，并在下一节中提供一种模型改进机制，同时给出模型改进迭代次数的界。对于给定的 $\epsilon > 0$ ，令 K_ϵ 为算法 1 的第一个满足 $\|\nabla \phi(x_k)\| \leq \epsilon$ 的迭代。我们

define the index sets

$$(2.3) \quad \begin{aligned} \mathcal{S}_\epsilon &:= \{k \in \{0, \dots, K_\epsilon - 1\} : \text{iteration } k \text{ is successful}\}. \\ \mathcal{M}_\epsilon &:= \{k \in \{0, \dots, K_\epsilon - 1\} : \text{iteration } k \text{ is model improving}\}. \\ \mathcal{U}_\epsilon &:= \{k \in \{0, \dots, K_\epsilon - 1\} : \text{iteration } k \text{ is unsuccessful}\}. \end{aligned}$$

We next present two lemmas that are key in the analysis of any trust region algorithm and specifically Algorithm 1. The first lemma establishes that once Δ_k is sufficiently small compared to the gradient norm, a successful step is ensured.

LEMMA 2.3 (small Δ_k implies successful step). *Under Assumption 2.2, if m_k is κ_{ef}, κ_{eg} -fully-linear and*

$$(2.4) \quad \sqrt{\frac{2\epsilon_f}{C_0}} \leq \Delta_k \leq C_1 \|\nabla \phi(x_k)\|, \text{ where } C_1 = \left(\max \left\{ \eta_2, \kappa_{bhm}, \frac{2\kappa_{ef} + C_0}{(1 - \eta_1)\kappa_{fcd}} \right\} + \kappa_{eg} \right)^{-1},$$

and C_0 is an arbitrary constant which we pick to equal $\max\{\eta_2, \kappa_{ef}\}$ for future convenience, then $\rho \geq \eta_1$, $\|g_k\| \geq \eta_2 \Delta_k$, thus iteration k is successful and $x_{k+1} = x_k + s_k$.

The proof is a simple modification of those in the stochastic trust region literature [6, 7]²

Proof. By the assumption that m_k is fully-linear, we have

$$\|\nabla \phi(x_k)\| \leq \|g_k\| + \kappa_{eg} \Delta_k.$$

From (2.4) we conclude that

$$\begin{aligned} (\max\{\kappa_{bhm}, \eta_2\} + \kappa_{eg}) \Delta_k &\leq \|\nabla \phi(x_k)\| \leq \|g_k\| + \kappa_{eg} \Delta_k \\ \max\{\kappa_{bhm}, \eta_2\} \Delta_k &\leq \|g_k\|. \end{aligned}$$

This implies that the first condition of the successful step, namely $\|g_k\| \geq \eta_2 \Delta_k$, is satisfied. From (2.2) we have $m_k(x_k) - m_k(x_k + s_k) \geq \kappa_{fcd} \|g_k\| \Delta_k / 2$. Thus, using the assumption that m_k is fully-linear again,

$$\begin{aligned} \rho_k &= \frac{m(x_k) - m(x_k + s_k) + (f(x_k) - m(x_k)) - (f(x_k + s_k) - m(x_k + s_k))}{m(x_k) - m(x_k + s_k)} \\ &= \frac{m(x_k) - m(x_k + s_k) + (\phi(x_k) - m(x_k)) - (\phi(x_k + s_k) - m(x_k + s_k))}{m(x_k) - m(x_k + s_k)} \\ &\quad + \frac{f(x_k) - \phi(x_k) + (f(x_k + s_k) - \phi(x_k + s_k))}{m(x_k) - m(x_k + s_k)} \\ &\geq 1 - \frac{\kappa_{ef} \Delta_k^2}{m(x_k) - m(x_k + s_k)} - \frac{2\epsilon_f}{m(x_k) - m(x_k + s_k)} \\ &\geq 1 - \frac{\kappa_{ef} \Delta_k^2}{\kappa_{fcd} \|g_k\| \Delta_k / 2} - \frac{2\epsilon_f}{\kappa_{fcd} \|g_k\| \Delta_k / 2} \geq 1 - \frac{(2\kappa_{ef} + 2\epsilon_f / \Delta_k^2) \Delta_k}{\kappa_{fcd} (\|\nabla \phi(x_k)\| - \kappa_{eg} \Delta_k)} \\ &\geq 1 - \frac{(2\kappa_{ef} + C_0) \Delta_k}{\kappa_{fcd} (\|\nabla \phi(x_k)\| - \kappa_{eg} \Delta_k)} \geq \eta_1, \end{aligned}$$

where the last inequality follows from (2.4) since $\|\nabla \phi(x_k)\| \geq (\frac{2\kappa_{ef}}{(1 - \eta_1)\kappa_{fcd}} + \kappa_{eg}) \Delta_k$. $\square$

LEMMA 2.4 (successful iteration implies function reduction). *Let Assumptions 1.2 and 2.2 hold. If the iteration k is successful, then*

$$(2.5) \quad \phi(x_k) - \phi(x_{k+1}) \geq C_2 \Delta_k^2 - 2\epsilon_f, \text{ where } C_2 = \frac{\eta_1 \eta_2 \kappa_{fcd}}{2} \min\left\{ \frac{\eta_2}{\kappa_{bhm}}, 1 \right\};$$

otherwise, we have $x_{k+1} = x_k$ and $\phi(x_k) - \phi(x_{k+1}) = 0$.

²This lemma was stated erroneously without proof in [11] with $4\epsilon_f$ instead of C_0 . The error is inconsequential, but we correct it here.

定义指标集

$$(2.3) \quad \begin{aligned} \mathcal{S}_\epsilon &:= \{k \in \{0, \dots, K_\epsilon - 1\} : \text{iteration } k \text{ is successful}\}. \\ \mathcal{M}_\epsilon &:= \{k \in \{0, \dots, K_\epsilon - 1\} : \text{iteration } k \text{ is model improving}\}. \\ \mathcal{U}_\epsilon &:= \{k \in \{0, \dots, K_\epsilon - 1\} : \text{iteration } k \text{ is unsuccessful}\}. \end{aligned}$$

接下来, 我们给出两个引理, 它们对于任何信赖域算法, 尤其是算法1的分析都至关重要。第一个引理表明, 一旦 Δ_k 相对于梯度范数足够小, 就能确保得到一个成功步。

引理 2.3 (小 Δ_k 蕴含成功步)。在假设2.2下, 如果 m_k 是 κ_{ef}, κ_{eg} -完全线性且

$$(2.4) \quad \sqrt{\frac{2\epsilon_f}{C_0}} \leq \Delta_k \leq C_1 \|\nabla \phi(x_k)\|, \text{ where } C_1 = \left(\max \left\{ \eta_2, \kappa_{bhm}, \frac{2\kappa_{ef} + C_0}{(1 - \eta_1)\kappa_{fcd}} \right\} + \kappa_{eg} \right)^{-1},$$

且 C_0 是一个任意常数, 为后续方便我们将其取为等于 $\max\{\eta_2, \kappa_{ef}\}$, 那么 $\rho \geq \eta_1 \|g_k\| \geq \eta_2 \Delta_k$, 因此迭代 k 是成功的, 且 $x_{k+1} = x_k + s_k$ 。

该证明是对随机信赖域文献中证明的简单修改 [6, 7]²

证明。根据 m_k 是完全线性的假设, 我们有

$$\|\nabla \phi(x_k)\| \leq \|g_k\| + \kappa_{eg} \Delta_k.$$

由 (2.4) 我们得出

$$\begin{aligned} (\max\{\kappa_{bhm}, \eta_2\} + \kappa_{eg}) \Delta_k &\leq \|\nabla \phi(x_k)\| \leq \|g_k\| + \kappa_{eg} \Delta_k \\ \max\{\kappa_{bhm}, \eta_2\} \Delta_k &\leq \|g_k\|. \end{aligned}$$

这意味着成功步的第一个条件, 即 $\|g_k\| \geq \eta_2 \Delta_k$, 得到了满足。由 (2.2) 我们有 $m_k(x_k) - m_k(x_k + s_k) \geq \kappa_{fcd} \|g_k\| \Delta_k / 2$ 。因此, 再次利用 m_k 是完全线性的这一假设,

$$\begin{aligned} \rho_k &= \frac{m(x_k) - m(x_k + s_k) + (f(x_k) - m(x_k)) - (f(x_k + s_k) - m(x_k + s_k))}{m(x_k) - m(x_k + s_k)} \\ &= \frac{m(x_k) - m(x_k + s_k) + (\phi(x_k) - m(x_k)) - (\phi(x_k + s_k) - m(x_k + s_k))}{m(x_k) - m(x_k + s_k)} \\ &\quad + \frac{f(x_k) - \phi(x_k) + (f(x_k + s_k) - \phi(x_k + s_k))}{m(x_k) - m(x_k + s_k)} \\ &\geq 1 - \frac{\kappa_{ef} \Delta_k^2}{m(x_k) - m(x_k + s_k)} - \frac{2\epsilon_f}{m(x_k) - m(x_k + s_k)} \\ &\geq 1 - \frac{\kappa_{ef} \Delta_k^2}{\kappa_{fcd} \|g_k\| \Delta_k / 2} - \frac{2\epsilon_f}{\kappa_{fcd} \|g_k\| \Delta_k / 2} \geq 1 - \frac{(2\kappa_{ef} + 2\epsilon_f / \Delta_k^2) \Delta_k}{\kappa_{fcd} (\|\nabla \phi(x_k)\| - \kappa_{eg} \Delta_k)} \\ &\geq 1 - \frac{(2\kappa_{ef} + C_0) \Delta_k}{\kappa_{fcd} (\|\nabla \phi(x_k)\| - \kappa_{eg} \Delta_k)} \geq \eta_1, \end{aligned}$$

其中最后一个不等式由 (2.4) 推出, 因为 $\|\nabla \phi(x_k)\| \geq (\frac{2\kappa_{ef}}{(1 - \eta_1)\kappa_{fcd}} + \kappa_{eg}) \Delta_k$ 。 $\square$

引理2.4 (成功迭代蕴含函数下降)。令假设 1.2 和 2.2 成立。如果迭代 k 成功, 则

$$(2.5) \quad \phi(x_k) - \phi(x_{k+1}) \geq C_2 \Delta_k^2 - 2\epsilon_f, \text{ where } C_2 = \frac{\eta_1 \eta_2 \kappa_{fcd}}{2} \min\left\{ \frac{\eta_2}{\kappa_{bhm}}, 1 \right\};$$

否则, 我们有 $x_{k+1} = x_k$ 和 $\phi(x_k) - \phi(x_{k+1}) = 0$ 。

²本引理在 [11] 中被错误地陈述, 且没有证明, 其中用 $4\epsilon_f$ 代替了 C_0 。这个错误无关紧要, 但我们在此予以更正。

This proof can be found in [11]. Thus, under the additional assumption that $\Delta_k \geq \sqrt{\frac{2\epsilon_f}{\tau C_2}}$ for some $\tau \in (0, 1)$ (2.5) can be further stated as

$$(2.6) \quad \phi(x_k) - \phi(x_{k+1}) \geq (1 - \tau)C_2\Delta_k^2, \text{ where } C_2 = \frac{\eta_1\eta_2\kappa_{fcd}}{2} \min\left\{\frac{\eta_2}{\kappa_{bhm}}, 1\right\}.$$

Lemmas 2.3 and 2.4 give us a bound on the total number of successful and unsuccessful iterations, as long as it can be ensured that Δ_k remains not smaller than $\sqrt{\frac{2\epsilon_f}{\min\{\tau C_2, C_0\}}}$. Henceforth, for simplicity we use $\tau = \frac{1}{2}$, also since we chose $C_0 \geq \eta_2$ and $\eta_2 \geq C_2$ the bound on Δ_k reduces to $\sqrt{\frac{4\epsilon_f}{C_2}}$. Due to Lemma 2.3 and the update mechanism for Δ_k the bound is ensured as long as $\|\nabla\phi(x_k)\| \geq \epsilon$ for ϵ sufficiently large.

LEMMA 2.5 (Lower bound on Δ_k). *For any $\epsilon > \sqrt{\frac{4\epsilon_f}{\gamma^2 C_2 C_1^2}}$, assuming that $\Delta_0 > \gamma C_1 \epsilon$, for all $k \in \{0, \dots, K_\epsilon - 1\}$ $\Delta_k \geq \gamma C_1 \epsilon$.*

Proof. According to Lemma 2.3, any iteration $k \in \{0, \dots, K_\epsilon - 1\}$ must be successful when $\Delta_k \leq C_1 \epsilon$ and m_k is κ_{ef}, κ_{eg} -fully linear. Thus no iteration can be unsuccessful when $\Delta_k \leq C_1 \epsilon$. Thus, given $\Delta_0 > \gamma C_1 \epsilon$, and by the mechanism of Algorithm 1 we must have $\Delta_k \geq \gamma C_1 \epsilon$ for all $k \in \{0, \dots, K_\epsilon - 1\}$. $\square$

The following bound holds under the result of Lemma 2.5.

LEMMA 2.6 (Bound of successful iterations). *For any $\epsilon > \sqrt{\frac{4\epsilon_f}{\gamma^2 C_2 C_1^2}}$, assuming $\Delta_0 > \gamma C_1 \epsilon$, we have*

$$|\mathcal{S}_\epsilon| \leq \frac{2(\phi(x_0) - \phi^*)}{C_2(\gamma C_1 \epsilon)^2}$$

Proof. Using Lemma 2.4 with $\tau = \frac{1}{2}$ and from $\Delta_k \geq \gamma C_1 \epsilon$ for all $k \in \{0, \dots, K_\epsilon - 1\}$ we have

$$\phi(x_0) - \phi^* \geq \sum_{k=0}^{K_\epsilon-1} \phi(x_k) - \phi(x_{k+1}) \geq \frac{1}{2} \sum_{k \in \mathcal{S}_\epsilon} C_2 \Delta_k^2 > \frac{1}{2} |\mathcal{S}_\epsilon| C_2 (\gamma C_1 \epsilon)^2$$

which gives the result of the lemma. $\square$

LEMMA 2.7. *For any $\epsilon > \sqrt{\frac{4\epsilon_f}{\gamma^2 C_2 C_1^2}}$, assuming that the initial trust-region radius $\Delta_0 > \gamma C_1 \epsilon$,*

$$|\mathcal{U}_\epsilon| \leq |\mathcal{S}_\epsilon| + \lceil \left(\log_\gamma \frac{C_1 \epsilon}{\Delta_0} \right) \rceil.$$

Proof. We observe that

$$\Delta_{K_\epsilon} = \gamma^{-|\mathcal{S}_\epsilon|} \gamma^{|\mathcal{U}_\epsilon|} \Delta_0 \geq C_1 \epsilon,$$

The last inequality follows from the fact that $\Delta_{K_\epsilon-1} \geq \gamma C_1 \epsilon$ and the $K_\epsilon - 1$ -th iteration must be successful. Thus the number of unsuccessful iterations can be bounded using the number of successful ones rearranging the terms and taking the logarithm. $\square$

THEOREM 2.8. *Let Assumptions 1.2 and 2.2 hold. For any $\epsilon > \sqrt{\frac{4\epsilon_f}{\gamma^2 C_2 C_1^2}}$, assuming that the initial trust-region radius $\Delta_0 > \gamma C_1 \epsilon$, where C_1 is defined as $\left(\max \left\{ \eta_2, \kappa_{bhm}, \frac{2\kappa_{ef} + \max\{\eta_2, \kappa_{ef}\}}{(1-\eta_1)\kappa_{fcd}} \right\} + \kappa_{eg} \right)^{-1}$, then we have the bound*

$$(2.7) \quad |\mathcal{S}_\epsilon| + |\mathcal{U}_\epsilon| \leq \frac{4(\phi(x_0) - \phi^*)}{C_2(\gamma C_1 \epsilon)^2} + \lceil \left(\log_\gamma \frac{C_1 \epsilon}{\Delta_0} \right) \rceil.$$

Constants C_1 and C_2 have a direct effect on the complexity and we will be using them (and their variations) throughout the paper. Let us pause here to understand their different components. Specifically, constants γ , η_1 and κ_{fcd} are usually chosen to be fixed in the algorithm and be close to 1 (say 0.9).

该证明可在 [11] 中找到。因此，在额外假设 $\Delta_k \geq \sqrt{\frac{2\epsilon_f}{\tau C_2}}$ 对某个 $\tau \in (0, 1)$ 成立的情况下，(2.5) 可进一步表述为

$$(2.6) \quad \phi(x_k) - \phi(x_{k+1}) \geq (1 - \tau)C_2\Delta_k^2, \text{ where } C_2 = \frac{\eta_1\eta_2\kappa_{fcd}}{2} \min\left\{\frac{\eta_2}{\kappa_{bhm}}, 1\right\}.$$

引理 2.3 和 2.4 给出了成功迭代和不成功迭代总数的界，只要能够确保 Δ_k 不小于 $\sqrt{\frac{2\epsilon_f}{\min\{\tau C_2, C_0\}}}$ 。此后，为简单起见，我们使用 $\tau = \frac{1}{2}$ ，同时由于我们选择了 $C_0 \geq \eta_2$ 和 $\eta_2 \geq C_2$ ， Δ_k 的界简化为 $\sqrt{\frac{4\epsilon_f}{C_2}}$ 。由于引理 2.3 和 Δ_k 的更新机制，只要 $\|\nabla\phi(x_k)\| \geq \epsilon$ 对于充分大的 ϵ 成立，就能确保该界。

引理 2.5 (关于 Δ_k 的下界)。对于任意 $\epsilon > \sqrt{\frac{4\epsilon_f}{\gamma^2 C_2 C_1^2}}$ ，假设 $\Delta_0 > \gamma C_1 \epsilon$ ，对于所有 $k \in \{0, \dots, K_\epsilon - 1\}$ $\Delta_k \geq \gamma C_1 \epsilon$ 。

证明。根据引理 2.3，当 $\Delta_k \leq C_1 \epsilon$ 和 m_k 为 κ_{ef}, κ_{eg} -完全线性时，任何迭代 $k \in \{0, \dots, K_\epsilon - 1\}$ 都必须成功。因此当 $\Delta_k \leq C_1 \epsilon$ 时，没有迭代会不成功。因此，给定 $\Delta_0 > \gamma C_1 \epsilon$ ，并且根据算法 1 的机制，我们必须对所有 $k \in \{0, \dots, K_\epsilon - 1\}$ 有 $\Delta_k \geq \gamma C_1 \epsilon$ 。 $\square$

以下界在引理 2.5 的结果下成立。

引理 2.6 (成功迭代的界)。对于任意 $\epsilon > \sqrt{\frac{4\epsilon_f}{\gamma^2 C_2 C_1^2}}$ ，假设 $\Delta_0 > \gamma C_1 \epsilon$ ，我们有

$$|\mathcal{S}_\epsilon| \leq \frac{2(\phi(x_0) - \phi^*)}{C_2(\gamma C_1 \epsilon)^2}$$

证明。利用引理 2.4，取 $\tau = \frac{1}{2}$ ，并由 $\Delta_k \geq \gamma C_1 \epsilon$ 对所有 $k \in \{0, \dots, K_\epsilon - 1\}$ ，我们有

$$\phi(x_0) - \phi^* \geq \sum_{k=0}^{K_\epsilon-1} \phi(x_k) - \phi(x_{k+1}) \geq \frac{1}{2} \sum_{k \in \mathcal{S}_\epsilon} C_2 \Delta_k^2 > \frac{1}{2} |\mathcal{S}_\epsilon| C_2 (\gamma C_1 \epsilon)^2$$

由此得到引理的结果。 $\square$

引理 2.7. 对于任意 $\epsilon > \sqrt{\frac{4\epsilon_f}{\gamma^2 C_2 C_1^2}}$ ，假设初始信赖域半径 $\Delta_0 > \gamma C_1 \epsilon$ ，

$$|\mathcal{U}_\epsilon| \leq |\mathcal{S}_\epsilon| + \lceil \left(\log_\gamma \frac{C_1 \epsilon}{\Delta_0} \right) \rceil.$$

证明。我们观察到

$$\Delta_{K_\epsilon} = \gamma^{-|\mathcal{S}_\epsilon|} \gamma^{|\mathcal{U}_\epsilon|} \Delta_0 \geq C_1 \epsilon,$$

最后一个不等式由 $\Delta_{K_\epsilon-1} \geq \gamma C_1 \epsilon$ 以及第 $K_\epsilon - 1$ 次迭代必须是成功迭代这一事实推出。因此，通过重新排列各项并取对数，可以用成功迭代的次数来界定不成功迭代的次数。 $\square$

定理 2.8. 假设 1.2 和 2.2 成立。对于任意 $\epsilon > \sqrt{\frac{4\epsilon_f}{\gamma^2 C_2 C_1^2}}$ ，假设初始信赖域半径 $\Delta_0 > \gamma C_1 \epsilon$ ，其中 C_1 定义为 $\left(\max \left\{ \eta_2, \kappa_{bhm}, \frac{2\kappa_{ef} + \max\{\eta_2, \kappa_{ef}\}}{\kappa_{fcd}} \right\} + \kappa_{eg} \right)^{-1} (1 - \eta_1)$ ，则我们得到界

$$(2.7) \quad |\mathcal{S}_\epsilon| + |\mathcal{U}_\epsilon| \leq \frac{4(\phi(x_0) - \phi^*)}{C_2(\gamma C_1 \epsilon)^2} + \lceil \left(\log_\gamma \frac{C_1 \epsilon}{\Delta_0} \right) \rceil.$$

常数 C_1 和 C_2 对复杂度有直接影响，我们将在整篇论文中使用它们（及其变体）。让我们在此暂停，以理解它们的不同组成部分。具体来说，常数 γ 、 η_1 和 κ_{fcd} 通常在算法中被选为固定值，并接近 1（例如 0.9）。

The key remaining constants are η_2 which is chosen in the algorithm and is used as specified, and κ_{ef} , κ_{eg} and κ_{bhm} which are all attributes of the constructed models and are not specified by the algorithm but are rather upper bounded by theory.

In what follows we will impose an upper bound on κ_{bhm} which will be dimension independent. Ideally κ_{bhm} should scale similarly to L , since the later is the bound on the norm of the true Hessian and the former is the bound on the model Hessian. Thus it is convenient to think of κ_{bhm} as $\sim \mathcal{O}(L)$, although it also can be zero if linear models are used but also can be large if allowed.

What remains is to derive concrete bounds on κ_{eg}, κ_{ef} with explicit dependence on dimension. These would depend on the manner in which the models are constructed.

The following standard lemma shows that a bound κ_{eg} (together with κ_{bhm}) implies a bound on κ_{ef} .

LEMMA 2.9 (Fully linear models). *Under Assumptions 1.2 and 2.2 if*

$$(2.8) \quad \|\nabla m(x) - \nabla \phi(x)\| \leq \kappa_{eg} \Delta$$

then $m_k(x_k + s)$ is a κ_{ef}, κ_{eg} -fully linear model of $\phi(x + s)$ on $B(x, \Delta)$ with $\kappa_{ef} = \kappa_{eg} + \frac{L + \kappa_{bhm}}{2}$.

Let us consider a concrete way of building fully-linear models and the complexity implications. From analysis of the finite difference gradient approximation error (see e.g.[4]), if one forms a gradient estimate via

$$(2.9) \quad g(x) = \sum_{i=1}^n \frac{f(x + \delta u_i) - f(x)}{\delta} u_i,$$

then one can derive a gradient approximation bound of

$$\|g(x) - \nabla \phi(x)\| \leq \frac{\sqrt{n}L\delta}{2} + \frac{2\sqrt{n}\epsilon_f}{\delta}.$$

Taking $\delta = \Delta_k$, at each iteration k , results in a κ_{ef}, κ_{eg} -fully linear model in $B(x_k, \Delta_k)$ with $\kappa_{eg} = \sqrt{n}L$ and $\kappa_{ef} = \frac{L + \kappa_{bhm}}{2} + \sqrt{n}L$, as long as $\Delta_k \geq 2\sqrt{\frac{\epsilon_f}{L}}$. To ensure this we add the lower bound $\gamma C_1 \epsilon \geq 2\sqrt{\frac{\epsilon_f}{L}}$ which translates to $\epsilon \geq \sqrt{\frac{4\epsilon_f}{\gamma^2 L C_1^2}}$.

The total bound on the number of function evaluations, i.e., oracle complexity, easily follows from Theorem 2.8 and from the fact that (2.9) gives a fully linear model at the cost of $n+1$ oracle calls. Thus there are no model improvement iterations and the final bound on the oracle complexity is derived via the bound on $(n+1)(|\mathcal{S}_\epsilon| + |\mathcal{U}_\epsilon|)$. As noted in [11], if η_2 is taken to be a constant independent of dimension, using this finite difference scheme, the total worse-case oracle complexity to achieve $\|\nabla \phi(x_k)\| \leq \epsilon$ for any $\epsilon > \sqrt{\frac{4\epsilon_f}{\gamma^2 \min\{C_2, L\} C_1^2}}$ is bounded by

$$\mathcal{C}_\epsilon \leq \mathcal{O}(n^2 \epsilon^{-2}).$$

In the following corollary, we show that this complexity is not optimal and can, in fact, be improved by taking η_2 to grow with the dimension of the problem by improving the bound on $|\mathcal{S}_\epsilon| + |\mathcal{U}_\epsilon|$ when κ_{ef} and κ_{eg} scale as $\mathcal{O}(\sqrt{n})$.

COROLLARY 2.10. *Under the same assumptions as Theorem 2.8, assuming that $g_k(x_k)$ is computed via (2.9) with $\delta = \Delta_k$ at each iteration, choosing $\eta_2 = \sqrt{n}$, and assuming $\kappa_{bhm} \leq \mathcal{O}(\sqrt{n})$, then for any $\epsilon > \sqrt{\frac{4\epsilon_f}{\gamma^2 \min\{C_2, L\} C_1^2}}$*

$$|\mathcal{S}_\epsilon| + |\mathcal{U}_\epsilon| \leq \mathcal{O}\left(\frac{\sqrt{n}}{\epsilon^2}\right)$$

and the total worst-case oracle complexity is bounded by

$$\mathcal{C}_\epsilon \leq \mathcal{O}\left(\frac{n^{3/2}}{\epsilon^2}\right)$$

其余的关键常数是 η_2 ，它在算法中被选定并按指定方式使用；以及 κ_{ef} 、 κ_{eg} 和 κ_{bhm} ，它们都是所构建模型的属性，并非由算法指定，而是由理论给出上界。

在下文中，我们将对 κ_{bhm} 施加一个与维数无关的上界。理想情况下， κ_{bhm} 应当以与 L 类似的方式缩放，因为后者是真实海森矩阵范数的界，而前者是模型海森矩阵的界。因此，将 κ_{bhm} 视为 $\sim \mathcal{O}(L)$ 是方便的，尽管如果使用线性模型，它也可以为零；但如果允许，它也可以很大。

剩下的是要得出关于 κ_{eg}, κ_{ef} 且显式依赖于维数的具体界。这些将取决于方式，其中模型是如何构造的。

The followin如下标准引理表明，一个界 κ_{eg} （连同 κ_{bhm} ）蕴含 κ_e 的一个界 f 。

引理 2.9（完全线性模型）。在假设 1.2 和 2.2 下，如果

$$(2.8) \quad \|\nabla m(x) - \nabla \phi(x)\| \leq \kappa_{eg} \Delta$$

那么 $m_k(x_k + s)$ 是 $\phi(x + s)$ 在 $B(x, \Delta)$ 上的一个 κ_{ef}, κ_{eg} -完全线性模型，其中 $\kappa_{ef} = \kappa_{eg} + \frac{L + \kappa_{bhm}}{2}$ 。

让我们考虑构建完全线性模型的一种具体方式及其复杂度影响。根据有限差分梯度近似误差的分析（例如参见 [4]），如果通过以下方式形成梯度估计

$$(2.9) \quad g(x) = \sum_{i=1}^n \frac{f(x + \delta u_i) - f(x)}{\delta} u_i,$$

那么可以推导出如下梯度近似界：

$$\|g(x) - \nabla \phi(x)\| \leq \frac{\sqrt{n}L\delta}{2} + \frac{2\sqrt{n}\epsilon_f}{\delta}.$$

取 $\delta = \Delta_k$ ，在每次迭代 k 时，会得到一个 κ_{ef}, κ_{eg} -完全线性模型，位于 $B(x_k, \Delta_k)$ 中，其中 $\kappa_{eg} = \sqrt{n}L$ 和 $\kappa_{ef} = \frac{L + \kappa_{bhm}}{2} + \sqrt{n}L$ ，只要 $\Delta_k \geq 2\sqrt{\frac{\epsilon_f}{L}}$ 。为确保这一点，我们添加下界 $\gamma C_1 \epsilon \geq 2\sqrt{\frac{\epsilon_f}{L}}$ ，其转化为 $\epsilon \geq \sqrt{\frac{4\epsilon_f}{\gamma^2 L C_1^2}}$ 。

函数求值次数的总界，即预言机复杂度，很容易由定理 2.8 以及以下事实推出：(2.9) 以 $n+1$ 次预言机调用为代价给出一个完全线性模型。因此不存在模型改进迭代，并且预言机复杂度的最终界是通过对 $(n+1)(|\mathcal{S}_\epsilon| + |\mathcal{U}_\epsilon|)$ 的界推导出来的。正如 [11]，中所指出的，如果 η_2 取为与维数无关的常数，则使用该有限差分格式时，为达到 $\|\nabla \phi(x_k)\| \leq \epsilon$ ，对于任意 $\epsilon > \sqrt{\frac{4\epsilon_f}{\gamma^2 \min\{C_2, L\} C_1^2}}$ ，总最坏情况预言机复杂度的上界为

$$\mathcal{C}_\epsilon \leq \mathcal{O}(n^2 \epsilon^{-2}).$$

在下面的推论中，我们表明该复杂度并非最优，事实上可以通过令 η_2 随问题维数增长，并在 κ_{ef} 和 κ_{eg} 按 $\mathcal{O}(\sqrt{n})$ 缩放时改进对 $|\mathcal{S}_\epsilon| + |\mathcal{U}_\epsilon|$ 的界来改善它。

推论 2.10. 在与定理 2.8 相同的假设下，假设 $g_k(x_k)$ 通过 (2.9) 使用 $\delta = \Delta_k$ 在每次迭代中计算，选择 $\eta_2 = \sqrt{n}$ ，并假设 $\kappa_{bhm} \leq \mathcal{O}(\sqrt{n})$ ，则对于任意 $\epsilon > \sqrt{\frac{4\epsilon_f}{\gamma^2 \min\{C_2, L\} C_1^2}}$

$$|\mathcal{S}_\epsilon| + |\mathcal{U}_\epsilon| \leq \mathcal{O}\left(\frac{\sqrt{n}}{\epsilon^2}\right)$$

并且总最坏情况预言机复杂度的上界为

$$\mathcal{C}_\epsilon \leq \mathcal{O}\left(\frac{n^{3/2}}{\epsilon^2}\right)$$

Proof. By the error bound from [4], we have $\kappa_{ef}, \kappa_{eg} = O(\sqrt{n})$. Recall $C_1^{-1} = \max \left\{ \eta_2, \kappa_{bhm}, \frac{2\kappa_{ef} + \max\{\eta_2, \kappa_{ef}\}}{(1-\eta_1)\kappa_{fcd}} \right\} + \kappa_{eg}$ and $C_2 = \frac{\eta_1\eta_2\kappa_{fcd}}{2} \min \left\{ \frac{\eta_2}{\kappa_{bhm}}, 1 \right\}$. Thus, under the assumptions of this corollary, we have $C_1^{-1} = \Theta(\sqrt{n})$ and $C_2 = \Theta(\sqrt{n})$. Thus, we can bound

$$\begin{aligned} |\mathcal{S}_\epsilon| + |\mathcal{U}_\epsilon| &\leq \frac{2(f(x_0) - f^\star)}{C_2(\gamma C_1 \epsilon)^2} + \log_\gamma \frac{C_1 \epsilon}{\Delta_0} \\ (2.10) \quad &= \mathcal{O} \left(\frac{\sqrt{n}(\phi(x_0) - \phi^\star)}{\epsilon^2} + \log \left(\frac{\sqrt{n}\Delta_0}{\epsilon} \right) \right) = \mathcal{O} \left(\frac{\sqrt{n}}{\epsilon^2} \right). \end{aligned}$$

The total oracle complexity follows as discussed before. $\square$

We note here that since we assume that L does not scale with n , while $C_1^{-1} = \Theta(\sqrt{n})$, then whether we choose η_2 to be constant or to equal $\sqrt{n}$ the lower bound on ϵ is $\Omega(\sqrt{n}\epsilon_f)$.

In the next section we show how fully-linear models can be constructed via polynomial interpolation, using less rigid sample sets than used in (2.9) and yet we are able to derive a competitive bound on κ_{eg} (and thus κ_{ef}).

3 Lagrange polynomials and fully-linear models. Before introducing the method we wish to analyze in this paper we need to discuss an important tool utilized by these algorithms - Lagrange polynomials. The concepts and the definitions below can be found in [13].

Lagrange polynomials and associated concepts will be defined with respect to a space of polynomials $\mathcal{P}$ of dimension p . Typically $\mathcal{P}$ is either the set of linear or quadratic polynomials, but it also can be a set of quadratic polynomials with a pre-defined Hessian sparsity pattern.

DEFINITION 3.1. Lagrange Polynomials

Given a space of polynomials $\mathcal{P}$ of dimension p and a set of points $\mathcal{Y} = \{y_1, \dots, y_p\} \subset \mathbb{R}^n$, a set of p polynomials $\ell_j(s)$ in $\mathcal{P}$ for $j = 1, \dots, p$, is called a basis of Lagrange polynomials associated with $\mathcal{Y}$, if

$$\ell_j(y_i) = \delta_{ij} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

If the basis of Lagrange polynomials exists for the given $\mathcal{Y}$ then $\mathcal{Y}$ is said to be poised.

DEFINITION 3.2. Λ -poisedness Given a space of polynomials $\mathcal{P}$ of dimension p , $\Lambda > 0$, and a set $\mathcal{B} \subset \mathbb{R}^n$. A poised set $\mathcal{Y} = \{y_1, \dots, y_p\}$ is said to be Λ -poised in $\mathcal{B}$ if $\mathcal{Y} \subset \mathcal{B}$ and for the basis of Lagrange polynomials associated with $\mathcal{Y}$, it holds that

$$\Lambda \geq \max_{j=1, \dots, p} \max_{s \in \mathcal{B}} |\ell_j(s)|.$$

We now show how Λ -poisedness of the interpolation set can ensure that related interpolation model is fully linear and derive corresponding constants κ_{ef} and κ_{eg} . Throughout this section we apply Assumptions 1.2 and 2.2. By Lemma 2.9, all we need is to ensure (2.8). For this we specify a way to construct the model $m(x)$.

Let $\mathcal{Y}_k$ be a set of points and $\mathcal{P}$ be a space of polynomials. Assuming that $\mathcal{Y}_k = \{y_1, \dots, y_p\}$ is poised in $B(0, \Delta)$ we define g_k and H_k to satisfy $\|H\| \leq \kappa_{bhm}$ and

$$(3.1) \quad g_k^\top y + \frac{1}{2} y^\top H_k y = \phi(x_k + y) - \phi(x_k), \quad \forall y \in \mathcal{Y}_k.$$

The model $m_k(x)$ is then defined as

$$(3.2) \quad m_k(x_k + s) = \phi(x_k) + g_k^\top s + \frac{1}{2} s^\top H_k s.$$

We now show an important result that establishes a bound on κ_{eg} when $p = n$ and $\mathcal{Y}_k = \{y_1, \dots, y_n\}$ is Λ -poised for homogeneous linear interpolation. This result is an extension of a similar result in [11] which allows m_k to include a quadratic term.

THEOREM 3.3. Let $\mathcal{Y} = \{y_1, \dots, y_n\}$ be such that $\mathcal{Y}$ is Λ -poised in $B(0, \Delta)$. Let g, H satisfy $\|H\| \leq \kappa_{bhm}$ and $g^\top y + \frac{1}{2} y^\top H y = \phi(x + y) - \phi(x)$, for each $y \in \mathcal{Y}$. Then

$$\|\nabla \phi(x) - g\| \leq \frac{1}{2} (L + \kappa_{bhm}) \Delta \sqrt{n} \sqrt{n(\Lambda^2 - 1) + 2}.$$

证明。根据来自 [4], 的误差界, 我们有 $\kappa_{ef}, \kappa_{eg} = O(\sqrt{n})$ 。回顾 $C_1^{-1} = \max \left\{ \eta_2, \kappa_{bhm}, \frac{2\kappa_{ef} + \max\{\eta_2, \kappa_{ef}\}}{\kappa_{fcd}} \right\} + \kappa_{eg}$ 以及 $C_2 = \frac{\eta_1\eta_2\kappa_{fcd}}{2} \min \left\{ \frac{\eta_2}{\kappa_{bhm}}, 1 \right\}$ 。因此, 在本推论的假设下, 我们有 $C_1^{-1} = \Theta(\sqrt{n})$ 以及 $C_2 = \Theta(\sqrt{n})$ 。因此, 我们可以界定

$$\begin{aligned} |\mathcal{S}_\epsilon| + |\mathcal{U}_\epsilon| &\leq \frac{2(f(x_0) - f^\star)}{C_2(\gamma C_1 \epsilon)^2} + \log_\gamma \frac{C_1 \epsilon}{\Delta_0} \\ (2.10) \quad &= \mathcal{O} \left(\frac{\sqrt{n}(\phi(x_0) - \phi^\star)}{\epsilon^2} + \log \left(\frac{\sqrt{n}\Delta_0}{\epsilon} \right) \right) = \mathcal{O} \left(\frac{\sqrt{n}}{\epsilon^2} \right). \end{aligned}$$

总预言机复杂度按前文所述方式得出。 $\square$

我们在此指出, 由于假设 L 不随 n 缩放, 而 $C_1^{-1} = \Theta(\sqrt{n})$, 因此无论我们选择 η_2 为常数还是等于 $\sqrt{n}$, ϵ 的下界均为 $\Omega(\sqrt{n}\epsilon_f)$ 。

在下一节中, 我们将展示如何通过多项式插值构造完全线性模型, 所使用的样本集比 (2.9) 中所用的更不严格, 但我们仍能推导出关于 κ_{eg} (从而 κ_{ef}) 的竞争性界。

3 拉格朗日多项式与完全线性模型。 在介绍本文希望分析的方法之前, 我们需要讨论这些算法所使用的一个重要工具——拉格朗日多项式。下文的概念和定义可见于 [13]。

拉格朗日多项式及相关概念将相对于一个维数为 p 的多项式空间 $\mathcal{P}$ 来定义。通常, $\mathcal{P}$ 要么是线性多项式集合, 要么是二次多项式集合, 但也可以是具有预定义 Hessian 稀疏模式的二次多项式集合。

定义3.1. 拉格朗日多项式

给定一个维数为 p 的多项式空间 $\mathcal{P}$ 和一个点集 $\mathcal{Y} = \{y_1, \dots, y_p\} \subset \mathbb{R}^n$, $\mathcal{P}$ 中的一组 p 个多项式 $\ell_j(s)$, 对于 $j = 1, \dots, p$, 称为与 $\mathcal{Y}$ 相关联的拉格朗日多项式基, 如果

$$\ell_j(y_i) = \delta_{ij} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

如果对给定的 $\mathcal{Y}$ 存在拉格朗日多项式基, 则称 $\mathcal{Y}$ 是适定的。

定义 3.2. Λ -适定性 给定一个维数为 p 的多项式空间 $\mathcal{P}$, $\Lambda > 0$, 以及一个集合 $\mathcal{B} \subset \mathbb{R}^n$ 。称一个适定集 $\mathcal{Y} = \{y_1, \dots, y_p\}$ 在 $\mathcal{B}$ 中是 Λ -适定的, 如果 $\mathcal{Y} \subset \mathcal{B}$, 并且对于与 $\mathcal{Y}$ 相关联的拉格朗日多项式基, 有

$$\Lambda \geq \max_{j=1, \dots, p} \max_{s \in \mathcal{B}} |\ell_j(s)|.$$

我们现在说明插值集的 Λ -适定性如何能够确保相关的插值模型是完全线性的, 并推导出相应的常数 κ_{ef} 和 κ_{ego} 。本节中我们始终应用假设 1.2 和 2.2。根据引理 2.9, 我们只需确保 (2.8)。为此, 我们给出一种构造模型 $m(x)$ 的方法。

令 $\mathcal{Y}_k$ 为一个点集, 且 $\mathcal{P}$ 为一个多项式空间。假设 $\mathcal{Y}_k = \{y_1, \dots, y_p\}$ 在 $B(0, \Delta)$ 中是适定的, 我们定义 g_k 和 H_k 满足 $\|H\| \leq \kappa_{bhm}$ 以及

$$(3.1) \quad g_k^\top y + \frac{1}{2} y^\top H_k y = \phi(x_k + y) - \phi(x_k), \quad \forall y \in \mathcal{Y}_k.$$

于是, 模型 $m_k(x)$ 定义为

$$(3.2) \quad m_k(x_k + s) = \phi(x_k) + g_k^\top s + \frac{1}{2} s^\top H_k s.$$

我们现在给出一个重要结果, 当 $p = n$ 且 $\mathcal{Y}_k = \{y_1, \dots, y_n\}$ 对于齐次线性插值是 Λ -适定时, 它建立了 κ_{eg} 的一个界。该结果是 [11] 中一个类似结果的推广, 后者允许 m_k 包含一个二次项。

定理 3.3. 设 $\mathcal{Y} = \{y_1, \dots, y_n\}$ 使得 $\mathcal{Y}$ 在 $B(0, \Delta)$ 中是 Λ -适定的。设 g, H 满足 $\|H\| \leq \kappa_{bhm}$ 和 $g^\top y + \frac{1}{2} y^\top H y = \phi(x + y) - \phi(x)$, 对每个 $y \in \mathcal{Y}$ 。则

$$\|\nabla \phi(x) - g\| \leq \frac{1}{2} (L + \kappa_{bhm}) \Delta \sqrt{n} \sqrt{n(\Lambda^2 - 1) + 2}.$$

In particular, if $\Lambda = 1 + O(1/n)$, then we will have $\|\nabla\phi(x) - g\| = O(\sqrt{n})\Delta$.

Proof. Let $\bar{\phi}(s) = \phi(x + s) - \phi(x)$, then $\bar{\phi}(0) = 0$ and $\nabla\phi(x) = \nabla\bar{\phi}(0)$. Let Y be the matrix with i th column equal to y_i , let D be the diagonal matrix such that $D_{ii} = \|y_i\|$, and let $\bar{\phi}(Y)$ be the vector with i th entry equal to $\bar{\phi}(y_i)$. The interpolation condition is

$$g^T y_i = \phi(x + y_i) - \phi(x) - \frac{1}{2} y_i^T H y_i, \quad i = 1, \dots, n$$

thus we have $g = Y^{-T} \bar{\phi}(Y) - Y^{-T} h(Y)$, where $h(Y)$ is a vector with components $\frac{1}{2} y_i^T H y_i$. Then we have $\|\nabla\phi(x) - g\| \leq \|\nabla\phi(x) - Y^{-T} \bar{\phi}(Y)\| + \|Y^{-T} h(Y)\|$. By the proof of Theorem 4.3 in [11], we have $\|Y^{-T} D\| \leq \sqrt{n(\Lambda^2 - 1) + 2}$ and

$$\|\nabla\phi(x) - Y^{-T} \bar{\phi}(Y)\| \leq \frac{1}{2} \sqrt{n} L \Delta \sqrt{n(\Lambda^2 - 1) + 2}$$

and we need only bound $\|Y^{-T} h(Y)\|$. We have $\|Y^{-T} h(Y)\| \leq \|Y^{-T} D\| \|D^{-1} h(Y)\|$. Observe that the i th element of $D^{-1} h(Y)$ is $\frac{1}{2} y_i^T H y_i / \|y_i\|$ and is therefore bounded by $\frac{1}{2} \kappa_{bhm} \Delta$. Now we have $\|D^{-1} h(Y)\| \leq \sqrt{n} \|D^{-1} h(Y)\|_\infty \leq \frac{1}{2} \sqrt{n} \kappa_{bhm} \Delta$. Thus $\|Y^{-T} h(Y)\| \leq \frac{1}{2} \sqrt{n} \kappa_{bhm} \Delta \sqrt{n(\Lambda^2 - 1) + 2}$, and in total

$$\|\nabla\phi(x) - g\| \leq \frac{1}{2} (L + \kappa_{bhm}) \Delta \sqrt{n} \sqrt{n(\Lambda^2 - 1) + 2}. \quad \square$$

This theorem allows us to conclude that our model is fully linear when our interpolation set is Λ -poised.

COROLLARY 3.4. *If m_k is defined as in (3.2) and if $\mathcal{Y}_k$ is Λ -poised, then m_k is κ_{ef}, κ_{eg} -fully-linear with*

$$\begin{aligned} \kappa_{eg} &= \frac{1}{2} (L + \kappa_{bhm}) \sqrt{n} \sqrt{n(\Lambda^2 - 1) + 2} \\ \kappa_{ef} &= \kappa_{eg} + \frac{L + \kappa_{bhm}}{2}. \end{aligned}$$

Proof. This follows from the definition of m_k in (3.2), Theorem 3.3 and Lemma 2.9. $\square$

We note that constructing the model to satisfy (3.1) is not possible unless $\phi(x + y) - \phi(x)$ can be computed exactly. In the case of inexact oracles we instead compute g and H from

$$(3.3) \quad g_k^\top y + \frac{1}{2} y^\top H_k y = f(x_k + y) - f(x_l) \quad \forall y \in \mathcal{Y}_k.$$

The previous theorem is modified as follows.

THEOREM 3.5. *Let $\mathcal{Y} = \{y_1, \dots, y_n\}$ be such that $\mathcal{Y}$ is Λ -poised in $B(0, \Delta)$. Let g, H be computed to satisfy (3.3) and $\|H\| \leq \kappa_{bhm}$. Then*

$$\|\nabla\phi(x) - g\| \leq \sqrt{n(\Lambda^2 - 1) + 2} \left(\frac{1}{2} (L + \kappa_{bhm}) \sqrt{n} \Delta + \sqrt{n} \frac{2\epsilon_f \Lambda}{\Delta} \right).$$

Proof. Define $\bar{\phi}$, D , $\bar{\phi}(Y)$, $h(Y)$ as in the previous proof. Diverging from that proof, the interpolation condition changes to

$$g^T y_i = \phi(x + y_i) - \phi(x) - \frac{1}{2} y_i^T H y_i + (f(x + y_i) - \phi(x + y_i)) - (f(x) - \phi(x)), \quad i = 1, \dots, n$$

thus we have $g = Y^{-T} \bar{\phi}(Y) - Y^{-T} h(Y) + Y^{-T} E$, where E is a vector with components $(f(x + y_i) - \phi(x + y_i)) - (f(x) - \phi(x))$. Then we can bound the error

$$\begin{aligned} \|\nabla\phi(x) - g\| &= \|Y^{-T} D(D^{-1} Y^T \nabla \bar{\phi}(0) - D^{-1} \bar{\phi}(Y) - D^{-1} E)\| + \|Y^{-T} h(Y)\| \\ &\leq \sqrt{n} \|Y^{-T} D\| \|D^{-1} Y^T \nabla \bar{\phi}(0) - D^{-1} \bar{\phi}(Y) - D^{-1} E\|_\infty + \|Y^{-T} h(Y)\| \\ &\leq \sqrt{n} \|Y^{-T} D\| (\|D^{-1} Y^T \nabla \bar{\phi}(0) - D^{-1} \bar{\phi}(Y)\|_\infty + \|D^{-1} E\|_\infty) + \|Y^{-T} h(Y)\| \\ &\leq \sqrt{n} \|Y^{-T} D\| (\|D^{-1} Y^T \nabla \bar{\phi}(0) - D^{-1} \bar{\phi}(Y)\|_\infty + \|D^{-1}\|_\infty \|E\|_\infty) + \|Y^{-T} h(Y)\|. \end{aligned}$$

特别地, 如果 $\Lambda = 1 + O(1/n)$, 那么我们将有 $\|\nabla\phi(x) - g\| = O(\sqrt{n})\Delta$.

证明. 令 $\phi(s) = \phi(x + s) - \phi(x)$, 则 $\phi(0) = 0$ 且 $\nabla\phi(x) = \nabla\phi(0)$. 令 Y 为第 i 列等于 y_i 的矩阵, 令 D 为满足 $D_{ii} = \|y_i\|$ 的对角矩阵, 并令 $\phi(Y)$ 为具有第 i 个元素等于 $\phi(y_i)$ 的向量。插值条件为

$$g^T y_i = \phi(x + y_i) - \phi(x) - \frac{1}{2} y_i^T H y_i, \quad i = 1, \dots, n$$

因此, 我们有 $g = Y^{-T} \bar{\phi}(Y) - Y^{-T} h(Y)$, 其中 $h(Y)$ 是分量为 $\frac{1}{2} y_i^T H y_i$ 的向量。于是我们有 $\|\nabla\phi(x) - g\| \leq \|\nabla\phi(x) - Y^{-T} \bar{\phi}(Y)\| + \|Y^{-T} h(Y)\|$ 。根据 [11], 中定理 4.3 的证明, 我们有 $\|Y^{-T} D\| \leq \sqrt{n(\Lambda^2 - 1) + 2}$ 且

$$\|\nabla\phi(x) - Y^{-T} \bar{\phi}(Y)\| \leq \frac{1}{2} \sqrt{n} L \Delta \sqrt{n(\Lambda^2 - 1) + 2}$$

而我们只需给出 $\|Y^{-T} h(Y)\|$ 的界。我们有 $\|Y^{-T} h(Y)\| \leq \|Y^{-T} D\| \|D^{-1} h(Y)\|$ 。注意到 $D^{-1} h(Y)$ 的第 i 个元素为 $\frac{1}{2} y_i^T H y_i / \|y_i\|$, 因此以 $\frac{1}{2} \kappa_{bhm} \Delta$ 为界。现在我们有 $\|D^{-1} h(Y)\| \leq \sqrt{n} \|D^{-1} h(Y)\|_\infty \leq \frac{1}{2} \sqrt{n} \kappa_{bhm} \Delta$ 。因此 $\|Y^{-T} h(Y)\| \leq \frac{1}{2} \sqrt{n} \kappa_{bhm} \Delta \sqrt{n(\Lambda^2 - 1) + 2}$, 总计

$$\|\nabla\phi(x) - g\| \leq \frac{1}{2} (L + \kappa_{bhm}) \Delta \sqrt{n} \sqrt{n(\Lambda^2 - 1) + 2}. \quad \square$$

该定理允许我们得出结论: 当我们的插值集为 Λ -适定时, 我们的模型是完全线性的。d.

推论 3.4. 如果 m_k 如 (3.2) 中所定义, 并且如果 $\mathcal{Y}_k$ 是 Λ -适定的, 那么 m_k 是 κ_{ef}, κ_{eg} -完全线性的, 其中

$$\begin{aligned} \kappa_{eg} &= \frac{1}{2} (L + \kappa_{bhm}) \sqrt{n} \sqrt{n(\Lambda^2 - 1) + 2} \\ \kappa_{ef} &= \kappa_{eg} + \frac{L + \kappa_{bhm}}{2}. \end{aligned}$$

证明. 这由 m_k 在 (3.2) 中的定义、定理 3.3 和引理 2.9 得出。 $\square$

我们注意到, 除非 $\phi(x + y) - \phi(x)$ 能够精确计算, 否则不可能构造出满足 (3.1) 的模型。在不精确神谕的情况下, 我们改为计算 g 和 H , 其来自

$$(3.3) \quad g_k^\top y + \frac{1}{2} y^\top H_k y = f(x_k + y) - f(x_l) \quad \forall y \in \mathcal{Y}_k.$$

前一定理修改如下。

定理 3.5. 设 $\mathcal{Y} = \{y_1, \dots, y_n\}$ 使得 $\mathcal{Y}$ 在 $B(0, \Delta)$ 中是 Λ -适定的。设 g, H 被计算为满足 (3.3) 和 $\|H\| \leq \kappa_{bhm}$ 。则

$$\|\nabla\phi(x) - g\| \leq \sqrt{n(\Lambda^2 - 1) + 2} \left(\frac{1}{2} (L + \kappa_{bhm}) \sqrt{n} \Delta + \sqrt{n} \frac{2\epsilon_f \Lambda}{\Delta} \right).$$

证明. 按前一个证明中的方式定义 ϕ , D , $\phi(Y)$, $h(Y)$ 。与那个证明不同, 插值条件变为

$$g^T y_i = \phi(x + y_i) - \phi(x) - \frac{1}{2} y_i^T H y_i + (f(x + y_i) - \phi(x + y_i)) - (f(x) - \phi(x)), \quad i = 1, \dots, n$$

因此, 我们有 $g = Y^{-T} \bar{\phi}(Y) - Y^{-T} h(Y) + Y^{-T} E$, 其中 E 是一个向量, 其分量为 $(f(x + y_i) - \phi(x + y_i)) - (f(x) - \phi(x))$ 。于是我们可以给出误差的界

$$\begin{aligned} \|\nabla\phi(x) - g\| &= \|Y^{-T} D(D^{-1} Y^T \nabla \bar{\phi}(0) - D^{-1} \bar{\phi}(Y) - D^{-1} E)\| + \|Y^{-T} h(Y)\| \\ &\leq \sqrt{n} \|Y^{-T} D\| \|D^{-1} Y^T \nabla \bar{\phi}(0) - D^{-1} \bar{\phi}(Y) - D^{-1} E\|_\infty + \|Y^{-T} h(Y)\| \\ &\leq \sqrt{n} \|Y^{-T} D\| (\|D^{-1} Y^T \nabla \bar{\phi}(0) - D^{-1} \bar{\phi}(Y)\|_\infty + \|D^{-1} E\|_\infty) + \|Y^{-T} h(Y)\| \\ &\leq \sqrt{n} \|Y^{-T} D\| (\|D^{-1} Y^T \nabla \bar{\phi}(0) - D^{-1} \bar{\phi}(Y)\|_\infty + \|D^{-1}\|_\infty \|E\|_\infty) + \|Y^{-T} h(Y)\|. \end{aligned}$$

We can bound $\|Y^{-T}D\|, \|D^{-1}Y^T\nabla\bar{\phi}(0) - D^{-1}\bar{\phi}(Y)\|_\infty, \|Y^{-T}h(Y)\|$ identically as in the previous proof. To bound $\|E\|_\infty$ observe that the condition that $|f(x+y) - \phi(x+y)| \leq \epsilon_f$ for $y \in \mathcal{Y} \cup \{0\}$ implies $\|E\|_\infty \leq 2\epsilon_f$. To bound $\|D^{-1}\|_\infty$ recall that $D_{ii} = \|y_i\|$. By the properties of Lagrange polynomials we have $\ell_i(y_i) = 1$. By linearity, we have $\ell_i(\Delta \frac{y_i}{\|y_i\|}) = \frac{\Delta}{\|y_i\|}$. By the poisedness condition we have $\left| \ell_i(\Delta \frac{y_i}{\|y_i\|}) \right| \leq \Lambda$. Combining these bounds we have $\frac{\Delta}{\|y_i\|} \leq \Lambda$ and thus $\frac{1}{\|y_i\|} \leq \frac{\Lambda}{\Delta}$. Thus $\|D^{-1}\|_\infty \leq \frac{\Lambda}{\Delta}$. The result follows. $\square$

This theorem allows us to conclude that our model is fully linear when our interpolation set is Λ -poised and when Δ is sufficiently large.

COROLLARY 3.6. *If m_k is defined as in (3.2), if $\mathcal{Y}_k$ is Λ -poised, and if $\Delta_k \geq \sqrt{\frac{4\epsilon_f\Lambda}{L+\kappa_{bhm}}}$, then m_k is κ_{ef}, κ_{eg} -fully-linear with*

$$\begin{aligned}\kappa_{eg} &= (L + \kappa_{bhm}) \sqrt{n} \sqrt{n(\Lambda^2 - 1) + 2} \\ \kappa_{ef} &= \kappa_{eg} + \frac{L + \kappa_{bhm}}{2}.\end{aligned}$$

Proof. This follows from the definition of m_k in (3.2), Theorem 3.5 and Lemma 2.9, since the bound on Δ_k implies that

$$\sqrt{n(\Lambda^2 - 1) + 2} \left(\frac{1}{2} (L + \kappa_{bhm}) \sqrt{n} \Delta_k + \sqrt{n} \frac{2\epsilon_f\Lambda}{\Delta_k} \right) \leq 2 \frac{1}{2} (L + \kappa_{bhm}) \Delta_k \sqrt{n} \sqrt{n(\Lambda^2 - 1) + 2}. \quad \square$$

In the next section we propose a concrete algorithm for constructing Λ -poised sample sets and deriving the bound on the number of iterations required to do so.

4 Model improving iterations based on Lagrange polynomials. In [11] a specific algorithm based on the framework of Algorithm 1 was proposed and its complexity analyzed. This algorithm maintains a set $\mathcal{Y}$ of n sample points and a set of linear Lagrange polynomials. At each model improving iteration the algorithm performs, what we call, a *geometry correcting step* by either replacing an interpolation point outside the trust region, if such a point exists, or replacing a point whose corresponding Lagrange polynomial violates the Λ -poisedness condition. If no such improvement is possible, then the set is Λ -poised and thus the model is fully-linear, by Theorems 3.3 and 3.5, hence the iteration is unsuccessful and the trust region radius is reduced. It is shown in [11] that the number of geometry correcting iterations between any two other iterations is at most $3n$. Each such iteration performs at most two function evaluations, while successful and unsuccessful iterations perform at most one. Thus using Theorem 2.8 one can derive the bound on $|\mathcal{M}_\epsilon|$ and consequently the bound on the total complexity of the algorithm.

The method in [11] fails to include several important practical elements. Firstly, the models that are being constructed by any successful model-based DFO method are quadratic and are usually based on quadratic interpolation. Secondly, since the function value is computed at the trial step, even if the step is not accepted as the new iterate, the step provides a new sample point which may improve the interpolation model. In fact it is likely to do so, since the reason the step is rejected, to begin with, is due to disagreement of the model and the function at the trial step. Thus adding this trial step to the new model provides new information. The improvement guaranteed by replacing some interpolation point by the trial step can be quantified by the value of the corresponding Lagrange polynomial at the trial step. A principled algorithm relying on this property, which is referred to as *self-correcting*, has been developed in [26] and shown to converge to a stationary point in the limit. This algorithm uses only such *self-correcting steps* for model improvement and does not perform geometry correction steps. It also includes the criticality step which is not practical but helps the analysis. No complexity bounds have been developed in [26]. Powell utilized both the self-correcting and geometry correcting steps in his algorithms [21, 23]. In what follows we present an algorithm that allows models to be constructed using quadratic interpolation and utilizes the self-correcting and geometry correcting steps. On the other hand it only maintains a set of linear Lagrange polynomials and ensures only a subset of interpolation points to be Λ -poised for linear interpolation. Combined with the results of the previous section this allows the algorithm to ensure eventual construction of fully-linear models. After we present the algorithm we state and prove the bound on the number of consecutive model improving iterations.

我们可以像之前的证明中那样完全相同地给出 $\|Y^{-T}D\|, \|D^{-1}Y^T\nabla\bar{\phi}(0) - D^{-1}\bar{\phi}(Y)\|_\infty, \|Y^{-T}h(Y)\|$ 的界。为给出 $\|E\|_\infty$ 的界，注意到 $|f(x+y) - \phi(x+y)| \leq \epsilon_f$ 对 $y \in \mathcal{Y} \cup \{0\}$ 成立这一条件蕴含 $\|E\|_\infty \leq 2\epsilon_f$ 。为给出 $\|D^{-1}\|_\infty$ 的界，回忆 $D_{ii} = \|y_i\|$ 。根据拉格朗日多项式的性质，我们有 $\ell_i(y_i) = 1$ 。根据线性，我们有 $\ell_i(\Delta \frac{y_i}{\|y_i\|}) = \frac{\Delta}{\|y_i\|}$ 。根据适定性条件，我们有 $\left| \ell_i(\Delta \frac{y_i}{\|y_i\|}) \right| \leq \Lambda$ 。结合这些界，我们有 $\frac{\Delta}{\|y_i\|} \leq \Lambda$ ，因此 $\frac{1}{\|y_i\|} \leq \frac{\Lambda}{\Delta}$ 。因此 $\|D^{-1}\|_\infty \leq \frac{\Lambda}{\Delta}$ 。结果随之成立。 $\square$

该定理使我们能够得出结论：当插值集是 Λ -适定的，并且当 Δ 足够大时，我们的模型是完全线性的。

推论 3.6. 如果 m_k 如 (3.2) 中所定义，如果 $\mathcal{Y}_k$ 是 Λ -适定的，并且如果 $\Delta_k \geq \sqrt{\frac{4\epsilon_f\Lambda}{L+\kappa_{bhm}}}$ ，那么 m_k 是 κ_{ef}, κ_{eg} -完全线性的，其中

$$\begin{aligned}\kappa_{eg} &= (L + \kappa_{bhm}) \sqrt{n} \sqrt{n(\Lambda^2 - 1) + 2} \\ \kappa_{ef} &= \kappa_{eg} + \frac{L + \kappa_{bhm}}{2}.\end{aligned}$$

证明. 这由 m_k 在 (3.2) 中的定义、定理 3.5 和引理 2.9 推出，因为对 Δ_k 的界意味着

$$\sqrt{n(\Lambda^2 - 1) + 2} \left(\frac{1}{2} (L + \kappa_{bhm}) \sqrt{n} \Delta_k + \sqrt{n} \frac{2\epsilon_f\Lambda}{\Delta_k} \right) \leq 2 \frac{1}{2} (L + \kappa_{bhm}) \Delta_k \sqrt{n} \sqrt{n(\Lambda^2 - 1) + 2}. \quad \square$$

在下一节中，我们提出一种具体算法，用于构造 Λ -适定的样本集，并推导出完成此操作所需迭代次数的界。

4 基于拉格朗日多项式的模型改进迭代。 在 [11] 中，提出了一种基于算法1框架的具体算法，并分析了其复杂度。该算法维护一个由 n 个样本点组成的集合 $\mathcal{Y}$ ，以及一组线性拉格朗日多项式。在每次模型改进迭代中，该算法执行我们所谓的几何校正步骤，其方式是：要么替换信赖域之外的一个插值点（如果这样的点存在），要么替换一个其对应的拉格朗日多项式违反 Λ -适定性条件的点。如果不可能有这样的改进，那么该集合是 Λ -适定的，因此根据定理 3.3 和定理 3.5，模型是完全线性的，所以该迭代是不成功迭代，并且信赖域半径被缩小。在 [11] 中表明，任意两个其他迭代之间的几何校正迭代次数至多为 $3n$ 。每个这样的迭代最多执行两次函数求值，而成功迭代和不成功迭代最多执行一次。因此，使用定理 2.8 可以推导出关于 $|\mathcal{M}_\epsilon|$ 的界，并由此推导出该算法总复杂度的界。

在 [11]中的方法未能包含若干重要的实践要素。首先，任何成功的基于模型的无导数优化方法所构建的模型都是二次模型，并且通常基于二次插值。其次，由于函数值在试探步处计算，即使该步未被接受为新迭代点，它仍提供了一个新的样本点，从而可能改进插值模型。事实上，这种情况很可能发生，因为一开始该步被拒绝的原因正是模型与函数在试探步处不一致。因此，将这个试探步加入新模型会提供新信息。用该试探步替换某个插值点所保证的改进，可由相应的拉格朗日多项式在该试探步处的值来量化。一种依赖这一性质的原则性算法（称为自校正）已在 [26] 中被提出，并被证明在极限情况下收敛到驻点。该算法仅使用此类自校正步骤进行模型改进，而不执行几何校正步。它还包括临界步，这一步骤并不实用，但有助于分析。在 [26]中尚未建立复杂度界。Powell在其算法 [21, 23]中同时利用了自校正步骤和几何校正步骤。接下来，我们提出一种算法，它允许使用二次插值构建模型，并利用自校正步骤和几何校正步骤。另一方面，它仅维护一组线性拉格朗日多项式，并确保只有插值点的一个子集对线性插值而言是 Λ -适定的。结合上一节的结果，这使算法能够确保最终构建完全线性模型。在我们给出该算法之后，我们陈述并证明连续模型改进迭代次数的界。

Algorithm 2 is the description of our proposed method that on the one hand tries to include most of the practical elements of Powell’s methods and on the other hand nearly matches the complexity of the simplified method in [11].

At each iteration the method maintains two sets of points - set $\mathcal{Y}$ of n points whose geometry is monitored and maintained by means of the associated linear Lagrange polynomials and another set $\mathcal{Z}$ of $p \leq (n-1)n/2$ points whose geometry is only monitored and maintained in terms of distance to the trust region center.

We construct the model by solving the following constrained least squares problem.

$$(4.1) \quad \min_{g, H} \sum_{z \in \mathcal{Z}_k} \left(f(x_k) + g^\top z + \frac{1}{2} z^\top H z - f(x_k + z) \right)^2$$

$$(4.2) \quad \text{subject to: } \|H\| \leq K$$

$$(4.3) \quad g^\top y + \frac{1}{2} y^\top H y = f(x_k + y) - f(x_k), \quad \forall y \in \mathcal{Y}_k.$$

Any solution to this problem will satisfy $\|H\| \leq \kappa_{bhm}$ with $\kappa_{bhm} \leq K$. If a quadratic model exists whose Hessian satisfies $\|H\| \leq \kappa_{bhm}$ with $\kappa_{bhm} \leq K$ and that interpolates all points in $\mathcal{Y}$ and $\mathcal{Z}$, then such model will be an optimal solution to this problem.³ When $p < \frac{n(n-1)}{2}$ then the problem may have multiple optimal solutions. In the case when this happens we can select the solution with the smallest Frobenius norm (see [13]), as long as it satisfies $\|H\| \leq K$. Alternatively, following Powell’s ideas from [21] we can select the solution for which H is the closest in Frobenius norm to the Hessian from the previous iteration. We will discuss both of these techniques in the computational section.

³We deliberately distinguish K and κ_{bhm} , since K is chosen by the algorithm as will be set to be large, while κ_{bhm} is the true bound that occurs during the algorithm and will depend on the problem.

算法2描述了我们提出的方法：一方面，它试图包含Powell方法的大部分实践要素；另一方面，它的复杂度几乎与在 [11]中的简化方法相匹配。

在每次迭代中，该方法维护两组点——集合 $\mathcal{Y}$ 包含 n 个点，其几何形态通过相关联的线性拉格朗日多项式进行监测和维护；另一集合 $\mathcal{Z}$ 包含 $p \leq (n-1)n/2$ 个点，其几何形态仅根据到信赖域中心的距离进行监测和维护。

我们通过求解以下约束最小二乘问题来构造模型。

$$(4.1) \quad \min_{g, H} \sum_{z \in \mathcal{Z}_k} \left(f(x_k) + g^\top z + \frac{1}{2} z^\top H z - f(x_k + z) \right)^2$$

$$(4.2) \quad \text{subject to: } \|H\| \leq K$$

$$(4.3) \quad g^\top y + \frac{1}{2} y^\top H y = f(x_k + y) - f(x_k), \quad \forall y \in \mathcal{Y}_k.$$

此问题的任何解都将满足 $\|H\| \leq \kappa_{bhm}$ 和 $\kappa_{bhm} \leq K$ 。如果存在一个二次模型，其 Hessian 矩阵满足 $\|H\| \leq \kappa_{bhm}$ 和 $\kappa_{bhm} \leq K$ ，并对 $\mathcal{Y}$ 和 $\mathcal{Z}$ 中的所有点进行插值，那么这样的模型将是此问题的最优解。³ 当 $p < \frac{n(n-1)}{2}$ 时，问题可能有多个最优解。在这种情况下，我们可以选择 Frobenius 范数最小的解（见 [13]），只要它满足 $\|H\| \leq K$ 。或者，按照 Powell 在 [21] 中提出的思路，我们可以选择使 H 在 Frobenius 范数下最接近上一迭代的 Hessian 矩阵的解。我们将在计算部分讨论这两种技术。

³我们有意区分 K 和 κ_{bhm} ，因为 K 由算法选择，并将被设得很大，而 κ_{bhm} 则是在算法运行期间出现的真实界，并且取决于问题。

Algorithm 2: Geometry-correcting algorithm

Inputs: A zeroth-order oracle $f(x) \approx \phi(x)$, $p = \frac{(n-1)n}{2}$, Δ_0 , x_0 , $\gamma \in (0, 1)$ $\eta_1 > 0$, $\eta_2 > 0$, $\Lambda > 1$, $\Lambda_{sc} \geq 1$.
Initialization An initial set $\mathcal{Y}_0$ such that $|\mathcal{Y}_0| = n$, an initial set $\mathcal{Z}_0$ such that $|\mathcal{Z}_0| \leq p$ and the function values $f(x_0)$, $f(x_0 + y_i)$, $y_i \in \mathcal{Y}_0$, $f(x_0 + z_i)$, $z_i \in \mathcal{Z}_0$. A set of Lagrange Polynomials $\{\ell_i(x), i = 1, \dots, n\}$ in $\mathcal{P}$ for the set $\mathcal{Y}_0$.
for $k = 0, 1, 2, \dots$ **do**
1 Build a quadratic model $m_k(x_k + s)$ as in (3.2) using $f(x_k)$ and $f(x_k + y_i)$, $y_i \in \mathcal{Y}_k$, $f(x_0 + z_i)$, $z_i \in \mathcal{Z}_k$.
2 Compute a trial step s_k and ratio ρ_k as in Algorithm 1.
3 *Successful iteration:* $\rho_k \geq \eta_1$ and $\|g_k\| \geq \eta_2 \Delta_k$.
 Set $x_{k+1} = x_k + s_k$, $\Delta_{k+1} = \gamma^{-1} \Delta_k$.

$$j_k^* = \arg \max_{j=1, \dots, n} \|y_j - s_k\|, \quad i_k^* = \arg \max_{i=1, \dots, p} \|z_i - s_k\|.$$
 If $\|y_{j_k^*} - s_k\| > \|z_{i_k^*} - s_k\|^a$ and $|\ell_{j_k^*}(s_k)| > 0$ then update set $\mathcal{Y}_{k+1} = (\mathcal{Y}_k \setminus \{y_{j_k^*}\} \cup \{0\}) - s_k$.
 Recompute Lagrange Polynomials for $\mathcal{Y}_{k+1}$. Otherwise update set $\mathcal{Z}_{k+1} = (\mathcal{Z}_k \setminus \{z_{i_k^*}\} \cup \{0\}) - s_k$ if $|\mathcal{Z}_k| = p$,
 or $\mathcal{Z}_{k+1} = \mathcal{Z}_k \cup \{0\} - s_k$ if $|\mathcal{Z}_k| < p$.
4 *Model improving or unsuccessful iteration:* $\rho_k < \eta_1$ or $\|g_k\| < \eta_2 \Delta_k$. Set $x_{k+1} = x_k$, $I_{imp} = 0$ and perform all the applicable steps below.
 (i) *Geometry correction by replacing a far point in $\mathcal{Y}$:* Let $j_k^* = \arg \max_{j=1, \dots, n} \|y_j\|$.
 If $\|y_{j_k^*}\| > \Delta_k \Rightarrow s^* = y_{j_k^*}$ $I_{imp} = 1$. If $|\ell_{j_k^*}(s_k)| > 0$, then $s_k^* = s_k$, otherwise $s_k^* = \arg \max_{s \in B(0, \Delta_k)} |\ell_{j_k^*}(s)|$, compute $f(x_k + s_k^*)$. Update $\mathcal{Y}_{k+1} = \mathcal{Y}_k \setminus \{y_{j_k^*}\} \cup \{s_k^*\}$ and Lagrange polynomials.
 (ii) *Self-correction by replacing a point in $\mathcal{Y}$ with a large Lagrange Polynomial value at the trial step:* If $I_{imp} = 0$,

$$j_k^* = \arg \max_{j=1, \dots, n} |\ell_j(s_k)|.$$
 If $|\ell_{j_k^*}(s_k)| > \Lambda_{sc}$, then $s^* = y_{j_k^*}$, update $\mathcal{Y}_{k+\frac{1}{2}} = \mathcal{Y}_k \setminus \{y_{j_k^*}\} \cup \{s_k\}$. Update the set of Lagrange Polynomials for $\mathcal{Y}_{k+\frac{1}{2}}$. Otherwise $s^* = s_k$, $\mathcal{Y}_{k+\frac{1}{2}} = \mathcal{Y}_k$.
 (iii) *Geometry correction of $\mathcal{Y}$ by replacing a "bad" point:* If $I_{imp} = 0$ compute

$$(j_k^*, s_k^*) = \arg \max_{j=1, \dots, p, s \in B(0, \Delta_k)} |\ell_j(s)|.$$
 If $|\ell_{j_k^*}(s_k^*)| > \Lambda$, compute $f(x_k + s_k^*)$, $\mathcal{Y}_{k+1} = \mathcal{Y}_{k+\frac{1}{2}} \setminus \{y_{j_k^*}\} \cup \{s_k^*\}$. Update the set of Lagrange Polynomials for $\mathcal{Y}_{k+1}$ and set $I_{imp} = 1$. Otherwise $\mathcal{Y}_{k+1} = \mathcal{Y}_{k+\frac{1}{2}}$.
 (iv) *Attempt to improve Z using available points.* For each defined $s \in \{s^*, s_k^*\}$ repeat:
 • *Improvement to Z by adding a point:* If $|\mathcal{Z}_k| < p$, $\mathcal{Z}_{k+1} = \mathcal{Z}_k \cup \{s\}$.
 • *Improvement to Z to by replacing a far point:* Else, let $i_k^* = \arg \max_{i=1, \dots, n} \|z_i\|$.
 If $\|z_{i_k^*}\| > \|s\| \Rightarrow \mathcal{Z}_{k+1} = \mathcal{Z}_k \setminus \{z_{i_k^*}\} \cup \{s\}$.
 • *Keep Z :* Otherwise $\mathcal{Z}_{k+1} = \mathcal{Z}_k$.
 Model improving iteration: If $I_{imp} = 1$, $\Delta_{k+1} = \Delta_k$.
 Unsuccessful iteration: If $I_{imp} = 0$, $\Delta_{k+1} = \gamma \Delta_k$.

^aIf $\mathcal{Z}_k \neq \emptyset$ check $\|y_{j_k^*} - s_k\| > 0$

We note that case (iii) of Step 4 of Algorithm 2 is equivalent to one iteration of the geometry correction

算法2：几何校正算法

输入：一个零阶预言机 $f(x) \approx \phi(x)$, $p = \frac{(n-1)n}{2}$, Δ_0 , x_0 , $\gamma \in (0, 1)$ $\eta_1 > 0$, $\eta_2 > 0$, $\Lambda > 1$, $\Lambda_{sc} \geq 1$.
初始化：一个初始集合 $\mathcal{Y}_0$ ，使得 $|\mathcal{Y}_0| = n$ ；一个初始集合 $\mathcal{Z}_0$ ，使得 $|\mathcal{Z}_0| \leq p$ ；以及函数值 $f(x_0)$ 、 $f(x_0 + y_i)$ 、 $y_i \in \mathcal{Y}_0$ 、 $f(x_0 + z_i)$ 、 $z_i \in \mathcal{Z}_0$ 。一个拉格朗日多项式集合 $\{\ell_i(x) \mid i = 1, \dots, n\}$ 在 $\mathcal{P}$ 中，用于集合 $\mathcal{Y}_0$ 。对于 $k = 0, 1, 2, \dots$ 执行 1 使用 $f(x_k)$ 和 $f(x_k + y_i)$ 、 $y_i \in \mathcal{Y}_k$ 、 $f(x_0 + z_i)$ 、 $z_i \in \mathcal{Z}_k$ ，按 (3.2) 构建二次模型 $m_k(x_k + s)$ 。2 按算法1计算试探步 s_k 和比率 ρ_k 。3 成功迭代： $\rho_k \geq \eta_1$ 和 $\|g_k\| \geq \eta_2 \Delta_k$ 。设置 $x_{k+1} = x_k + s_k$, $\Delta_{k+1} = \gamma^{-1} \Delta_k$ 。如果 $\|y_{j_k^*} - s_k\| > \|z_{i_k^*} - s_k\|^a$ 且 $|\ell_{j_k^*}(s_k)| > 0$ ，则更新集合 $\mathcal{Y}_{k+1} = (\mathcal{Y}_k \setminus \{y_{j_k^*}\} \cup \{0\}) - s_k$ 。为 $\mathcal{Y}_{k+1}$ 重新计算拉格朗日多项式。否则，若 $|\mathcal{Z}_k| = p$ ，则更新集合 $\mathcal{Z}_{k+1} = (\mathcal{Z}_k \setminus \{z_{i_k^*}\} \cup \{0\}) - s_k$ ；或者若 $|\mathcal{Z}_k| < p$ ，则 $\mathcal{Z}_{k+1} = \mathcal{Z}_k \cup \{0\} - s_k$ 。4 模型改进或不成功迭代： $\rho_k < \eta_1$ 或 $\|g_k\| < \eta_2 \Delta_k$ 。设置 $x_{k+1} = x_k$, $I_{imp} = 0$ ，并执行下面所有适用的步骤。(i) 通过替换 $\mathcal{Y}$ 中的远点进行几何校正：令 $j_k^* = \arg \max_{j=1, \dots, n} \|y_j\|$ 。如果 $\|y_{j_k^*}\| > \Delta_k \Rightarrow s^* = y_{j_k^*}$ $I_{imp} = 1$ 。如果 $|\ell_{j_k^*}(s_k)| > 0$ ，则 $s_k^* = s_k$ ，否则 $s_k^* = \arg \max_{s \in B(0, \Delta_k)} |\ell_{j_k^*}(s)|$ ，计算 $f(x_k + s_k^*)$ 。更新 $\mathcal{Y}_{k+1} = \mathcal{Y}_k \setminus \{y_{j_k^*}\} \cup \{s_k^*\}$ 和拉格朗日多项式。(ii) 通过用在试探步处具有较大拉格朗日多项式值的点替换 $\mathcal{Y}$ 中的点进行自校正：如果 $I_{imp} = 0$ ，如果 $|\ell_{j_k^*}(s_k)| > \Lambda_{sc}$ ，则 $s^* = y_{j_k^*}$ ，更新 $\mathcal{Y}_{k+\frac{1}{2}} = \mathcal{Y}_k \setminus \{y_{j_k^*}\} \cup \{s_k\}$ 。为 $\mathcal{Y}_{k+\frac{1}{2}}$ 更新拉格朗日多项式集合。否则 $s^* = s_k$, $\mathcal{Y}_{k+\frac{1}{2}} = \mathcal{Y}_k$ 。(iii) 通过替换一个“坏”点对 $\mathcal{Y}$ 进行几何校正：如果 $I_{imp} = 0$ ，计算；如果 $|\ell_{j_k^*}(s_k^*)| > \Lambda$ ，计算 $f(x_k + s_k^*)$, $\mathcal{Y}_{k+1} = \mathcal{Y}_{k+\frac{1}{2}} \setminus \{y_{j_k^*}\} \cup \{s_k^*\}$ 。为 $\mathcal{Y}_{k+1}$ 更新拉格朗日多项式集合，并设置 $I_{imp} = 1$ 。否则 $\mathcal{Y}_{k+1} = \mathcal{Y}_{k+\frac{1}{2}}$ 。(iv) 尝试使用可用点改进 Z 。对于每个已定义的 $s \in \{s^*, s_k^*\}$ ，重复：

$$j_k^* = \arg \max_{j=1, \dots, n} |\ell_j(s_k)|.$$

$$(j_k^*, s_k^*) = \arg \max_{j=1, \dots, p, s \in B(0, \Delta_k)} |\ell_j(s)|.$$

- 通过添加一个点来改进 Z ：如果 $|\mathcal{Z}_k| < p$, $\mathcal{Z}_{k+1} = \mathcal{Z}_k \cup \{s\}$ 。
- 改进 Z （通过替换远点）：否则，令 $i_k^* = \arg \max_{i=1, \dots, n} \|z_i\|$ 。如果 $\|z_{i_k^*}\| > \|s\| \Rightarrow \mathcal{Z}_{k+1} = \mathcal{Z}_k \setminus \{z_{i_k^*}\} \cup \{s\}$ 。
- 保留 Z ：否则 $\mathcal{Z}_{k+1} = \mathcal{Z}_k$ 。

模型改进迭代：如果 $I_{imp} = 1$, $\Delta_{k+1} = \Delta_k$ 。

不成功迭代：如果 $I_{imp} = 0$, $\Delta_{k+1} = \gamma \Delta_k$ 。

^a如果 $\mathcal{Z}_k \neq \emptyset$ 检查 $\|y_{j_k^*} - s_k\| > 0$ 。我们注意到，算法2步骤4中的情形(iii)等价于几何校正的一次迭代。

procedure described in [13].

In cases (ii) and (iii) of Step 4 point $y_{j_k^*}$ in the current $\mathcal{Y}$ set is replaced by a new point, let's call it $\tilde{s}$. The update for deriving the new Lagrange Polynomial set ℓ^+ can be carried out via the following formulae:

$$(4.4) \quad \ell_{j_k^*}^+(x) = \frac{\ell_{j_k^*}(x)}{\ell_{j_k^*}(\tilde{s})}$$

$$(4.5) \quad \ell_i^+(x) = \ell_i(x) - \ell_i(\tilde{s})\ell_{j_k^*}^+(x) \quad i \neq j_k^*.$$

Note that since we assume that $\mathcal{Y}_0$ is poised then so are all consequent $\mathcal{Y}_k$ sets by construction.

We now provide results that allow us to bound the number of model improving iterations $\mathcal{M}_\epsilon$.

THEOREM 4.1. *Let $\mathcal{P}$ be the set of linear polynomials (with dimension $p = n$). Then the number of oracle calls in any sequence of consecutive model improving iterations; i.e. such that $k \in \mathcal{M}_\epsilon$ is at most $4n \log n + 8n + 2n|\log \log(\Lambda)|$.*

We break the proof of this theorem into the following two lemmas, the first of which derives a bound on the number of consecutive model improving iterations required to obtain a desired Λ -poised set starting from a Λ_0 -poised set for an arbitrary Λ_0 . The second lemma shows that after the first $2n$ consecutive model improving iterations Λ_0 -poised set is obtained with a particular bound on Λ_0 .

LEMMA 4.2. *Let $\mathcal{P}$ be the set of linear polynomials (with dimension $p = n$). If $\mathcal{Y}_k$ is Λ_0 -poised in $B(0, \Delta)$, then there will be at most $\lceil n \log n + n|\log \log \Lambda_0| + n|\log \log(\Lambda)| \rceil$ additional consecutive model improving iterations.*

Proof. Assume for simplicity and w.l.o.g that $\Delta = 1$. Having a Λ_0 -poised set implies by Hadamard's inequality that we have

$$|\det(Y^{-T})| \leq \prod_i^n \|(Y^{-T})_i\| \leq \Lambda_0^n.$$

since $\max_{x \in B(0,1)} |\ell_i(x)| = \|(Y^{-T})_i\|$. Thus $|\det(Y)| \geq \Lambda_0^{-n}$. We have by Cramer's rule that replacing y_i with s results in a matrix Y^+ which satisfies

$$|\det(Y^+)| = |\det(Y)| |\ell_i(s)|.$$

Combining this with the fact that $|\det(Y)| \leq 1$, we have that $\max_{i \in [n]} \max_{x \in B(0,1)} |\ell_i(x)| \leq |\det(Y)|^{-1}$. Thus $\mathcal{Y}$ is Λ -poised if $|\det(Y)| \geq \Lambda^{-1}$. All that remains is to show that $|\det(Y)|$ must increase quickly.

After “Geometry correction of $\mathcal{Y}$ by replacing a "bad" point” resulting in a matrix Y^+ , we have

$$|\det(Y^+)| = |\det(Y)| \left(\max_{i \in [n]} \max_{x \in B(0,1)} |\ell_i(x)| \right) = |\det(Y)| \max_{i \in [n]} \|(Y^{-T})_i\|.$$

Observe by the AM-GM inequality

$$\max_{i \in [n]} \|(Y^{-T})_i\|^2 \geq \frac{1}{n} \sum_i \|(Y^{-T})_i\|^2 = \frac{1}{n} \|Y^{-T}\|_F^2 = \frac{1}{n} \sum_i \sigma_i(Y^{-T})^2 \geq \sqrt[n]{\sigma_i(Y^{-T})^2} = |\det(Y)|^{-\frac{2}{n}}.$$

Thus we have $\max_{i \in [n]} \max_{x \in B(0,1)} |\ell_i(x)| = \max_{i \in [n]} \|(Y^{-T})_i\| \geq |\det(Y)|^{-\frac{1}{n}}$ and hence

$$|\det(Y^+)| \geq |\det(Y)| |\det(Y)|^{-\frac{1}{n}} = |\det(Y)|^{(1-\frac{1}{n})}.$$

Taking the logarithm of both sides, we have

$$\log |\det(Y^+)| \geq \left(1 - \frac{1}{n}\right) \log |\det(Y)|.$$

Thus, $-\log |\det(Y)|$ shrinks exponentially.

If in a single iteration we first do “Geometry correction by replacing a point in $\mathcal{Y}$ with a Lagrange Polynomial value” to get Y^+ , and then do “Geometry correction of $\mathcal{Y}$ by replacing a "bad" point” to get Y^{++} we have

$$\log |\det(Y^{++})| \geq \left(1 - \frac{1}{n}\right) \log |\det(Y^+)| \geq \left(1 - \frac{1}{n}\right) \log (|\det(Y)| |\ell_i(s_k^*)|) \geq \left(1 - \frac{1}{n}\right) \log |\det(Y)|.$$

在 [13] 中描述的过程。

在步骤4的情形 (ii) 和 (iii) 中, 当前 $\mathcal{Y}$ 集合中的点 $y_{j_k^*}$ 被一个新点替代, 我们称之为 $\tilde{s}$ 。用于推导新的拉格朗日多项式集 ℓ^+ 的更新可以通过以下公式进行:

$$(4.4) \quad \ell_{j_k^*}^+(x) = \frac{\ell_{j_k^*}(x)}{\ell_{j_k^*}(\tilde{s})}$$

$$(4.5) \quad \ell_i^+(x) = \ell_i(x) - \ell_i(\tilde{s})\ell_{j_k^*}^+(x) \quad i \neq j_k^*.$$

注意 由于我们假设 $\mathcal{Y}_0$ 是适定的, 那么根据构造, 所有后续的 $\mathcal{Y}_k$ 集合也都是适定的 n .

我们现在给出一些结果, 使我们能够对模型改进迭代 $\mathcal{M}_\epsilon$ 的次数给出界。

定理4.1. 令 $\mathcal{P}$ 为线性多项式集合 (维数为 $p = n$)。那么, 在任何连续模型改进迭代序列中, 预言机调用的次数; 即, 使得 $k \in \mathcal{M}_\epsilon$ 至多为 $4n \log n + 8n + 2n|\log \log(\Lambda)|$ 。

我们将这个定理的证明分解为以下两个引理, 第一个引理推导出对任意 Λ_0 , 从 Λ_0 -适定集出发, 获得期望的 Λ -适定集所需的连续模型改进迭代次数的界。第二个引理表明, 在前 $2n$ 次连续模型改进迭代之后, 可获得 Λ_0 -适定集, 且 Λ_0 具有特定的界。

引理 4.2. 设 $\mathcal{P}$ 为线性多项式集合 (维数为 $p = n$)。若 $\mathcal{Y}_k$ 在 $B(0, \Delta)$ 中是 Λ_0 -适定的, 则最多会有 $\lceil n \log n + n|\log \log \Lambda_0| + n|\log \log(\Lambda)| \rceil$ 额外的连续模型改进迭代。

证明。 为简单起见且不失一般性, 假设 $\Delta = 1$ 。拥有一个 Λ_0 -适定集意味着, 由哈达马不等式可知, 我们有

$$|\det(Y^{-T})| \leq \prod_i^n \|(Y^{-T})_i\| \leq \Lambda_0^n.$$

因为 $\max_{x \in B(0,1)} |\ell_i(x)| = \|(Y^{-T})_i\|$ 。因此 $|\det(Y)| \geq \Lambda_0^{-n}$ 。由克莱姆法则可知, 将 y_i 替换为 s 会得到一个矩阵 Y^+ , 其满足

$$|\det(Y^+)| = |\det(Y)| |\ell_i(s)|.$$

将此与 $|\det(Y)| \leq 1$ 这一事实结合起来, 可得 $\max_{i \in [n]} \max_{x \in B(0,1)} |\ell_i(x)| \leq |\det(Y)|^{-1}$ 。因此, 若 $|\det(Y)| \geq \Lambda^{-1}$, 则 $\mathcal{Y}$ 是 Λ -适定的。剩下只需证明 $|\det(Y)|$ 必须快速增加。

在 “Geo metry 校正 $\mathcal{Y}$, 通过替换一个 ‘坏’ 点” 得到矩阵 Y^+ , 我们有

$$|\det(Y^+)| = |\det(Y)| \left(\max_{i \in [n]} \max_{x \in B(0,1)} |\ell_i(x)| \right) = |\det(Y)| \max_{i \in [n]} \|(Y^{-T})_i\|.$$

由算术-几何平均不等式可观察到

$$\max_{i \in [n]} \|(Y^{-T})_i\|^2 \geq \frac{1}{n} \sum_i \|(Y^{-T})_i\|^2 = \frac{1}{n} \|Y^{-T}\|_F^2 = \frac{1}{n} \sum_i \sigma_i(Y^{-T})^2 \geq \sqrt[n]{\sigma_i(Y^{-T})^2} = |\det(Y)|^{-\frac{2}{n}}.$$

因此, 我们有 $\max_{i \in [n]} \max_{x \in B(0,1)} |\ell_i(x)| = \max_{i \in [n]} \|(Y^{-T})_i\| \geq |\det(Y)|^{-\frac{1}{n}}$, 从而

$$|\det(Y^+)| \geq |\det(Y)| |\det(Y)|^{-\frac{1}{n}} = |\det(Y)|^{(1-\frac{1}{n})}.$$

对两边取对数, 我们有

$$\log |\det(Y^+)| \geq \left(1 - \frac{1}{n}\right) \log |\det(Y)|.$$

因此, $-\log |\det(Y)|$ 呈指数级收缩。

如果在单次迭代中, 我们首先进行 “通过将 $\mathcal{Y}$ 中的一个点替换为拉格朗日多项式值来进行几何校正” 以得到 Y^+ , 然后进行 “通过替换一个 ‘坏’ 点来对 $\mathcal{Y}$ 进行几何校正” 以得到 Y^{++} , 则我们有

$$\log |\det(Y^{++})| \geq \left(1 - \frac{1}{n}\right) \log |\det(Y^+)| \geq \left(1 - \frac{1}{n}\right) \log (|\det(Y)| |\ell_i(s_k^*)|) \geq \left(1 - \frac{1}{n}\right) \log |\det(Y)|.$$

Thus, in either case, $-\log |\det(Y)|$ shrinks exponentially.

Recall that we have $|\det(Y)| \geq \Lambda_0^{-n}$ and thus $\log |\det(Y)| \geq -n \log(\Lambda_0)$. Also, we will have achieved Λ -poisedness if $|\det(Y)| \geq \Lambda^{-1}$ or equivalently $\log |\det(Y)| \geq -\log(\Lambda)$. Thus, the number of additional iterations is bounded by

$$\left\lceil \frac{\log \left(\frac{-n \log(\Lambda_0)}{-\log(\Lambda)} \right)}{\log \left(1 - \frac{1}{n} \right)} \right\rceil \leq \left\lceil n \log \left(\frac{n \log(\Lambda_0)}{\log(\Lambda)} \right) \right\rceil \leq \lceil n \log n + n |\log \log \Lambda_0| + n |\log \log(\Lambda)| \rceil.$$

□

LEMMA 4.3. *Let $\mathcal{P}$ be the set of linear polynomials (with dimension $p = n$). Then $\mathcal{Y}_k$ is 14^n -poised after at most $2n$ consecutive model improving iterations.*

Proof. Because the first step of a model improving iteration replaces any point in $\mathcal{Y}_k$ that is outside $B(0, \Delta_k)$ then after at most n such iterations (and n oracle calls), $\mathcal{Y}_k$ contains n points all of which have norm at most Δ_k . For simplicity, we assume $\Delta_k = 1$. We show that in at most n additional iterations, the set $\mathcal{Y}_k$ becomes 14^n -poised.

For a subspace S , we say that a set of points $\mathcal{Y}$ is Λ -poised in S if for the Lagrange polynomials ℓ_i , we have $\max_{x \in B(0,1) \cap S} |\ell_i(x)| \leq \Lambda$. Note that if $y_1 \in \mathcal{Y}_k$ is a unit vector, then $\mathcal{Y}$ is 1-poised in $S = \text{span}(\{y_1\})$. Below, we prove two claims regarding poisedness in a subspace.

First, we claim that if $\mathcal{Y}$ is Λ -poised in S , then after step (ii): “Self-correction by replacing a point in $\mathcal{Y}$ with a large Lagrange Polynomial value”, we will have that the new set of points is 2Λ -poised in S .

Let s_k be the point we are adding and let $j_k^* = \arg \max_{j=1, \dots, n} |\ell_j(s_k)|$. For the replacement to take place, we must have $|\ell_{j_k^*}(s_k)| \geq 1$. Let ℓ_i^+ denote the Lagrange polynomials after replacement. Then from (4.4) we have

$$\max_{x \in B(0,1) \cap S} |\ell_{j_k^*}^+(x)| = \max_{x \in B(0,1) \cap S} \frac{|\ell_{j_k^*}(x)|}{|\ell_{j_k^*}(s_k)|} \leq \frac{\Lambda}{1} \leq 2\Lambda.$$

For $i \neq j_k^*$, we have

$$\begin{aligned} \max_{x \in B(0,1) \cap S} |\ell_i^+(x)| &= \max_{x \in B(0,1) \cap S} \left| \ell_i(x) - \frac{\ell_i(s_k) \ell_{j_k^*}(x)}{\ell_{j_k^*}(s_k)} \right| \\ &\leq \max_{x \in B(0,1) \cap S} |\ell_i(x)| + \left| \frac{\ell_i(s_k)}{\ell_{j_k^*}(s_k)} \right| \left(\max_{x \in B(0,1) \cap S} |\ell_{j_k^*}(x)| \right) \leq 2\Lambda. \end{aligned}$$

The last inequality is because $|\ell_i(s_k)| \leq |\ell_{j_k^*}(s_k)|$. This completes the proof of this first claim.

Second, we claim that if $\mathcal{Y}$ is Λ -poised in a proper subspace S , then after (iii): “Geometry correction of $\mathcal{Y}$ by replacing a "bad" point”, we will have that the new set of points is 7Λ -poised in S^+ , where S^+ is a subspace of one dimension greater than S .

Let $(j_k^*, s_k^*) = \arg \max_{j=1, \dots, p, s \in B(0, \Delta_k)} |\ell_j(s)|$. We will replace $y_{j_k^*}$ with s_k^* ; let ℓ_i^+ denote the Lagrange polynomials after replacement and apply (4.4).

If $\mathcal{Y}$ is 2Λ -poised in the whole space, then we have

$$\max_{x \in B(0,1)} |\ell_{j_k^*}^+(x)| = \max_{x \in B(0,1)} \frac{|\ell_{j_k^*}(x)|}{|\ell_{j_k^*}(s_k^*)|} = 1 \leq 7\Lambda.$$

Furthermore, for $i \neq j_k^*$, we have

$$\begin{aligned} \max_{x \in B(0,1)} |\ell_i^+(x)| &= \max_{x \in B(0,1)} \left| \ell_i(x) - \frac{\ell_i(s_k^*) \ell_{j_k^*}(x)}{\ell_{j_k^*}(s_k^*)} \right| \\ &\leq \max_{x \in B(0,1)} |\ell_i(x)| + \left| \frac{\ell_i(s_k^*)}{\ell_{j_k^*}(s_k^*)} \right| \left(\max_{x \in B(0,1)} |\ell_{j_k^*}(x)| \right) \leq 4\Lambda \leq 7\Lambda. \end{aligned}$$

Thus, we have that the points are 7Λ poised in the whole space and we can take S^+ to be any subspace of dimension one greater than S . Thus, the claim is shown if $\mathcal{Y}$ is 2Λ -poised in the whole space. For the remainder of the proof of the claim we may assume $\mathcal{Y}$ is not 2Λ -poised in the whole space.

因此, 无论哪种情况, $-\log |\det(Y)|$ 都会呈指数级缩小。

回顾一下, 我们有 $|\det(Y)| \geq \Lambda_0^{-n}$, 因此 $\log |\det(Y)| \geq -n \log(\Lambda_0)$ 。此外, 如果 $|\det(Y)| \geq \Lambda^{-1}$, 或者等价地 $\log |\det(Y)| \geq -\log(\Lambda)$, 我们将达到 Λ -适定性。因此, 额外迭代的次数上界为

$$\left\lceil \frac{\log \left(\frac{-n \log(\Lambda_0)}{-\log(\Lambda)} \right)}{\log \left(1 - \frac{1}{n} \right)} \right\rceil \leq \left\lceil n \log \left(\frac{n \log(\Lambda_0)}{\log(\Lambda)} \right) \right\rceil \leq \lceil n \log n + n |\log \log \Lambda_0| + n |\log \log(\Lambda)| \rceil.$$

□

引理 4.3. 设 $\mathcal{P}$ 为线性多项式集合 (维数为 $p = n$)。那么, $\mathcal{Y}_k$ 在至多 $2n$ 次连续模型改进迭代之后是 14^n -适定的。

证明。因为模型改进迭代的第一步会替换 $\mathcal{Y}_k$ 中位于 $B(0, \Delta_k)$ 之外的任意点, 所以在至多 n 次这样的迭代 (以及 n 次预言机调用) 之后, $\mathcal{Y}_k$ 包含 n 个点, 且所有这些点的范数至多为 Δ_k 。为简单起见, 我们假设 $\Delta_k = 1$ 。我们证明, 在至多 n 个额外迭代之后, 集合 $\mathcal{Y}_k$ 变为 14^n -适定的。

对于子空间 S , 如果对于拉格朗日多项式 ℓ_i 有 $\max_{x \in B(0,1) \cap S} |\ell_i(x)| \leq \Lambda$, 我们就称点集 $\mathcal{Y}$ 在 S 中是 Λ -适定的。注意, 如果 $y_1 \in \mathcal{Y}_k$ 是单位向量, 那么 $\mathcal{Y}$ 在 $S = \text{span}(\{y_1\})$ 中是 1-适定的。下面, 我们证明两个关于子空间中适定性的断言。

首先, 我们断言, 如果 $\mathcal{Y}$ 在 S 中是 Λ -适定的, 那么在执行步骤 (ii): “通过用具有较大拉格朗日多项式值的点替换 $\mathcal{Y}$ 中的一个点来进行自校正”之后, 新点集将在 S 中是 2Λ -适定的。

令 s_k 为我们正在添加的点, 并令 $j_k^* = \arg \max_{j=1, \dots, n} |\ell_j(s_k)|$ 。为了使替换发生, 必须有 $|\ell_{j_k^*}(s_k)| \geq 1$ 。令 ℓ_i^+ 表示替换后的拉格朗日多项式。那么由 (4.4) 我们有

$$\max_{x \in B(0,1) \cap S} |\ell_{j_k^*}^+(x)| = \max_{x \in B(0,1) \cap S} \frac{|\ell_{j_k^*}(x)|}{|\ell_{j_k^*}(s_k)|} \leq \frac{\Lambda}{1} \leq 2\Lambda.$$

对于 $i \neq j_k^*$, 我们有

$$\begin{aligned} \max_{x \in B(0,1) \cap S} |\ell_i^+(x)| &= \max_{x \in B(0,1) \cap S} \left| \ell_i(x) - \frac{\ell_i(s_k) \ell_{j_k^*}(x)}{\ell_{j_k^*}(s_k)} \right| \\ &\leq \max_{x \in B(0,1) \cap S} |\ell_i(x)| + \left| \frac{\ell_i(s_k)}{\ell_{j_k^*}(s_k)} \right| \left(\max_{x \in B(0,1) \cap S} |\ell_{j_k^*}(x)| \right) \leq 2\Lambda. \end{aligned}$$

最后一个不等式成立是因为 $|\ell_i(s_k)| \leq |\ell_{j_k^*}(s_k)|$ 。这就完成了第一个断言的证明。

其次, 我们断言: 如果 $\mathcal{Y}$ 在真子空间 S 中是 Λ -适定的, 那么在执行 (iii): “通过替换一个“坏”点来对 $\mathcal{Y}$ 进行几何校正”之后, 我们将有新的点集在 S^+ 中是 7Λ -适定的, 其中 S^+ 是维数比 S 大一维的子空间。

令 $(j_k^*, s_k^*) = \arg \max_{j=1, \dots, p, s \in B(0, \Delta_k)} |\ell_j(s)|$ 。我们将用 s_k^* 替换 $y_{j_k^*}$; 令 ℓ_i^+ 表示替换后的拉格朗日多项式, 并应用 (4.4)。

如果 $\mathcal{Y}$ 在全空间中是 2Λ -适定的, 那么我们有

$$\max_{x \in B(0,1)} |\ell_{j_k^*}^+(x)| = \max_{x \in B(0,1)} \frac{|\ell_{j_k^*}(x)|}{|\ell_{j_k^*}(s_k^*)|} = 1 \leq 7\Lambda.$$

此外, 对于 $i \neq j_k^*$, 我们有

$$\begin{aligned} \max_{x \in B(0,1)} |\ell_i^+(x)| &= \max_{x \in B(0,1)} \left| \ell_i(x) - \frac{\ell_i(s_k^*) \ell_{j_k^*}(x)}{\ell_{j_k^*}(s_k^*)} \right| \\ &\leq \max_{x \in B(0,1)} |\ell_i(x)| + \left| \frac{\ell_i(s_k^*)}{\ell_{j_k^*}(s_k^*)} \right| \left(\max_{x \in B(0,1)} |\ell_{j_k^*}(x)| \right) \leq 4\Lambda \leq 7\Lambda. \end{aligned}$$

因此, 我们得到这些点在全空间中是 7Λ 适定的, 并且可以取 S^+ 为任意一个维数比 S 大一维的子空间。因此, 如果 $\mathcal{Y}$ 在全空间中是 2Λ -适定的, 则该断言得证。在该断言的证明剩余部分中, 我们可以假设 $\mathcal{Y}$ 在全空间中不是 2Λ -适定的。

Let P_S and $P_S^\perp$ denote the projections onto S and its orthogonal complement, respectively. By linearity, we have

$$2\Lambda < \|\ell(s_k^*)\|_\infty = \|\ell(P_S^\perp s_k^*) + \ell(P_S s_k^*)\|_\infty \leq \|\ell(P_S^\perp s_k^*)\|_\infty + \|\ell(P_S s_k^*)\|_\infty \leq \|\ell(P_S^\perp s_k^*)\|_\infty + \Lambda.$$

Thus, $\|\ell(P_S^\perp s_k^*)\|_\infty > \Lambda$. By homogeneity and optimality we have,

$$\frac{\|\ell(P_S^\perp s_k^*)\|_\infty}{\|P_S^\perp s_k^*\|} = \left\| \ell \left(\frac{P_S^\perp s_k^*}{\|P_S^\perp s_k^*\|} \right) \right\|_\infty \leq \|\ell(s_k^*)\|_\infty \leq \|\ell(P_S^\perp s_k^*)\|_\infty + \Lambda.$$

Thus we have

$$\|P_S^\perp s_k^*\| \geq \frac{\|\ell(P_S^\perp s_k^*)\|_\infty}{\|\ell(P_S^\perp s_k^*)\|_\infty + \Lambda} \geq \frac{1}{2}$$

since $\|\ell(P_S^\perp s_k^*)\|_\infty > \Lambda$.

By an argument identical to that used in the proof of the first claim, we have $\max_{x \in B(0,1) \cap S} \|\ell^+(x)\|_\infty \leq 2\Lambda$.

Next we wish to bound $\left\| \ell^+ \left(\frac{P_S^\perp s_k^*}{\|P_S^\perp s_k^*\|} \right) \right\|_\infty$. Noting that $\ell^+(s_k^*) = e_{j_k^*}$, we have

$$\|\ell^+(P_S^\perp s_k^*)\|_\infty = \|\ell^+(s_k^*) - \ell^+(P_S s_k^*)\|_\infty \leq \|\ell^+(s_k^*)\|_\infty + \|\ell^+(P_S s_k^*)\|_\infty = 1 + 2\Lambda \leq 3\Lambda.$$

Then we have

$$\left\| \ell^+ \left(\frac{P_S^\perp s_k^*}{\|P_S^\perp s_k^*\|} \right) \right\|_\infty = \frac{\|\ell^+(P_S^\perp s_k^*)\|_\infty}{\|P_S^\perp s_k^*\|} \leq 6\Lambda.$$

Let us define $S^+ = S \oplus \text{span}(\{P_S^\perp s_k^*\})$. We can represent any vector x in $B(0,1) \cap S^+$, as $\alpha x' + \beta \frac{P_S^\perp s_k^*}{\|P_S^\perp s_k^*\|}$ where $x' \in B(0,1) \cap S$ and $\alpha^2 + \beta^2 \leq 1$. Then we have

$$\left\| \ell^+ \left(\alpha x' + \beta \frac{P_S^\perp s_k^*}{\|P_S^\perp s_k^*\|} \right) \right\|_\infty \leq \alpha \|\ell^+(x')\|_\infty + \beta \left\| \ell^+ \left(\frac{P_S^\perp s_k^*}{\|P_S^\perp s_k^*\|} \right) \right\|_\infty \leq |\alpha|2\Lambda + |\beta|6\Lambda \leq \left\| \begin{bmatrix} 2 \\ 6 \end{bmatrix} \right\| \Lambda \leq 7\Lambda.$$

This completes the proof of the second claim.

For $t \leq n$, we will show by induction that after at most t iterations after removing far points, $\mathcal{Y}_k$ is 14^t -poised in a subspace of dimension at least t .

For the base case, when “Geometry correction of $\mathcal{Y}_k$ by replacing a "bad" point” is performed for the first time, it is done by maximizing a linear Lagrange polynomial over a ball. Since the maximum of a linear function over the unit ball is always attained on the boundary, the new point added to $\mathcal{Y}_k$ is on the boundary, thus $\mathcal{Y}_k$ contains at least one unit vector. Recall that if $y_1 \in \mathcal{Y}_k$ is a unit vector, then $\mathcal{Y}$ is 1-poised in $S = \text{span}(\{y_1\})$. Thus, $\mathcal{Y}_k$ is 1-poised in a one-dimensional subspace, establishing the base case.

Now let us assume that after the t th iteration after removing far points, $\mathcal{Y}_k$ is 14^t -poised in a subspace, S , of dimension at least t , and let us show that after an additional iteration, $\mathcal{Y}_k$ is 14^{t+1} -poised in a subspace of dimension at least $t+1$. In the t th-iteration, the algorithm either performs (ii) “Self-correction by replacing a point in $\mathcal{Y}_k$ with a large Lagrange Polynomial value” and (iii) “Geometry correction of $\mathcal{Y}_k$ by replacing a "bad" point”, or just the latter. If the algorithm performs both corrections, then after the first correction we have that the points are $2 \cdot 14^t$ -poised in S , by the first claim. Then after the second correction, by the second claim, the points are $7 \cdot 2 \cdot 14^t = 14^{t+1}$ -poised in S^+ which is of dimension at least one greater than S . Alternatively, if the algorithm only performs the latter correction, then by the second claim, the points are $7 \cdot 14^t \leq 14^{t+1}$ -poised in S^+ which is of dimension at least one greater than S . Thus, in either case, the inductive argument is complete. Thus, after at most n iterations after removing far points, $\mathcal{Y}_k$ is 14^n -poised in a subspace of dimension at least n (i.e. the whole space). Thus, after at most $2n$ total iterations (including removal of far away points), $\mathcal{Y}_k$ is 14^n -poised. $\square$

Proof of Theorem 4.1. We can combine Lemmas 4.3 and 4.2 by letting $\Lambda_0 = 14^n$. Thus the total number of iterations can be bounded as follows:

$$\begin{aligned} & 2n + \lceil n \log n + n \log \log 14^n \rceil + n \log \log(\Lambda) \rceil \\ &= 2n + \lceil 2n \log n + n \log \log 14 \rceil + n \log \log(\Lambda) \rceil \\ &\leq 2n \log n + 4n + n \log \log(\Lambda). \end{aligned}$$

Since each iteration uses at most 2 oracle calls, the result follows. $\square$

令 P_S 和 $P_S^\perp$ 分别表示到 S 及其正交补上的投影。根据线性，我们有

$$2\Lambda < \|\ell(s_k^*)\|_\infty = \|\ell(P_S^\perp s_k^*) + \ell(P_S s_k^*)\|_\infty \leq \|\ell(P_S^\perp s_k^*)\|_\infty + \|\ell(P_S s_k^*)\|_\infty \leq \|\ell(P_S^\perp s_k^*)\|_\infty + \Lambda.$$

因此， $\|\ell(P_S^\perp s_k^*)\|_\infty > \Lambda$ 。由齐次性和最优性可得，

$$\frac{\|\ell(P_S^\perp s_k^*)\|_\infty}{\|P_S^\perp s_k^*\|} = \left\| \ell \left(\frac{P_S^\perp s_k^*}{\|P_S^\perp s_k^*\|} \right) \right\|_\infty \leq \|\ell(s_k^*)\|_\infty \leq \|\ell(P_S^\perp s_k^*)\|_\infty + \Lambda.$$

因此我们有

$$\|P_S^\perp s_k^*\| \geq \frac{\|\ell(P_S^\perp s_k^*)\|_\infty}{\|\ell(P_S^\perp s_k^*)\|_\infty + \Lambda} \geq \frac{1}{2}$$

由于 $\|\ell(P_S^\perp s_k^*)\|_\infty > \Lambda$ 。

采用与第一个断言证明中相同的论证，我们有 $\max_{x \in B(0,1) \cap S} \|\ell^+(x)\|_\infty \leq 2\Lambda$ 。接下来，我们希望给出

$\left\| \ell^+ \left(\frac{P_S^\perp s_k^*}{\|P_S^\perp s_k^*\|} \right) \right\|_\infty$ 的界。注意到 $\ell^+(s_k^*) = e_{j_k^*}$ ，我们有

$$\|\ell^+(P_S^\perp s_k^*)\|_\infty = \|\ell^+(s_k^*) - \ell^+(P_S s_k^*)\|_\infty \leq \|\ell^+(s_k^*)\|_\infty + \|\ell^+(P_S s_k^*)\|_\infty = 1 + 2\Lambda \leq 3\Lambda.$$

于是我们有

$$\left\| \ell^+ \left(\frac{P_S^\perp s_k^*}{\|P_S^\perp s_k^*\|} \right) \right\|_\infty = \frac{\|\ell^+(P_S^\perp s_k^*)\|_\infty}{\|P_S^\perp s_k^*\|} \leq 6\Lambda.$$

我们定义 $S^+ = S \oplus \text{span}(\{P_S^\perp s_k^*\})$ 。我们可以将任意向量 x 在 $B(0,1) \cap S^+$ 中表示为 $\alpha x' + \beta \frac{P_S^\perp s_k^*}{\|P_S^\perp s_k^*\|}$ ，

其中 $x' \in B(0,1) \cap S$ ，且 $\alpha^2 + \beta^2 \leq 1$ 。于是我们有

$$\left\| \ell^+ \left(\alpha x' + \beta \frac{P_S^\perp s_k^*}{\|P_S^\perp s_k^*\|} \right) \right\|_\infty \leq \alpha \|\ell^+(x')\|_\infty + \beta \left\| \ell^+ \left(\frac{P_S^\perp s_k^*}{\|P_S^\perp s_k^*\|} \right) \right\|_\infty \leq |\alpha|2\Lambda + |\beta|6\Lambda \leq \left\| \begin{bmatrix} 2 \\ 6 \end{bmatrix} \right\| \Lambda \leq 7\Lambda.$$

这就完成了第二个断言的证明。

对于 $t \leq n$ ，我们将通过归纳法证明：在移除远点后至多经过 t 次迭代， $\mathcal{Y}_k$ 在维数至少为 t 的子空间中是 14^t -适定的。

对于基础情形，当首次执行“通过替换一个‘坏’点对 $\mathcal{Y}_k$ 进行几何校正”时，它是通过在一个球上最大化一个线性拉格朗日多项式来完成的。由于线性函数在单位球上的最大值总在边界上取得，添加到 $\mathcal{Y}_k$ 的新点位于边界上，因此 $\mathcal{Y}_k$ 至少包含一个单位向量。回顾一下，如果 $y_1 \in \mathcal{Y}_k$ 是一个单位向量，那么 $\mathcal{Y}$ 在 $S = \text{span}(\{y_1\})$ 中是 1-适定的。因此， $\mathcal{Y}_k$ 在一维子空间中是 1-适定的，从而建立了基础情形。

现在假设，在移除远点后的第 t 次迭代之后， $\mathcal{Y}_k$ 在子空间 S 中是 14^t -适定的，且该子空间的维数至少为 t ；下面证明，在经过额外一次迭代后， $\mathcal{Y}_k$ 在维数至少为 $t+1$ 的子空间中是 14^{t+1} -适定的。在第 t 次迭代中，算法要么执行 (ii) “通过用较大的拉格朗日多项式值替换 $\mathcal{Y}_k$ 中的一个点来进行自校正”和 (iii) “通过替换一个‘坏’点对 $\mathcal{Y}_k$ 进行几何校正”，要么只执行后者。如果算法执行这两种校正，那么在第一次校正之后，根据第一个断言，这些点在 S 中是 $2 \cdot 14^t$ -适定的。然后，在第二次校正之后，根据第二个断言，这些点在 S^+ 中是 $7 \cdot 2 \cdot 14^t = 14^{t+1}$ -适定的，而 S^+ 的维数至少比 S 大 1。或者，如果算法只执行后一种校正，那么根据第二个断言，这些点在 S^+ 中是 $7 \cdot 14^t \leq 14^{t+1}$ -适定的，而 S^+ 的维数至少比 S 大 1。因此，无论哪种情况，归纳论证都完成了。因此，在移除远点后至多经过 n 次迭代， $\mathcal{Y}_k$ 在维数至少为 n 的子空间中是 14^n -适定的（即全空间）。因此，在至多 $2n$ 次总迭代之后（包括移除远点）， $\mathcal{Y}_k$ 是 14^n -适定的。 $\square$

定理 4.1 的证明。我们可以通过令 $\Lambda_0 = 14^n$ 来结合引理 4.3 和 4.2。因此，总迭代次数可以如下界定：

$$\begin{aligned} & 2n + \lceil n \log n + n \log \log 14^n \rceil + n \log \log(\Lambda) \rceil \\ &= 2n + \lceil 2n \log n + n \log \log 14 \rceil + n \log \log(\Lambda) \rceil \\ &\leq 2n \log n + 4n + n \log \log(\Lambda). \end{aligned}$$

由于每次迭代最多使用 2 次预言机调用，故结论成立。 $\square$

With these results, we may now present a total complexity theorem for Algorithm 2. Recall that Theorem 2.8 provides a bound for $|\mathcal{S}_\epsilon| + |\mathcal{U}_\epsilon|$. As we have just demonstrated the number of consecutive model improving iterations cannot be larger than $2n \log n + 4n + n \lfloor \log \log(\Lambda) \rfloor$ until set $\mathcal{Y}_k$ is Λ -poised in $B(x_k, \Delta_k)$. Let $\Lambda = 1 + \frac{1}{n}$, then the number of consecutive iterations can be bounded by $\mathcal{O}(n \log n)$. By Corollary 3.6 we have, for any $\Delta_k \geq \sqrt{\frac{4\epsilon_f \Lambda}{L + \kappa_{bhm}}}$,

$$\begin{aligned}\kappa_{eg} &= (L + \kappa_{bhm}) \sqrt{n} \sqrt{n(\Lambda^2 - 1) + 2} = \Theta(\sqrt{n}) \\ \kappa_{ef} &= \kappa_{eg} + \frac{L + \kappa_{bhm}}{2} = \Theta(\sqrt{n}).\end{aligned}$$

Thus we have the following corollary.

COROLLARY 4.4. *Under the same assumptions as Theorem 2.8, for any $\epsilon > \sqrt{\frac{4\epsilon_f}{\gamma^2 \min\{C_2, L + \kappa_{bhm}\} C_1^2}}$ (i.e. $\epsilon \geq \Omega(\sqrt{n\epsilon_f})$) the total oracle complexity of Algorithm 2 is bounded as*

$$\mathcal{C}_\epsilon \leq \mathcal{O}\left(\frac{n^{3/2} \log n}{\epsilon^2}\right)$$

if $\eta_2 = \sqrt{n}$ and as

$$\mathcal{C}_\epsilon \leq \mathcal{O}\left(\frac{n^2 \log n}{\epsilon^2}\right)$$

if η_2 is constant.

5 Model based trust region methods in subspaces. We now consider a trust region method where a model $m(x)$ is built and optimized in a random low-dimensional subspace of $\mathbb{R}^n$. The idea of using random subspace embeddings within derivative-free methods has gained a lot of popularity in the literature lately. It was shown in [16] that applying direct search methods in a low-dimensional subspace reduces the oracle complexity from $\mathcal{O}(n^2\epsilon^{-2})$ to $\mathcal{O}(n\epsilon^{-2})$ (with dependence on the subspace dimensions suppressed). A random subspace version of a model-based TR method was first studied in [8] with the use of Johnson-Lindenstrauss (JL) subspace embeddings which achieved complexity $\mathcal{O}(n^2\epsilon^{-2})$. Later in [14] this approach was combined with a stochastic model-based trust region method. Recently it was shown in [11] that a subspace model-based trust region method achieves an improved $\mathcal{O}(n\epsilon^{-2})$ complexity when the model is based on a random projection rather than a JL embedding. The key difference lies in the scaling of the projected gradient that is being estimated. A recent note [9] confirms that by rescaling the JL transformation in [8] the rate $\mathcal{O}(n\epsilon^{-2})$ can be achieved.

The works [8, 11] do not consider noisy function values. In [14] the noise in the function values is stochastic and is assumed to be reducible to any desired accuracy, dictated by the trust region radius, which is allowed to shrink to an arbitrarily small value. Here we extend the analysis of a subspace trust region method from [11] to accommodate fixed (deterministic) noise in the function oracle. The fundamental difficulty of doing so is that the algorithmic framework with random subspaces does not by itself guarantee a lower bound on the trust region radius, which is necessary for the analysis of the noisy function oracles, as we have seen in the sections above. As a consequence we need to introduce two algorithmic modifications - a relaxed step acceptance criterion and an enforced lower bound on the trust region radius.

We begin by introducing the subspace embedding of our problem. Given a matrix $Q \in \mathbb{R}^{n \times q}$, with $q \leq n$ and orthonormal columns, $QQ^T \nabla \phi(x)$ is an orthogonal projection of $\nabla \phi(x)$ onto a subspace spanned by the columns of Q (we will call it a subspace induced by Q). We also define a reduction of $\phi(x)$ to the subspace, given by Q around x : $\hat{\phi}(v) = \phi(x + Qv)$, $v \in \mathbb{R}^q$, which implies $Q \nabla \hat{\phi}(0) = QQ^T \nabla \phi(x)$. Similarly we define $\hat{m}(v) = m(x + Qv)$, $v \in \mathbb{R}^q$, which implies $Q \nabla \hat{m}(0) = QQ^T \nabla m(x)$.

We now present a modified trust-region algorithm that constructs models and computes steps in the subspace. At each iteration $k \in \{0, 1, \dots\}$ the algorithm chooses $Q_k \in \mathbb{R}^{n \times q}$ with orthonormal columns. The model m_k is defined as

$$(5.1) \quad m_k(x_k + Q_k v) = \phi(x_k) + g_k^T Q_k v + \frac{1}{2} v^T Q_k^T H_k Q_k v.$$

有了这些结果，我们现在可以给出算法2的总复杂度定理。回忆定理 2.8 为 $|\mathcal{S}_\epsilon| + |\mathcal{U}_\epsilon|$ 提供了一个界。正如我们刚刚展示的，连续模型改进迭代的次数不可能大于 $2n \log n + 4n + n \lfloor \log \log(\Lambda) \rfloor$ ，直到集合 $\mathcal{Y}_k$ 在 $B(x_k, \Delta_k)$ 中是 Λ -适定的。令 $\Lambda = 1 + \frac{1}{n}$ ，则连续迭代的次数可由 $\mathcal{O}(n \log n)$ 界定。由推论3.6，我们有，对任意 $\Delta_k \geq \sqrt{\frac{4\epsilon_f \Lambda}{L + \kappa_{bhm}}}$ ，

$$\begin{aligned}\kappa_{eg} &= (L + \kappa_{bhm}) \sqrt{n} \sqrt{n(\Lambda^2 - 1) + 2} = \Theta(\sqrt{n}) \\ \kappa_{ef} &= \kappa_{eg} + \frac{L + \kappa_{bhm}}{2} = \Theta(\sqrt{n}).\end{aligned}$$

因此，我们有以下推论。

推论 4.4. 在与定理 2.8 相同的假设下，对于任意 $\epsilon > \sqrt{\frac{4\epsilon_f}{\gamma^2 \min\{C_2, L + \kappa_{bhm}\} C_1^2}}$ (即 $\epsilon \geq \Omega(\sqrt{n\epsilon_f})$)，

算法2的总预言机复杂度有界如下

$$\mathcal{C}_\epsilon \leq \mathcal{O}\left(\frac{n^{3/2} \log n}{\epsilon^2}\right)$$

如果 $\eta_2 = \sqrt{n}$ 且当

$$\mathcal{C}_\epsilon \leq \mathcal{O}\left(\frac{n^2 \log n}{\epsilon^2}\right)$$

如果 η_2 为常数。

5 子空间中的基于模型的信赖域方法。 我们现在考虑一种信赖域方法，其中在 $\mathbb{R}^n$ 的一个随机低维子空间中构建并优化模型 $m(x)$ 。近年来，在无导数方法中使用随机子空间嵌入的想法在文献中已非常流行。[16] 中表明，在低维子空间中应用直接搜索方法可将预言机复杂度从 $\mathcal{O}(n^2\epsilon^{-2})$ 降低到 $\mathcal{O}(n\epsilon^{-2})$ （其中对子空间维数的依赖被抑制）。基于模型的信赖域方法的随机子空间版本最早在 [8] 中研究，其中使用了 Johnson-Lindenstrauss (JL) 子空间嵌入，达到了复杂度 $\mathcal{O}(n^2\epsilon^{-2})$ 。随后在 [14] 中，该方法与基于随机模型的信赖域方法相结合。最近，[11] 中表明，当模型基于随机投影而非 JL 嵌入时，基于子空间模型的信赖域方法可实现改进的 $\mathcal{O}(n\epsilon^{-2})$ 复杂度。关键区别在于所估计的投影梯度的缩放。最近的一篇注记 [9] 证实，通过重新缩放 [8] 中的 JL 变换，可以达到速率 $\mathcal{O}(n\epsilon^{-2})$ 。

文献 [8, 11] 未考虑带噪函数值。在 [14] 中，函数值中的噪声是随机的，并假定可降低到任意期望精度，该精度由信赖域半径决定，而信赖域半径可以收缩到任意小。在此，我们将来自 [11] 的子空间信赖域方法的分析进行扩展，以容纳函数预言机中的固定（确定性）噪声。这样做的基本困难在于，带有随机子空间的算法框架本身并不能保证信赖域半径的下界，而正如上文各节所见，这对于带噪函数预言机的分析是必要的。因此，我们需要引入两项算法修改——放宽的步接受准则和信赖域半径的强制下界。

我们首先引入问题的子空间嵌入。给定矩阵 $Q \in \mathbb{R}^{n \times q}$ ，其具有 $q \leq n$ 和标准正交列， $QQ^T \nabla \phi(x)$ 是 $\nabla \phi(x)$ 到由 Q 的列张成的子空间上的正交投影（我们将称其为由 Q 诱导的子空间）。我们还定义 $\phi(x)$ 到该子空间的约化，由 Q 在 x 附近给出： $\hat{\phi}(v) = \phi(x + Qv)$ ， $v \in \mathbb{R}^q$ ，这意味着 $Q \nabla \hat{\phi}(0) = QQ^T \nabla \phi(x)$ 。类似地，我们定义 $\hat{m}(v) = m(x + Qv)$ ， $v \in \mathbb{R}^q$ ，这意味着 $Q \nabla \hat{m}(0) = QQ^T \nabla m(x)$ 。

我们现在提出一种改进信赖域算法，该算法在子空间中构造模型并计算步长。在每次迭代 $k \in \{0, 1, \dots\}$ 中，该算法选择具有标准正交列的 $Q_k \in \mathbb{R}^{n \times q}$ 。模型 m_k 定义为

$$(5.1) \quad m_k(x_k + Q_k v) = \phi(x_k) + g_k^T Q_k v + \frac{1}{2} v^T Q_k^T H_k Q_k v.$$

For any vector v , $g_k^T Q_k v = g_k^T Q_k Q_k^T Q_k v$, thus without loss of generality, we will assume that $Q_k Q_k^T g_k = g_k$, in other words, g_k lies in the subspace induced by Q_k . We define the trust region in the subspace induced by Q_k as $B_{Q_k}(x_k, \Delta_k) = \{z : z = x_k + Q_k v, \|v\| \leq \Delta_k\}$.

We will also need the definition of a fully linear model with respect to the subspace. We use the following definition which is the same as in [11].

DEFINITION 5.1 (Fully-linear model in a subspace). *Given a matrix with orthonormal columns $Q \in \mathbb{R}^{n \times q}$, let $B_Q(x, \Delta) = \{z : z = x + Qv, \|v\| \leq \Delta\}$. Let*

$$(5.2) \quad m(x + Qv) = \phi(x) + g^T Qv + \frac{1}{2} v^T Q^T H Q v$$

and $\hat{m}(v) = m(x + Qv)$, $v \in \mathbb{R}^q$. We say that model $m(x + Qv)$ is a κ_{ef}, κ_{eg} -fully linear model of $\phi(x + Qv)$ on $B_Q(x, \Delta)$ if

$$(5.3) \quad \|\nabla \hat{m}(0) - \nabla \hat{\phi}(0)\| \leq \kappa_{eg} \Delta$$

and

$$(5.4) \quad |\hat{m}(v) - \hat{\phi}(v)| \leq \kappa_{ef} \Delta^2$$

for all $\|v\| \leq \Delta$.

Here too (5.3) implies (5.4) with a specific value of κ_{ef} .

LEMMA 5.2 (Lemma 6.6 from [11]). *Under Assumptions 1.2 and 2.2, if (5.3) holds then m is κ_{ef}, κ_{eg} -fully linear on $B_Q(x, \Delta)$ with*

$$\kappa_{ef} = \kappa_{eg} + \frac{L_Q + \kappa_{bhm}}{2}$$

where L_Q is the Lipschitz constant of $QQ^T \nabla \phi(x)$.

对于任意向量 v , $g_k^T Q_k v = g_k^T Q_k Q_k^T Q_k v$, 因此, 不失一般性, 我们将假设 $Q_k Q_k^T g_k = g_k$, 换言之, g_k 位于由 Q_k 诱导的子空间中。我们将由 Q_k 诱导的子空间中的信赖域定义为 $B_{Q_k}(x_k, \Delta_k) = \{z : z = x_k + Q_k v, \|v\| \leq \Delta_k\}$ 。

我们还需要关于子空间的完全线性模型的定义。我们使用以下定义, 它与 [11] 中的定义相同。

定义 5.1 (子空间中的完全线性模型)。给定一个具有标准正交列的矩阵 $Q \in \mathbb{R}^{n \times q}$, 令 $B_Q(x, \Delta) = \{z : z = x + Qv, \|v\| \leq \Delta\}$ 。令

$$(5.2) \quad m(x + Qv) = \phi(x) + g^T Qv + \frac{1}{2} v^T Q^T H Q v$$

以及 $\hat{m}(v) = m(x + Qv)$, $v \in \mathbb{R}^q$ 。我们说模型 $m(x + Qv)$ 是 $\phi(x + Qv)$ 在 $B_Q(x, \Delta)$ 上的一个 κ_{ef}, κ_{eg} -完全线性模型, 如果

$$(5.3) \quad \|\nabla \hat{m}(0) - \nabla \hat{\phi}(0)\| \leq \kappa_{eg} \Delta$$

and

$$(5.4) \quad |\hat{m}(v) - \hat{\phi}(v)| \leq \kappa_{ef} \Delta^2$$

对所有 $\|v\| \leq \Delta$ 。

这里同样, (5.3) 以 κ_{ef} 的一个特定值蕴含 (5.4)。

引理 5.2 (引理6.6, 来自 [11])。在假设 1.2 和 2.2 下, 如果 (5.3) 成立, 则 m 在 $B_Q(x, \Delta)$ 上是 κ_{ef}, κ_{eg} -完全线性的, 其中

$$\kappa_{ef} = \kappa_{eg} + \frac{L_Q + \kappa_{bhm}}{2}$$

其中 L_Q 是 $QQ^T \nabla \phi(x)$ 的利普希茨常数。

Algorithm 3: Trust region method based on fully-linear models in subspace	
Inputs: Inexact zeroth order oracle $ f(x) - \phi(x) \leq \epsilon_f$, minimum radius $\Delta_{\min}$, initial x_0 , $\Delta_0 \geq \Delta_{\min}$, initial matrix $Q_0 \in \mathbb{R}^{n \times q}$ with orthonormal columns and $\eta_1 \in (0, 1)$, $\eta_2 > 0$, and $\gamma \in (0, 1)$.	
for $k = 0, 1, 2, \dots$ do	
1	For the current Q_k compute model m_k as in (5.1).
2	Compute a trial step $x_k + s_k$ where $s_k = Q_k v_k$ with $v_k \approx \arg \min_v \{m_k(x_k + Q_k v) : \ v\ \leq \Delta_k\}$.
3	Compute the ratio ρ_k as
	$\rho_k = \frac{f(x_k) - f(x_k + s_k) + 2\epsilon_f}{m_k(x_k) - m_k(x_k + s_k)}.$
4	Update the iterate and the TR radius as
	$(x_{k+1}, \Delta_{k+1}) \leftarrow \begin{cases} (x_k + s_k, \gamma^{-1} \Delta_k) & \text{if } \rho_k \geq \eta_1 \text{ and } \ g_k\ \geq \eta_2 \Delta_k, \\ (x_k, \Delta_k) & \text{else, if the model is not fully-linear in } B_{Q_k}(x_k, \Delta_k). \\ (x_k, \max\{\gamma \Delta_k, \Delta_{\min}\}) & \text{otherwise.} \end{cases}$
5	Update the subspace
	$Q_{k+1} \in \mathbb{R}^{n \times q} \leftarrow \begin{cases} \text{random} & \text{if } \rho_k \geq \eta_1 \text{ and } \ g_k\ \geq \eta_2 \Delta_k \text{ or if a model is fully-linear} \\ Q_k & \text{otherwise.} \end{cases}$
6	Perform some model improvement steps.

算法 3: 基于子空间完全线性模型的信赖域方法	
输入: 不精确零阶预言机 $ f(x) - \phi(x) \leq \epsilon_f$, 最小半径 $\Delta_{\min}$, 初始 x_0 , $\Delta_0 \geq \Delta_{\min}$, 初始矩阵 $Q_0 \in \mathbb{R}^{n \times q}$ 具有标准正交列, 且 $\eta_1 \in (0, 1)$ 、 $\eta_2 > 0$ 和 $\gamma \in (0, 1)$ 。	
对于 $k = 0, 1, 2, \dots$ 执行	
1	对于当前 Q_k , 按 (5.1) 计算模型 m_k 。
2	计算一个试探步 $x_k + s_k$, 其中 $s_k = Q_k v_k$, 且 $v_k \approx \arg \min_v \{m_k(x_k + Q_k v) : \ v\ \leq \Delta_k\}$ 。
3	计算比率 ρ_k 如下
	$\rho_k = \frac{f(x_k) - f(x_k + s_k) + 2\epsilon_f}{m_k(x_k) - m_k(x_k + s_k)}.$
4	按如下方式更新迭代点和信赖域半径
	$(x_{k+1}, \Delta_{k+1}) \leftarrow \begin{cases} (x_k + s_k, \gamma^{-1} \Delta_k) & \text{如果 } \rho_k \geq \eta_1 \text{ 且 } \ g_k\ \geq \eta_2 \Delta_k, \\ (x_k, \Delta_k) & \text{否则, 如果模型在 } B_{Q_k}(x_k, \Delta_k) \text{ 中不是完全线性的。} \\ (x_k, \max\{\gamma \Delta_k, \Delta_{\min}\}) & \text{否则。} \end{cases}$
5	更新子空间
	$Q_{k+1} \in \mathbb{R}^{n \times q} \leftarrow \begin{cases} \text{random} & \text{if } \rho_k \geq \eta_1 \text{ and } \ g_k\ \geq \eta_2 \Delta_k \text{ or if a model is fully-linear} \\ Q_k & \text{otherwise.} \end{cases}$
6	执行一些模型改进步骤。

We will assume, as before, that Assumption 2.2 holds. The two key modifications of Algorithm 3 compared to Algorithm 1 are the additional term $2\epsilon_f$ in the definition of ρ and the imposed lower bound on the trust region radius Δ_{min} . Both of these modifications are needed because the noisy zeroth-order oracle makes it essential for Δ_k to remain sufficiently positive to ensure fully linear models. Such a lower bound on Δ_k is ensured in the deterministic framework of Algorithm 1 for sufficiently large ϵ . But this is not the case when random subspaces are used because unsuccessful iterations can occur even if Δ_k is small, due to the subspace not being chosen well. We note that since ϵ_f is an upper bound on the error in the zeroth order oracle, any upper estimate of it can be used in the algorithm. Of course, unnecessarily large values will have an adverse effect on the resulting best achievable accuracy ϵ . Similarly, we will see that the choice of ϵ_f dictates the best choice for Δ_{min} .

We now introduce a definition from [11] of a measure of how well the subspace induced by Q aligns with the current gradient.

DEFINITION 5.3 (Well aligned subspace). *The subspace spanned by columns of Q is κ_g -well aligned with $\nabla\phi(x)$ for a given x if*

$$(5.5) \quad \|QQ^T\nabla\phi(x) - \nabla\phi(x)\| \leq \kappa_g\|\nabla\phi(x)\|$$

for some $\kappa_g \in [0, 1)$.

While condition (5.5) involves the gradient $\|\nabla\phi(x)\|$ it ultimately reduces to the properties of the subspace. Essentially, it requires that the gradient is not too close to being orthogonal to the subspace induced by Q . Similar conditions and terminology have been used in [8, 11].

We also recall the following related lemma (recalling that $g_k = Q_k Q_k^T g_k$).

LEMMA 5.4 (Lemma 6.3 from [11]). *On iteration k , Q_k is κ_g -well aligned with $\nabla\phi(x_k)$ if and only if*

$$(5.6) \quad \|Q_k Q_k^T \nabla\phi(x_k)\|^2 \geq (1 - \kappa_g^2) \|\nabla\phi(x_k)\|^2.$$

Also, if $m(x_k + s)$ is κ_{ef}, κ_{eg} -fully linear model of $\phi(x_k + s)$ on $B_{Q_k}(x_k, \Delta_k)$ then

$$(5.7) \quad \|g_k - Q_k Q_k^T \nabla\phi(x_k)\| \leq \kappa_{eg} \Delta_k$$

We can now show the following lemma which is analogous to Lemma 6.4 from [11].

LEMMA 5.5 (sufficient condition for a successful step). *Under Assumptions 1.2 and 2.2, if Q_k is κ_g -well aligned with $\nabla\phi(x_k)$, $m_k(x_k + s)$ is a κ_{ef}, κ_{eg} -fully linear model of $\phi(x_k + s)$ on $B_{Q_k}(x_k, \Delta_k)$ and if*

$$(5.8) \quad \Delta_k \leq \sqrt{1 - \kappa_g^2} \tilde{C}_1 \|\nabla\phi(x_k)\|$$

where

$$\tilde{C}_1 = (\max \left\{ \eta_2, \kappa_{bhm}, \frac{2\kappa_{ef}}{(1 - \eta_1)\kappa_{fcd}} \right\} + \kappa_{eg})^{-1}$$

then $\rho_k \geq \eta_1$, $\|g_k\| \geq \eta_2 \Delta_k$, and $x_{k+1} = x_k + s_k$, i.e. the iteration k is successful.

Proof. Due to Lemma 5.4, specifically (5.7) by triangle inequality,

$$\|Q_k Q_k^T \nabla\phi(x_k)\| \leq \|g_k\| + \kappa_{eg} \Delta_k$$

and also due to Lemma 5.4

$$(5.9) \quad \Delta_k \leq \sqrt{1 - \kappa_g^2} \tilde{C}_1 \|\nabla\phi(x_k)\| \leq \tilde{C}_1 \|Q_k Q_k^T \nabla\phi(x_k)\|$$

By (5.9) we have

$$(\max \{ \kappa_{bhm}, \eta_2, \frac{2\kappa_{ef}}{(1 - \eta_1)\kappa_{fcd}} \} + \kappa_{eg}) \Delta_k \leq (\|g_k\| + \kappa_{eg} \Delta_k)$$

which implies

$$\max \{ \kappa_{bhm}, \eta_2 \} \Delta_k \leq \|g_k\|.$$

我们将像之前一样假设假设2.2成立。与算法1相比，算法3的两个关键修改是：在 ρ 的定义中增加额外的项 $2\epsilon_f$ ，以及对信赖域半径 Δ_{min} 施加的下界。这两项修改都是必要的，因为带噪零阶预言机使得 Δ_k 必须保持足够正，以确保完全线性模型。在算法1的确定性框架中，对于足够大的 ϵ ，可以确保 Δ_k 有这样的下界。但是当使用随机子空间时情况并非如此，因为即使 Δ_k 很小，也可能由于子空间选择不佳而发生不成功迭代。我们注意到，由于 ϵ_f 是零阶预言机中误差的上界，因此它的任何上界估计都可以在算法中使用。当然，不必要的过大值会对由此得到的最佳可达精度 ϵ 产生不利影响。类似地，我们将看到， ϵ_f 的选择决定了 Δ_{min} 的最佳选择。

我们现在引入一个定义，来自 [11]，给出一种度量，衡量由 Q 诱导的子空间与当前梯度的对齐程度。

定义5.3（良对齐子空间）。对于给定的 x ，由 Q 的列张成的子空间与 $\nabla\phi(x)$ 是 κ_g -良对齐的，如果

$$(5.5) \quad \|QQ^T\nabla\phi(x) - \nabla\phi(x)\| \leq \kappa_g\|\nabla\phi(x)\|$$

对于某个 $\kappa_g \in [0, 1)$ 。

虽然条件 (5.5) 涉及梯度 $\|\nabla\phi(x)\|$ ，但最终归结为子空间的性质。本质上，它要求梯度与由 Q 诱导的子空间不能过于接近正交。类似的条件和术语已在 [8, 11] 中使用。

我们同样回顾以下相关引理（回顾 $g_k = Q_k Q_k^T g_k$ ）。

引理5.4（引理6.3来自 [11]）。在第 k 次迭代， Q_k 与 $\nabla\phi(x_k)$ 是 κ_g -良好对齐的，当且仅当

$$(5.6) \quad \|Q_k Q_k^T \nabla\phi(x_k)\|^2 \geq (1 - \kappa_g^2) \|\nabla\phi(x_k)\|^2.$$

此外，如果 $m(x_k + s)$ 是 $\phi(x_k + s)$ 在 $B_{Q_k}(x_k, \Delta_k)$ 上的 κ_{ef}, κ_{eg} -完全线性模型，则

$$(5.7) \quad \|g_k - Q_k Q_k^T \nabla\phi(x_k)\| \leq \kappa_{eg} \Delta_k$$

我们现在可以展示以下引理，该引理与来自 [11] 的引理6.4 类似。

引理5.5（成功步的充分条件）。在假设 1.2 和 2.2 下，如果 Q_k 与 $\nabla\phi(x_k)$ 是 κ_g -良好对齐的， $m_k(x_k + s)$ 是 $\phi(x_k + s)$ 在 $B_{Q_k}(x_k, \Delta_k)$ 上的 κ_{ef}, κ_{eg} -完全线性模型，并且如果

$$(5.8) \quad \Delta_k \leq \sqrt{1 - \kappa_g^2} \tilde{C}_1 \|\nabla\phi(x_k)\|$$

其中

$$\tilde{C}_1 = (\max \left\{ \eta_2, \kappa_{bhm}, \frac{2\kappa_{ef}}{(1 - \eta_1)\kappa_{fcd}} \right\} + \kappa_{eg})^{-1}$$

则 $\rho_k \geq \eta_1$ 、 $\|g_k\| \geq \eta_2 \Delta_k$ 和 $x_{k+1} = x_k + s_k$ ，即迭代 k 成功。

证明。根据引理5.4，特别地，由三角不等式得到(5.7)，

$$\|Q_k Q_k^T \nabla\phi(x_k)\| \leq \|g_k\| + \kappa_{eg} \Delta_k$$

并且同样根据引理5.4

$$(5.9) \quad \Delta_k \leq \sqrt{1 - \kappa_g^2} \tilde{C}_1 \|\nabla\phi(x_k)\| \leq \tilde{C}_1 \|Q_k Q_k^T \nabla\phi(x_k)\|$$

由(5.9)可得

$$(\max \{ \kappa_{bhm}, \eta_2, \frac{2\kappa_{ef}}{(1 - \eta_1)\kappa_{fcd}} \} + \kappa_{eg}) \Delta_k \leq (\|g_k\| + \kappa_{eg} \Delta_k)$$

这意味着

$$\max \{ \kappa_{bhm}, \eta_2 \} \Delta_k \leq \|g_k\|.$$

This establishes that $\|g_k\| \geq \eta_2 \Delta_k$ and also $m_k(x_k) - m_k(x_k + s_k) \geq \kappa_{fcd} \|g_k\| \Delta_k / 2$ by Assumption 2.2. Then, using the fact that $|f(x) - \phi(x)| \leq \epsilon_f$ and the fully linear assumption on m_k in the subspace induced by Q_k and recalling that $s_k = Q_k v_k$ we have

$$\begin{aligned} \rho_k &= \frac{m_k(x_k) - m_k(x_k + s_k) + (f(x_k) - m_k(x_k)) - (f(x_k + s_k) - m_k(x_k + s_k)) + 2\epsilon_f}{m_k(x_k) - m(x_k + s_k)} \\ &\geq \frac{m_k(x_k) - m_k(x_k + s_k) + (\phi(x_k) - m_k(x_k)) - (\phi(x_k + s_k) - m_k(x_k + s_k))}{m_k(x_k) - m_k(x_k + s_k)} \\ &\geq 1 - \frac{\kappa_{ef} \Delta_k^2}{m_k(x_k) - m_k(x_k + s_k)} \geq 1 - \frac{\kappa_{ef} \Delta_k^2}{\kappa_{fcd} \|g_k\| \Delta_k / 2} \\ &\geq 1 - \frac{2\kappa_{ef} \Delta_k}{\kappa_{fcd} (\|Q_k Q_k^T \nabla \phi(x_k)\| - \kappa_{eg} \Delta_k)} \geq \eta_1, \end{aligned}$$

where the last step is true because $\|Q_k Q_k^T \nabla \phi(x_k)\| \geq (\frac{2\kappa_{ef}}{(1-\eta_1)\kappa_{fcd}} + \kappa_{eg}) \Delta_k$ follows from (5.9). $\square$

Now we show a lower bound on progress made in successful iterations similar to Lemma 4.3 of [7].

LEMMA 5.6 (Progress made in a successful iteration). *Under Assumptions 1.2 and 2.2, in Algorithm 3, if $\rho_k \geq \eta_1$ and $\|g_k\| \geq \eta_2 \Delta_k$, then*

$$\phi(x_k) - \phi(x_{k+1}) \geq C_2 \Delta_k^2 - 4\epsilon_f,$$

where $C_2 = \frac{1}{2} \eta_1 \eta_2 \kappa_{fcd} \min \left\{ \frac{\eta_2}{\kappa_{bhm}}, 1 \right\}$.

Proof. Since $\rho_k \geq \eta_1$, we have

$$\eta_1 \leq \frac{f(x_k) - f(x_k + s_k) + 2\epsilon_f}{m_k(x_k) - m_k(x_k + s_k)} \leq \frac{\phi(x_k) - \phi(x_k + s_k) + 4\epsilon_f}{m_k(x_k) - m_k(x_k + s_k)},$$

which we can rearrange as $\phi(x_k) - \phi(x_k + s_k) \geq \eta_1 (m_k(x_k) - m_k(x_k + s_k)) - 4\epsilon_f$. Since $\|g_k\| \geq \eta_2 \Delta_k$, then by (2.2), we have

$$\eta_1 (m_k(x_k) - m_k(x_k + s_k)) \geq \frac{\eta_1 \kappa_{fcd}}{2} \|g_k\| \min \left\{ \frac{\|g_k\|}{\kappa_{bhm}}, \Delta_k \right\} \geq C_2 \Delta_k^2. \quad \square$$

From these results one could bound the number of needed iterations if one had a guarantee of having well-aligned subspaces, however we wish to cover the random case where the subspace is well-aligned only with some probability. Accordingly, we study the random subspace case next.

5.1 Complexity analysis under random subspace selection. Observe that the matrix Q_k , and the trust region radius, Δ_k do not change on model improving iterations. We define Q_t to be the t th random matrix Q_k and similarly Δ_t and x_t are the trust region radius and iterate which correspond to Q_t . In other words, index t counts iterations that follow either a successful or an unsuccessful iteration. In what follows we will essentially derive a bound for $|\mathcal{S}_\epsilon| + |\mathcal{U}_\epsilon|$ by examining what happens to the objective function and trust region radius after these iterations (since we know that after model improving iterations neither the TR radius nor the function value change).

We now define some relevant stochastic processes:

$$\begin{aligned} I_t &= \mathbb{1}\{Q_t \text{ is } \kappa_g\text{-well aligned with } \nabla \phi(x_t)\}, \\ A_t &= \mathbb{1}\{Q_t \text{ leads to a successful iteration i.e., } \Delta_{t+1} = \gamma^{-1} \Delta_t\}, \\ B_t &= \mathbb{1}\{\Delta_t > \hat{C}_1 \|\nabla \phi(x_t)\|\}, \end{aligned}$$

where $\hat{C}_1 = \sqrt{1 - \kappa_g^2} \tilde{C}_1$ for $\tilde{C}_1$ as defined in Lemma 5.5.

We will say that matrix Q_t is "true" if $I_t = 1$. Let T_ϵ be the first t such that the iterate x_t produced by Algorithm 3 achieves $\|\nabla \phi(x_t)\| \leq \epsilon$. Let $\mathcal{F}_{t-1}$ denote the σ -algebra generated by the first t matrices, $\mathcal{F}_{t-1} = \sigma(Q_0, Q_1, \dots, Q_{t-1})$. We note that the random variables x_t and Δ_t are measurable with respect to $\mathcal{F}_{t-1}$. We define m_t , s_t , ρ_t to be the last model, proposed step, and ratio, respectively, which correspond to the matrix Q_t . The random variables m_t , s_t and ρ_t are measurable with respect to $\mathcal{F}_t$. The random variable $T_\epsilon = \min\{t : \|\nabla \phi(x_t)\| \leq \epsilon\}$ is a stopping time adapted to the filtration $\{\mathcal{F}_{t-1}\}$.

这证明了 $\|g_k\| \geq \eta_2 \Delta_k$ ，并且还证明了 $m_k(x_k) - m_k(x_k + s_k) \geq \kappa_{fcd} \|g_k\| \Delta_k / 2$ ，这由假设2.2可得。然后，利用 $|f(x) - \phi(x)| \leq \epsilon_f$ 这一事实，以及关于 m_k 在由 Q_k 诱导的子空间中的完全线性假设，并回想 $s_k = Q_k v_k$ ，我们有

$$\begin{aligned} \rho_k &= \frac{m_k(x_k) - m_k(x_k + s_k) + (f(x_k) - m_k(x_k)) - (f(x_k + s_k) - m_k(x_k + s_k)) + 2\epsilon_f}{m_k(x_k) - m(x_k + s_k)} \\ &\geq \frac{m_k(x_k) - m_k(x_k + s_k) + (\phi(x_k) - m_k(x_k)) - (\phi(x_k + s_k) - m_k(x_k + s_k))}{m_k(x_k) - m_k(x_k + s_k)} \\ &\geq 1 - \frac{\kappa_{ef} \Delta_k^2}{m_k(x_k) - m_k(x_k + s_k)} \geq 1 - \frac{\kappa_{ef} \Delta_k^2}{\kappa_{fcd} \|g_k\| \Delta_k / 2} \\ &\geq 1 - \frac{2\kappa_{ef} \Delta_k}{\kappa_{fcd} (\|Q_k Q_k^T \nabla \phi(x_k)\| - \kappa_{eg} \Delta_k)} \geq \eta_1, \end{aligned}$$

其中最后一步成立，因为 $\|Q_k Q_k^T \nabla \phi(x_k)\| \geq (\frac{2\kappa_{ef}}{(1-\eta_1)\kappa_{fcd}} + \kappa_{eg}) \Delta_k$ 由 (5.9) 推出。 $\square$

现在，我们给出成功迭代中取得的进展的一个下界，类似于 [7] 的引理 4.3。

引理5.6（成功迭代中取得的进展）。在假设 1.2 和 2.2 下，在算法 3 中，如果 $\rho_k \geq \eta_1$ 且 $\|g_k\| \geq \eta_2 \Delta_k$ ，则

$$\phi(x_k) - \phi(x_{k+1}) \geq C_2 \Delta_k^2 - 4\epsilon_f,$$

其中 $C_2 = \frac{1}{2} \eta_1 \eta_2 \kappa_{fcd} \min \left\{ \frac{\eta_2}{\kappa_{bhm}}, 1 \right\}$ 。

证明。由于 $\rho_k \geq \eta_1$ ，我们有

$$\eta_1 \leq \frac{f(x_k) - f(x_k + s_k) + 2\epsilon_f}{m_k(x_k) - m_k(x_k + s_k)} \leq \frac{\phi(x_k) - \phi(x_k + s_k) + 4\epsilon_f}{m_k(x_k) - m_k(x_k + s_k)},$$

我们可以将其重新排列为 $\phi(x_k) - \phi(x_k + s_k) \geq \eta_1 (m_k(x_k) - m_k(x_k + s_k)) - 4\epsilon_f$ 。由于 $\|g_k\| \geq \eta_2 \Delta_k$ ，于是根据 (2.2)，我们有

$$\eta_1 (m_k(x_k) - m_k(x_k + s_k)) \geq \frac{\eta_1 \kappa_{fcd}}{2} \|g_k\| \min \left\{ \frac{\|g_k\|}{\kappa_{bhm}}, \Delta_k \right\} \geq C_2 \Delta_k^2. \quad \square$$

根据这些结果，如果能够保证子空间良好对齐，人们就可以对所需迭代次数给出界；然而，我们希望覆盖子空间仅以一定概率良好对齐的随机情形。因此，我们接下来研究随机子空间情形。

5.1 随机子空间选择下的复杂度分析。 注意，矩阵 Q_k 和信赖域半径 Δ_k 在模型改进迭代中不会改变。我们定义 Q_t 为第 t 个随机矩阵 Q_k ，类似地， Δ_t 和 x_t 是对应于 Q_t 的信赖域半径和迭代点。换句话说，索引 t 计数的是成功迭代或不成功迭代之后的迭代。在下文中，我们将通过考察这些迭代之后目标函数和信赖域半径发生的变化，来本质上推导出 $|\mathcal{S}_\epsilon| + |\mathcal{U}_\epsilon|$ 的一个界（因为我们知道，在模型改进迭代之后，信赖域半径和函数值都不变）。

我们现在定义一些相关的随机过程：

$$\begin{aligned} I_t &= \mathbb{1}\{Q_t \text{ is } \kappa_g\text{-well aligned with } \nabla \phi(x_t)\}, \\ A_t &= \mathbb{1}\{Q_t \text{ leads to a successful iteration i.e., } \Delta_{t+1} = \gamma^{-1} \Delta_t\}, \\ B_t &= \mathbb{1}\{\Delta_t > \hat{C}_1 \|\nabla \phi(x_t)\|\}, \end{aligned}$$

其中， $\hat{C}_1 = \sqrt{1 - \kappa_g^2} \tilde{C}_1$ 对于 $\tilde{C}_1$ ，如引理 5.5 所定义。

若 $I_t = 1$ ，则称矩阵 Q_t 为“真”。令 T_ϵ 为第一个 t ，使得由算法 3 产生的迭代点 x_t 达到 $\|\nabla \phi(x_t)\| \leq \epsilon$ 。令 $\mathcal{F}_{t-1}$ 表示由前 t 个矩阵， $\mathcal{F}_{t-1} = \sigma(Q_0, Q_1, \dots, Q_{t-1})$ 生成的 σ -代数。我们注意到随机变量 x_t 和 Δ_t 关于 $\mathcal{F}_{t-1}$ 可测。我们定义 m_t 、 s_t 、 ρ_t 分别为最后一个模型、建议步和比率，它们对应于矩阵 Q_t 。随机变量 m_t 、 s_t 和 ρ_t 关于 $\mathcal{F}_t$ 可测。随机变量 $T_\epsilon = \min\{t : \|\nabla \phi(x_t)\| \leq \epsilon\}$ 是适应于滤子 $\{\mathcal{F}_{t-1}\}$ 的停时。

ASSUMPTION 5.7. *There exists a $\theta \in (\frac{1}{2}, 1]$ such that*

$$\mathbb{P}\{I_t = 1 | \mathcal{F}_{t-1}\} \geq \theta.$$

By Lemma 6.7 of [11], if we take Q_t such that the subspace it induces is uniformly distributed, we can take $\theta \geq \frac{243}{443} > 1/2$ and $\kappa_g = \sqrt{1 - \frac{q}{10n}}$.

Note that $\sigma(B_t) \subset \mathcal{F}_{t-1}$ and $\sigma(A_t) \subset \mathcal{F}_t$, that is the random variable B_t is fully determined by matrices $Q_0, \dots, Q_{t-1}$ produced by the algorithm, while A_t is fully determined by the matrices $Q_0, \dots, Q_t$. The stochastic process described here has essentially the same dynamics as the process analyzed in [10] and [11] enabling us to reuse the results. The only differences are the presence of lower bound $\Delta_{\min}$ and the possible increase of $\phi(x_k)$ on some successful iterations. The lower bound does not alter the main properties of the dynamics of Δ_k , since by fixing ϵ to be sufficiently large with respect to $\Delta_{\min}$ we ensure that $\Delta_{\min} < \hat{C}_1 \|\nabla \phi(x_t)\|$ for $t = 0, 1, \dots, T_\epsilon - 1$. Thus we retain the key property which follows from Lemma 5.5:

$$A_t \geq I_t(1 - B_t),$$

in other words, if matrix Q_t is true and the trust region radius is sufficiently small, then the iteration is successful.

The increase of objective function on certain iterations is due to the relaxed definition of ρ_k and the error in the zeroth order oracle. Such situations have been previously analyzed for line (step) search in [5, 18] and trust-region method in [7]. The analysis here is simpler but the key idea is that the increase is bounded by $4\epsilon_f$ and occurs on iterations whose number is not too large compared to the number of iterations where function decreases. By ensuring that the decrease is sufficiently large to compensate for the increase, the results are derived. Below we present the analysis.

To bound the total number of successful and unsuccessful iterations we first bound the number of matrices that lead to successful iterations with large Δ . For that let $\bar{B}_t = \mathbb{1}\{\Delta_t \geq \gamma \hat{C}_1 \epsilon\}$ (note that $B_t = 1 \Rightarrow \bar{B}_t = 1$). Then from the dynamics of Algorithm 3 we have the bound similar to [10] (also used in [11]).

LEMMA 5.8. *Suppose $\epsilon > \sqrt{\frac{8\epsilon_f}{C_2\gamma^2\hat{C}_1^2}}$ For any $l \in \{0, \dots, T_\epsilon - 1\}$ and for all realizations of Algorithm 3, we have*

$$\sum_{t=0}^l \bar{B}_t I_t A_t \leq \sum_{t=0}^l \bar{B}_t A_t \leq \frac{\phi(x_0) - \phi^* + 4\epsilon_f(\sum_{t=0}^l (1 - \bar{B}_t)A_t)}{\frac{1}{2}C_2(\gamma(\hat{C}_1\epsilon))^2},$$

Proof. Since $\epsilon > \sqrt{\frac{8\epsilon_f}{C_2\gamma^2\hat{C}_1^2}}$, $\Delta_t \geq \gamma \hat{C}_1 \epsilon$ implies that $\Delta_t \geq \sqrt{\frac{8\epsilon_f}{C_2}}$ which in turn implies that for large successful iterations (corresponding to $\bar{B}_t A_t$), $\phi(x_t) - \phi(x_{t+1}) \geq \frac{1}{2}C_2\Delta_t^2$ by Lemma 5.6. For small successful iterations (corresponding to $(1 - \bar{B}_t)A_t$), by the same lemma, we have $\phi(x_t) - \phi(x_{t+1}) \geq -4\epsilon_f$. The result follows. $\square$

A useful lemma that easily follows from the dynamics is as follows.

LEMMA 5.9. *Suppose $\Delta_0 \geq \hat{C}_1 \epsilon$ and $\Delta_{\min} \leq \gamma \hat{C}_1 \epsilon$. For any $l \in \{0, \dots, T_\epsilon - 1\}$ and for all realizations of Algorithm 3, we have*

$$\sum_{t=0}^l B_t(1 - A_t) \leq \sum_{t=0}^l \bar{B}_t A_t + \log_\gamma \left(\frac{\hat{C}_1 \epsilon}{\Delta_0} \right).$$

The following result is shown in [10] under Assumption 5.7,

$$\mathbb{E} \left(\sum_{t=0}^{T_\epsilon-1} \bar{B}_t(1 - I_t) \right) \leq \frac{1 - \theta}{\theta} \mathbb{E} \left(\sum_{t=0}^{T_\epsilon-1} \bar{B}_t I_t \right),$$

from which the following lemma is derived.

LEMMA 5.10. *Let Assumption 5.7 hold. Under the condition that $\theta > 1/2$, $\Delta_0 \geq \hat{C}_1 \epsilon$, and $\Delta_{\min} \leq \gamma \hat{C}_1 \epsilon$, we have*

$$\mathbb{E} \left(\sum_{t=0}^{T_\epsilon-1} B_t \right) \leq \frac{1}{2\theta - 1} \left(\sum_{t=0}^{T_\epsilon-1} \bar{B}_t A_t + \log_\gamma \left(\frac{\hat{C}_1 \epsilon}{\Delta_0} \right) \right).$$

假设 5.7. 存在一个 $\theta \in (\frac{1}{2}, 1]$ 使得

$$\mathbb{P}\{I_t = 1 | \mathcal{F}_{t-1}\} \geq \theta.$$

根据 [11], 的引理 6.7, 如果我们取 Q_t 使得其诱导的子空间是均匀分布的, 则我们可以取 $\theta \geq \frac{243}{443} > 1/2$ 和

$$\kappa_g = \sqrt{1 - \frac{q}{10n}}$$

注意, $\sigma(B_t) \subset \mathcal{F}_{t-1}$ 且 $\sigma(A_t) \subset \mathcal{F}_t$, 即随机变量 B_t 完全由算法生成的矩阵 $Q_0, \dots, Q_{t-1}$ 确定, 而 A_t 完全由矩阵 $Q_0, \dots, Q_t$ 确定. 这里描述的随机过程与在 [10] 和 [11] 中分析的过程具有基本相同的动力学, 这使我们能够复用其结果. 唯一的区别在于下界 $\Delta_{\min}$ 的存在, 以及在若干成功迭代中 $\phi(x_k)$ 可能增加. 下界不会改变 Δ_k 的动力学的主要性质, 因为通过将 ϵ 固定为相对于 $\Delta_{\min}$ 足够大, 我们确保 $\Delta_{\min} < C_1 \|\nabla \phi(x_t)\|$ 对于 $t = 0, 1, \dots, T_\epsilon - 1$ 成立. 因此, 我们保留由引理 5.5 得出的关键性质:

$$A_t \geq I_t(1 - B_t),$$

换句话说, 如果矩阵 Q_t 为真且信赖域半径足够小, 则该次迭代为成功迭代.

目标函数在某些迭代上的增加是由于对 ρ_k 的放宽定义以及零阶预言机中的误差. 此类情形此前已在 [5, 18] 的线 (步长) 搜索和 [7] 的信赖域方法中分析过. 这里的分析更简单, 但关键思想是: 增加量由 $4\epsilon_f$ 界定, 并且其发生的迭代次数与目标函数下降的迭代次数相比不算太多. 通过确保下降量足够大以补偿增加量, 这些结果得以导出. 下面我们给出分析.

为了界定成功迭代和不成功迭代的总数, 我们首先界定导致具有较大 Δ 的成功迭代的矩阵数量. 为此, 令 $\bar{B}_t = \mathbb{1}\{\Delta_t \geq \gamma \hat{C}_1 \epsilon\}$ (注意 $B_t = 1 \Rightarrow \bar{B}_t = 1$). 然后, 由算法 3 的动力学, 我们得到与 [10] 类似的界 (在 [11] 中也有使用)。

引理 5.8. 假设 $\epsilon > \sqrt{\frac{8\epsilon_f}{C_2\gamma^2\hat{C}_1^2}}$, 对于任意 $l \in \{0, \dots, T_\epsilon - 1\}$, 并且对于算法 3 的所有实现, 我们有

$$\sum_{t=0}^l \bar{B}_t I_t A_t \leq \sum_{t=0}^l \bar{B}_t A_t \leq \frac{\phi(x_0) - \phi^* + 4\epsilon_f(\sum_{t=0}^l (1 - \bar{B}_t)A_t)}{\frac{1}{2}C_2(\gamma(\hat{C}_1\epsilon))^2},$$

证明. 由于 $\epsilon > \sqrt{\frac{8\epsilon_f}{C_2\gamma^2\hat{C}_1^2}}$, $\Delta_t \geq \gamma \hat{C}_1 \epsilon$ 蕴含 $\Delta_t \geq \sqrt{\frac{8\epsilon_f}{C_2}}$, 这又意味着对于大成功迭代 (对应于 $\bar{B}_t A_t$), 根据引理 5.6, $\phi(x_t) - \phi(x_{t+1}) \geq \frac{1}{2}C_2\Delta_t^2$. 对于小成功迭代 (对应于 $(1 - \bar{B}_t)A_t$), 根据同一引理, 我们有 $\phi(x_t) - \phi(x_{t+1}) \geq -4\epsilon_f$. 结论得证. $\square$

一个容易由动力学推出的有用引理如下.

引理 5.9. 假设 $\Delta_0 \geq \hat{C}_1 \epsilon$ 且 $\Delta_{\min} \leq \gamma \hat{C}_1 \epsilon$. 对于任意 $l \in \{0, \dots, T_\epsilon - 1\}$ 以及对算法 3 的所有实现, 我们有

$$\sum_{t=0}^l B_t(1 - A_t) \leq \sum_{t=0}^l \bar{B}_t A_t + \log_\gamma \left(\frac{\hat{C}_1 \epsilon}{\Delta_0} \right).$$

在假设 5.7 下, 以下结果在 [10] 中给出,

$$\mathbb{E} \left(\sum_{t=0}^{T_\epsilon-1} \bar{B}_t(1 - I_t) \right) \leq \frac{1 - \theta}{\theta} \mathbb{E} \left(\sum_{t=0}^{T_\epsilon-1} \bar{B}_t I_t \right),$$

由此推导出以下引理.

引理 5.10. 设假设 5.7 成立. 在 $\theta > 1/2$, $\Delta_0 \geq \hat{C}_1 \epsilon$, 以及 $\Delta_{\min} \leq \gamma \hat{C}_1 \epsilon$ 的条件下, 我们有

$$\mathbb{E} \left(\sum_{t=0}^{T_\epsilon-1} B_t \right) \leq \frac{1}{2\theta - 1} \left(\sum_{t=0}^{T_\epsilon-1} \bar{B}_t A_t + \log_\gamma \left(\frac{\hat{C}_1 \epsilon}{\Delta_0} \right) \right).$$

Finally the following lemma is shown in [10] for the stochastic processes I_t, A_t and B_t since $A_t \geq I_t(1 - B_t)$ and by the dynamics of Δ_t .

LEMMA 5.11. *Let Assumption 5.7 hold.*

$$\mathbb{E} \left(\sum_{t=0}^{T_\epsilon-1} (1 - B_t) \right) \leq \frac{1}{2\theta} \mathbb{E} [T_\epsilon].$$

Putting these lemmas together we obtain the final expected complexity result.

THEOREM 5.12. *Let Assumption 1.2, Assumption 2.2 and Assumption 5.7 hold. Then for any $\epsilon > \sqrt{\frac{16\epsilon_f}{(2\theta-1)^2 C_2 \hat{C}_1^2 \gamma^2}}$, assuming an initial trust-region radius $\Delta_0 \geq \hat{C}_1 \epsilon$, and $\Delta_{\min} \leq \gamma \hat{C}_1 \epsilon$, let T_ϵ be the random stopping time for the event $\{\|\nabla \phi(x_t)\| \leq \epsilon\}$. We have the bound*

$$\mathbb{E} [T_\epsilon] \leq \frac{4\theta}{(2\theta-1)^2} \left(\frac{\phi(x_0) - \phi^\star}{\frac{1}{2} C_2 (\gamma \hat{C}_1 \epsilon)^2} + \log_\gamma \left(\frac{\hat{C}_1 \epsilon}{\Delta_0} \right) \right)$$

where $\hat{C}_1 = \sqrt{1 - \kappa_g^2} \tilde{C}_1$ for $\tilde{C}_1$ as in (5.8) and C_2 as in Lemma 5.6.

Proof. Observe that $\sum_{t=0}^{T_\epsilon-1} (1 - \bar{B}_t) A_t \leq \sum_{t=0}^{T_\epsilon-1} (1 - B_t)$. Thus, from Lemmas 5.8 and 5.10, we have

$$\mathbb{E} \left(\sum_{t=0}^{T_\epsilon-1} B_t \right) \leq \frac{1}{2\theta-1} \left(\frac{\phi(x_0) - \phi^\star}{\frac{1}{2} C_2 (\gamma \hat{C}_1 \epsilon)^2} + \log_\gamma \left(\frac{\hat{C}_1 \epsilon}{\Delta_0} \right) \right) + C_3 \left(\sum_{t=0}^{T_\epsilon-1} (1 - B_t) \right),$$

where $C_3 = \frac{4\epsilon_f}{(2\theta-1)^{\frac{1}{2}} C_2 (\gamma \hat{C}_1 \epsilon)^2}$. It follows that

$$\mathbb{E} [T_\epsilon] \leq \frac{1}{2\theta-1} \left(\frac{\phi(x_0) - \phi^\star}{\frac{1}{2} C_2 (\gamma \hat{C}_1 \epsilon)^2} + \log_\gamma \left(\frac{\hat{C}_1 \epsilon}{\Delta_0} \right) \right) + (1 + C_3) \mathbb{E} \left(\sum_{t=0}^{T_\epsilon-1} (1 - B_t) \right).$$

Combining with Lemma 5.11, we obtain

$$\mathbb{E} [T_\epsilon] \leq \frac{1}{2\theta-1} \left(\frac{\phi(x_0) - \phi^\star}{\frac{1}{2} C_2 (\gamma \hat{C}_1 \epsilon)^2} + \log_\gamma \left(\frac{\hat{C}_1 \epsilon}{\Delta_0} \right) \right) + (1 + C_3) \frac{1}{2\theta} \mathbb{E} [T_\epsilon].$$

The lower bound on ϵ implies that $C_3 \leq \theta - \frac{1}{2}$ and so $\frac{1+C_3}{2\theta} \leq \frac{2\theta+1}{4\theta}$. Thus we have

$$\frac{2\theta-1}{4\theta} \mathbb{E} [T_\epsilon] \leq \frac{1}{2\theta-1} \left(\frac{\phi(x_0) - \phi^\star}{\frac{1}{2} C_2 (\gamma \hat{C}_1 \epsilon)^2} + \log_\gamma \left(\frac{\hat{C}_1 \epsilon}{\Delta_0} \right) \right). \quad \square$$

In order to use this theorem for effective complexity bounds, we must specify how to form models in a subspace. In the following subsections we discuss the two different approaches we used in the full space case - finite differences and interpolation based Λ -poised sets. The key difference now is in the lower bound on Δ_k imposed by $\Delta_{\min}$ rather than occurring automatically.

5.2 Building models in a subspace One can form a gradient estimate via a subspace version of (2.9). This can take the following form given in [11]:

$$(5.10) \quad \hat{g}(0) = \sum_{i=1}^q \frac{f(x + \delta Q u_i) - f(x)}{\delta} u_i$$

where u_i is the i th column of an orthogonal $q \times q$ matrix. Let us define $g(x) = Q \hat{g}(0)$.

By similar analysis as in [4], we can derive the bound

$$\left\| \nabla \hat{m}(0) - \nabla \hat{\phi}(0) \right\| \leq \frac{\sqrt{q} L \delta}{2} + \frac{2\sqrt{q} \epsilon_f}{\delta}.$$

最后, 下述引理在 [10] 中针对随机过程 I_t, A_t 和 B_t 得到证明, 因为 $A_t \geq I_t(1 - B_t)$, 并且根据 Δ_t 的动力学。

引理 5.11. 令假设5.7成立。

$$\mathbb{E} \left(\sum_{t=0}^{T_\epsilon-1} (1 - B_t) \right) \leq \frac{1}{2\theta} \mathbb{E} [T_\epsilon].$$

综合这些引理, 我们得到最终的期望复杂度结果。

定理5.12. 令假设1.2、假设2.2和假设5.7成立。那么, 对于任意 $\epsilon > \sqrt{\frac{16\epsilon_f}{(2\theta-1)^2 C_2 (\gamma \hat{C}_1 \epsilon)^2}}$, 假设初始信赖域半径为 $\Delta_0 \geq \gamma \hat{C}_1 \epsilon$, 且 $\Delta_{\min} \leq \gamma \hat{C}_1 \epsilon$, 令 T_ϵ 为事件 $\{\|\nabla \phi(x_t)\| \leq \epsilon\}$ 的随机停时。我们有如下界

$$\mathbb{E} [T_\epsilon] \leq \frac{4\theta}{(2\theta-1)^2} \left(\frac{\phi(x_0) - \phi^\star}{\frac{1}{2} C_2 (\gamma \hat{C}_1 \epsilon)^2} + \log_\gamma \left(\frac{\hat{C}_1 \epsilon}{\Delta_0} \right) \right)$$

其中 $\hat{C}_1 = \sqrt{1 - \kappa_g^2} \tilde{C}_1$ 对于 $\tilde{C}_1$, 如 (5.8) 中那样, 且 C_2 如引理

5.6 中那样。

证明。观察到 $\sum_{t=0}^{T_\epsilon-1} (1 - \bar{B}_t) A_t \leq \sum_{t=0}^{T_\epsilon-1} (1 - B_t)$ 。因此, 由引理 5.8 和 5.10, 我们有

$$\mathbb{E} \left(\sum_{t=0}^{T_\epsilon-1} B_t \right) \leq \frac{1}{2\theta-1} \left(\frac{\phi(x_0) - \phi^\star}{\frac{1}{2} C_2 (\gamma \hat{C}_1 \epsilon)^2} + \log_\gamma \left(\frac{\hat{C}_1 \epsilon}{\Delta_0} \right) \right) + C_3 \left(\sum_{t=0}^{T_\epsilon-1} (1 - B_t) \right),$$

其中 $C_3 = \frac{4\epsilon_f}{(2\theta-1)^{\frac{1}{2}} C_2 (\gamma \hat{C}_1 \epsilon)^2}$ 。由此可得

$$\mathbb{E} [T_\epsilon] \leq \frac{1}{2\theta-1} \left(\frac{\phi(x_0) - \phi^\star}{\frac{1}{2} C_2 (\gamma \hat{C}_1 \epsilon)^2} + \log_\gamma \left(\frac{\hat{C}_1 \epsilon}{\Delta_0} \right) \right) + (1 + C_3) \mathbb{E} \left(\sum_{t=0}^{T_\epsilon-1} (1 - B_t) \right).$$

结合引理5.11, 可得

$$\mathbb{E} [T_\epsilon] \leq \frac{1}{2\theta-1} \left(\frac{\phi(x_0) - \phi^\star}{\frac{1}{2} C_2 (\gamma \hat{C}_1 \epsilon)^2} + \log_\gamma \left(\frac{\hat{C}_1 \epsilon}{\Delta_0} \right) \right) + (1 + C_3) \frac{1}{2\theta} \mathbb{E} [T_\epsilon].$$

ϵ 的下界意味着 $C_3 \leq \theta - \frac{1}{2}$, 从而 $\frac{1+C_3}{2\theta} \leq \frac{2\theta+1}{4\theta}$ 。因此我们有

$$\frac{2\theta-1}{4\theta} \mathbb{E} [T_\epsilon] \leq \frac{1}{2\theta-1} \left(\frac{\phi(x_0) - \phi^\star}{\frac{1}{2} C_2 (\gamma \hat{C}_1 \epsilon)^2} + \log_\gamma \left(\frac{\hat{C}_1 \epsilon}{\Delta_0} \right) \right). \quad \square$$

为了利用该定理得到有效复杂度界, 我们必须说明如何构造子空间模型。在以下小节中, 我们讨论在全空间情形中使用的两种不同方法——有限差分 and 基于插值的 Λ -适定集。现在的关键区别在于, Δ_k 的下界是由 $\Delta_{\min}$ 施加的, 而不是自动出现的。

5.2 构建子空间模型 可以通过 (2.9) 的子空间版本构造梯度估计。这可以采取 [11] 中给出的以下形式:

$$(5.10) \quad \hat{g}(0) = \sum_{i=1}^q \frac{f(x + \delta Q u_i) - f(x)}{\delta} u_i$$

其中 u_i 是正交 $q \times q$ 矩阵的第 i 列。我们定义 $g(x) = Q \hat{g}(0)$ 。

通过与 [4], 中类似的分析, 我们可以推导出该界

$$\left\| \nabla \hat{m}(0) - \nabla \hat{\phi}(0) \right\| \leq \frac{\sqrt{q} L \delta}{2} + \frac{2\sqrt{q} \epsilon_f}{\delta}.$$

Choosing $\delta = \Delta_k$, we have that m_k is κ_{ef}, κ_{eg} -fully linear model for

$$\kappa_{eg} = \frac{\sqrt{q}L}{2} + \frac{2\sqrt{q}\epsilon_f}{\Delta_{\min}^2}, \quad \kappa_{ef} = \kappa_{eg} + \frac{LQ + \kappa_{bhm}}{2}.$$

In this case there are no model improving iterations. Thus all iterations are either successful or unsuccessful and each iteration requires either q or $q + 1$ function evaluations. With these specifics we can give a final complexity bound for Algorithm 3. For simplicity of the presentation we will give the final bounds in terms of the key components, such as $n, \epsilon, \epsilon_f, L$ and $\Delta_{\min}$.

THEOREM 5.13. *Let Assumption 1.2 and 2.2 hold. When randomizing, take Q_t such that the subspace it induces is uniformly distributed with $q \geq 3$. For all $k = 0, 1, \dots, K_\epsilon - 1$, define $m_k(x_k + s)$ with g_k as in (5.10). Let the parameters $\eta_1, \eta_2, \kappa_{bhm}, \gamma$ be constants and assume $L \geq 1$. Then for $\epsilon > \Omega(\sqrt{n}(L + \frac{\epsilon_f}{\Delta_{\min}^2})(\sqrt{\epsilon_f} + \Delta_{\min}))$, assuming an initial trust-region radius $\Delta_0 \geq \Omega(\sqrt{\epsilon_f} + \Delta_{\min})$, let K_ϵ be the random stopping time for the event $\{\|\nabla\phi(x_k)\| \leq \epsilon\}$. We have the bound*

$$\mathbb{E}[K_\epsilon] \leq \mathcal{O}\left(\left(\frac{n}{\epsilon^2}\right)\left(L + \frac{\epsilon_f}{\Delta_{\min}^2}\right)^2\right)$$

where the “big- $\mathcal{O}$ ” notation suppresses constant factors and an additive logarithmic term.

Proof. Since we use (5.10) in every iteration, we have that k and t are equivalent. From the definition of $\tilde{C}_1$ and the bound on κ_{ef} , we have $\tilde{C}_1^{-1} = \Theta(\kappa_{eg}) = \Theta(\sqrt{q}(L + \frac{\epsilon_f}{\Delta_{\min}^2}))$. By Lemma 6.7 of [11], we have $\sqrt{1 - \kappa_g^2} = \Theta(\sqrt{\frac{q}{n}})$. Thus we have $\hat{C}_1^{-1} = \Theta(\sqrt{n}(L + \frac{\epsilon_f}{\Delta_{\min}^2}))$. Then the condition that $\epsilon > \sqrt{\frac{16\epsilon_f}{(2\theta-1)^2 C_2^2 \hat{C}_1^2 \gamma^2}}$ from Theorem 5.12 becomes that $\epsilon > \Omega(\sqrt{n}(L + \frac{\epsilon_f}{\Delta_{\min}^2})\sqrt{\epsilon_f})$. The condition that $\Delta_{\min} \leq \gamma \hat{C}_1 \epsilon$, becomes $\epsilon > \Omega(\sqrt{n}(L + \frac{\epsilon_f}{\Delta_{\min}^2})\Delta_{\min})$. Combining these two bounds results in the condition $\epsilon > \Omega(\sqrt{n}(L + \frac{\epsilon_f}{\Delta_{\min}^2})(\sqrt{\epsilon_f} + \Delta_{\min}))$. Finally the condition that $\Delta_0 \geq \hat{C}_1 \epsilon$ becomes $\Delta_0 \geq \mathcal{O}(\sqrt{\epsilon_f} + \Delta_{\min})$. The expected iteration bound then follows directly from Theorem 5.12. $\square$

Here we note that the lower bound on $\epsilon, \Omega(\sqrt{n}(L + \frac{\epsilon_f}{\Delta_{\min}^2})(\sqrt{\epsilon_f} + \Delta_{\min}))$, can be approximately optimized by taking $\Delta_{\min} = \Theta(\sqrt{\epsilon_f})$. The lower bound then becomes $\epsilon \geq \Omega(\sqrt{n\epsilon_f})$ with a rate of

$$\mathbb{E}[K_\epsilon] \leq \mathcal{O}\left(\frac{n}{\epsilon^2}\right).$$

We also note that each iteration requires only $\mathcal{O}(q)$ function evaluations. Thus the total expected complexity rate is

$$\mathbb{E}[\mathcal{C}_\epsilon] \leq \mathcal{O}\left(\frac{nq}{\epsilon^2}\right).$$

5.3 Geometry-correcting algorithm in subspaces. We now describe a geometry-correcting version of Algorithm 3. This algorithm performs model improving steps of Algorithm 2 until either successful step is achieved or a fully linear model in the subspace is formed. At that point it terminates the work in that subspace and regenerates a new subspace as well as restarts the models using the initial sample sets $\mathcal{Y}_0, \mathcal{Z}_0$. This initialization choice is somewhat arbitrary and can be replaced by different initial sets. Each time, however, this requires computation of new function values for all points in the “initial” sample set. Our computational results show that this is quite expensive, if we use $\mathcal{Y}_0$ and $\mathcal{Z}_0$ that contain q points in each. We can delay resampling the random subspace until several successful or unsuccessful steps have been encountered and extend the theory to such strategies. However our computational results so far do not support an advantage of this approach. We can also choose $\mathcal{Y}_0$ to contain only 1 point and $\mathcal{Z}_0$ to be empty by modifying model improvement step and Lagrange polynomial computation to allow for incomplete sets. This modification is simple from the theory point of view but whether it can be practically competitive is yet unclear. Thus we retain the simplest approach for our analysis.

For this algorithm we can show the following complexity rate.

COROLLARY 5.14. *Under the same assumptions as Theorem 5.13, letting $\Lambda = 1 + \frac{1}{q}$ and $\Delta_{\min} = \sqrt{\epsilon_f}$, and $|\mathcal{Z}_0| = q$, for any $\epsilon > \Omega(\sqrt{n\epsilon_f})$, the expected total oracle complexity of Algorithm 4 is bounded as*

$$\mathcal{O}\left(\frac{nq \log q}{\epsilon^2}\right).$$

选取 $\delta = \Delta_k$ 后, 我们有 m_k 是 κ_{ef}, κ_{eg} -完全线性模型, 对于

$$\kappa_{eg} = \frac{\sqrt{q}L}{2} + \frac{2\sqrt{q}\epsilon_f}{\Delta_{\min}^2}, \quad \kappa_{ef} = \kappa_{eg} + \frac{LQ + \kappa_{bhm}}{2}.$$

在这种情况下, 不存在模型改进迭代。因此, 所有迭代要么成功, 要么不成功, 且每次迭代需要 q 或 $q + 1$ 次函数求值。基于这些具体情况, 我们可以给出算法 3 的最终复杂度界。为表述简便, 我们将以关键组成部分来表示最终界, 例如 $n, \epsilon, \epsilon_f, L$ 和 $\Delta_{\min}$ 。

定理5.13. 设假设1.2和假设2.2成立。在随机化时, 取 Q_t , 使得其诱导的子空间与 $q \geq 3$ 均匀分布。对于所有 $k = 0, 1, \dots, K_\epsilon - 1$, 定义 $m_k(x_k + s)$, 其中 g_k 如 (5.10) 中那样。设参数 $\eta_1, \eta_2, \kappa_{bhm}, \gamma$ 为常数, 并假设 $L \geq 1$ 。则对于 $\epsilon > \Omega(\sqrt{n}(L + \frac{\epsilon_f}{\Delta_{\min}^2})(\sqrt{\epsilon_f} + \Delta_{\min}))$, 假设初始信赖域半径为 $\Delta_0 \geq \Omega(\sqrt{\epsilon_f} + \Delta_{\min})$, 令 K_ϵ 为事件 $\{\|\nabla\phi(x_k)\| \leq \epsilon\}$ 的随机停时。我们有如下界

$$\mathbb{E}[K_\epsilon] \leq \mathcal{O}\left(\left(\frac{n}{\epsilon^2}\right)\left(L + \frac{\epsilon_f}{\Delta_{\min}^2}\right)^2\right)$$

其中“大 $\mathcal{O}$ ”记号省略了常数因子和一个加性对数项。

证明。 由于我们在每次迭代中都使用 (5.10), 因此, k 和 t 等价。由 C_1 的定义和 κ_{ef} 的界, 可得 $\tilde{C}_1^{-1} = \Theta(\kappa_{eg}) = \Theta(\sqrt{q}(L + \frac{\epsilon_f}{\Delta_{\min}^2}))$ 。由 [11], 中的引理 6.7, 我们有 $\sqrt{1 - \kappa_g^2} = \Theta(\sqrt{\frac{q}{n}})$ 。因此, 我们有 $\hat{C}_1^{-1} = \Theta(\sqrt{n}(L + \frac{\epsilon_f}{\Delta_{\min}^2}))$ 。于是, 由定理5.12给出的条件 $\epsilon > \sqrt{\frac{16\epsilon_f}{(2\theta-1)^2 C_2^2 \hat{C}_1^2 \gamma^2}}$ 变为 $\epsilon > \Omega(\sqrt{n}(L + \frac{\epsilon_f}{\Delta_{\min}^2})\sqrt{\epsilon_f})$ 。条件 $\Delta_{\min} \leq \gamma \hat{C}_1 \epsilon$ 变为 $\epsilon > \Omega(\sqrt{n}(L + \frac{\epsilon_f}{\Delta_{\min}^2})\Delta_{\min})$ 。结合这两个界得到条件 $\epsilon > \Omega(\sqrt{n}(L + \frac{\epsilon_f}{\Delta_{\min}^2})(\sqrt{\epsilon_f} + \Delta_{\min}))$ 。最后, 条件 $\Delta_0 \geq \hat{C}_1 \epsilon$ 变为 $\Delta_0 \geq \mathcal{O}(\sqrt{\epsilon_f} + \Delta_{\min})$ 。于是, 期望迭代次数界直接由定理5.12得出。 $\square$

这里我们注意到, ϵ 的下界 $\Omega(\sqrt{n}(L + \frac{\epsilon_f}{\Delta_{\min}^2})(\sqrt{\epsilon_f} + \Delta_{\min}))$ 可以通过取 $\Delta_{\min} = \Theta(\sqrt{\epsilon_f})$ 来近似优化。此时下界变为 $\epsilon \geq \Omega(\sqrt{n\epsilon_f})$, 其速率为

$$\mathbb{E}[K_\epsilon] \leq \mathcal{O}\left(\frac{n}{\epsilon^2}\right).$$

我们还注意到, 每次迭代只需要 $\mathcal{O}(q)$ 次函数求值。因此, 总期望复杂度速率是

$$\mathbb{E}[\mathcal{C}_\epsilon] \leq \mathcal{O}\left(\frac{nq}{\epsilon^2}\right).$$

5.3 子空间中的几何校正算法。 我们现在描述算法 3 的几何校正版本。该算法执行算法2的模型改进步骤, 直到实现成功步, 或者在子空间中形成完全线性模型。此时, 它终止该子空间中的工作, 重新生成一个新的子空间, 并使用初始样本集 $\mathcal{Y}_0, \mathcal{Z}_0$ 重新启动模型。这种初始化选择有些随意, 可以替换为不同的初始集合。然而, 每次这样做都需要为“初始”样本集中的所有点计算新的函数值。我们的计算结果表明, 如果我们使用 $\mathcal{Y}_0$ 和 $\mathcal{Z}_0$, 且它们各自包含 q 个点, 那么这会相当昂贵。我们可以推迟对随机子空间的重新采样, 直到遇到若干成功步或不成功步, 并将理论扩展到此类策略。然而, 我们迄今为止的计算结果并不支持这种方法具有优势。我们还可以通过修改模型改进步骤和拉格朗日多项式计算以允许不完全集, 从而选择 $\mathcal{Y}_0$ 仅包含 1 个点, 并选择 $\mathcal{Z}_0$ 为空。从理论角度看, 这种修改很简单, 但它是否能在实践中具有竞争力尚不清楚。因此, 我们在分析中保留最简单的做法。

对于该算法, 我们可以展示以下复杂度速率。

推论 5.14. 在与定理 5.13 相同的假设下, 令 $\Lambda = 1 + \frac{1}{q}$ 和 $\Delta_{\min} = \sqrt{\epsilon_f}$, 且 $|\mathcal{Z}_0| = q$, 对于任意 $\epsilon > \Omega(\sqrt{n\epsilon_f})$, 算法 4 的期望总预言机复杂度的界为

$$\mathcal{O}\left(\frac{nq \log q}{\epsilon^2}\right).$$

Algorithm 4: Geometry-correcting algorithm in subspace

Inputs: Inexact zeroth order oracle $|f(x) - \phi(x)| \leq \epsilon_f$, minimum radius $\Delta_{\min}$, initial x_0 , $\Delta_0 \geq \Delta_{\min}$, initial matrix $Q_0 \in \mathbb{R}^{n \times q}$ with orthonormal columns and $\eta_1 \in (0, 1)$, $\eta_2 > 0, \gamma \in (0, 1)$, $\Lambda > 1$, $\Lambda_{sc} \geq 1$.
Initialization Initial sets $\mathcal{Y}_0, \mathcal{Z}_0 \subset \mathbb{R}^q$ and the function values $f(x_0), f(x_0 + Q_0 y_i), y_i \in \mathcal{Y}_0, f(x_0 + Q_0 z_i), z_i \in \mathcal{Z}_0$. A set of Lagrange Polynomials $\{\ell_i(v), i = 1, \dots, q\}$ in $\mathcal{P}$ for the set $\mathcal{Y}_0$.
for $k = 0, 1, 2, \dots$ **do**
1 For the current Q_k , build a quadratic model $\hat{m}_k(v) = m_k(x_k + Q_k v)$ as in (3.2) using $f(x_k)$ and $f(x_k + Q_k y_i), y_i \in \mathcal{Y}_k, f(x_k + Q_k z_i), z_i \in \mathcal{Z}_k$.
2 Compute a trial step $x_k + s_k$ where $s_k = Q_k v_k$ with $v_k \approx \arg \min_v \{m_k(x_k + Q_k v) : \|v\| \leq \Delta_k\}$.
3 Compute the ratio ρ_k as
$$\rho_k = \frac{f(x_k) - f(x_k + s_k) + 2\epsilon_f}{m_k(x_k) - m_k(x_k + s_k)}.$$

4 Update the iterate and the TR radius as
$$(x_{k+1}, \Delta_{k+1}) \leftarrow \begin{cases} (x_k + s_k, \gamma^{-1} \Delta_k) & \text{if } \rho_k \geq \eta_1 \text{ and } \|g_k\| \geq \eta_2 \Delta_k, \\ (x_k, \Delta_k) & \text{else, if } \mathcal{Y}_k \text{ is not } \Lambda\text{-poised,} \\ (x_k, \max\{\gamma \Delta_k, \Delta_{\min}\}) & \text{otherwise.} \end{cases}$$

5 Update the subspace
$$Q_{k+1} \in \mathbb{R}^{n \times q} \leftarrow \begin{cases} \text{random} & \text{if } \rho \geq \eta_1 \text{ and } \|g_k\| \geq \eta_2 \Delta_k \text{ or if } \mathcal{Y}_k \text{ is } \Lambda\text{-poised} \\ Q_k & \text{otherwise.} \end{cases}$$

6 Update the interpolation sets
$$\mathcal{Y}_{k+1}, \mathcal{Z}_{k+1} \leftarrow \begin{cases} \mathcal{Y}_0, \mathcal{Z}_0 & \text{if } \rho \geq \eta_1 \text{ and } \|g_k\| \geq \eta_2 \Delta_k \text{ or if } \mathcal{Y}_k \text{ is } \Lambda\text{-poised} \\ \text{Update as in Step 4 of Algorithm 2} & \text{otherwise.} \end{cases}$$

Proof. By Theorem 3.5, since we chose $\Lambda = 1 + \frac{1}{q}$, we have the error bound

$$\|\nabla \hat{m}_k(0) - \nabla \hat{\phi}(0)\| \leq \mathcal{O}\left(\sqrt{q} \left(L \Delta_k + \frac{\epsilon_f}{\Delta_k}\right)\right).$$

Thus for iterations when $\mathcal{Y}_k$ is Λ -poised, we have that m_k is κ_{ef}, κ_{eg} -fully linear model for

$$\kappa_{eg}, \kappa_{ef} = \mathcal{O}\left(\sqrt{q} \left(L + \frac{\epsilon_f}{\Delta_{\min}^2}\right)\right).$$

From this, Theorem 5.12, and the arguments from the proof of Theorem 5.13, we can bound the number of non-geometry-correcting iterations (i.e. iterations where Q_k is resampled) by $\mathcal{O}(\frac{n}{\epsilon^2})$. For these iterations, we use only one oracle call to evaluate $f(x_k + s_k)$, however for the iteration which immediately follows a successful or an unsuccessful iteration, we use $\mathcal{O}(q)$ oracle calls to evaluate $f(x_{k+1} + Q_{k+1} y_i)$ and $f(x_{k+1} + Q_{k+1} z_i)$. For all other iterations, as in Algorithm 2, we use at most 2 oracle calls. Finally, by Theorem 4.1, the maximum number of consecutive geometry correcting iterations is $\mathcal{O}(q \log q)$. The result follows. $\square$

6 Numerical Implementations and Results In this section we propose an implementation of Algorithm 2, which incorporates all its elements such as the self-correcting and geometry-correcting steps but in addition includes several practical features. Some of these features are borrowed from Powell's algorithms and some are new. As we will discuss, all these additional features improve practical performance but make the analysis more cumbersome. However, ultimately the order of the worst-case complexity of the algorithm is preserved.

算法 4: 子空间中的几何校正算法

输入: 不精确零阶预言机 $|f(x) - \phi(x)| \leq \epsilon_f$, 最小半径 $\Delta_{\min}$, 初始 x_0 , $\Delta_0 \geq \Delta_{\min}$, 初始矩阵 $Q_0 \in \mathbb{R}^{n \times q}$ 具有标准正交列, 以及 $\eta_1 \in (0, 1)$, $\eta_2 > 0, \gamma \in (0, 1)$, $\Lambda > 1$, $\Lambda_{sc} \geq 1$.
初始化 初始集合 $\mathcal{Y}_0, \mathcal{Z}_0 \subset \mathbb{R}^q$ 以及函数值 $f(x_0), f(x_0 + Q_0 y_i), y_i \in \mathcal{Y}_0, f(x_0 + Q_0 z_i), z_i \in \mathcal{Z}_0$. 一组拉格朗日多项式 $\{\ell_i(v) \mid i = 1, \dots, q\}$ 在 $\mathcal{P}$ 中针对集合 $\mathcal{Y}_0$.
对于 $k = 0, 1, 2, \dots$ **执行**
1 对于当前的 Q_k , 按照式 (3.2) 构建二次模型 $\hat{m}_k(v) = m_k(x_k + Q_k v)$, 使用 $f(x_k)$ 和 $f(x_k + Q_k y_i), y_i \in \mathcal{Y}_k, f(x_k + Q_k z_i), z_i \in \mathcal{Z}_k$.
2 计算试探步 $x_k + s_k$, 其中 $s_k = Q_k v_k$, 且 $v_k \approx \arg \min_v \{m_k(x_k + Q_k v) : \|v\| \leq \Delta_k\}$.
3 计算比率 ρ_k 如下
$$\rho_k = \frac{f(x_k) - f(x_k + s_k) + 2\epsilon_f}{m_k(x_k) - m_k(x_k + s_k)}.$$

4 将迭代点和信赖域半径更新如下
$$(x_{k+1}, \Delta_{k+1}) \leftarrow \begin{cases} (x_k + s_k, \gamma^{-1} \Delta_k) & \text{如果 } \rho_k \geq \eta_1 \text{ 且 } \|g_k\| \geq \eta_2 \Delta_k, \\ (x_k, \Delta_k) & \text{否则, 如果 } \mathcal{Y}_k \text{ 不是 } \Lambda\text{-适定的,} \\ (x_k, \max\{\gamma \Delta_k, \Delta_{\min}\}) & \text{否则.} \end{cases}$$

5 更新子空间
$$Q_{k+1} \in \mathbb{R}^{n \times q} \leftarrow \begin{cases} \text{random} & \text{if } \rho \geq \eta_1 \text{ and } \|g_k\| \geq \eta_2 \Delta_k \text{ or if } \mathcal{Y}_k \text{ is } \Lambda\text{-poised} \\ Q_k & \text{otherwise.} \end{cases}$$

6 更新插值集
$$\mathcal{Y}_{k+1}, \mathcal{Z}_{k+1} \leftarrow \begin{cases} \mathcal{Y}_0, \mathcal{Z}_0 & \text{if } \rho \geq \eta_1 \text{ and } \|g_k\| \geq \eta_2 \Delta_k \text{ or if } \mathcal{Y}_k \text{ is } \Lambda\text{-poised} \\ \text{Update as in Step 4 of Algorithm 2} & \text{otherwise.} \end{cases}$$

证明。根据定理 3.5, 由于我们选择了 $\Lambda = 1 + \frac{1}{q}$, 我们有误差界

$$\|\nabla \hat{m}_k(0) - \nabla \hat{\phi}(0)\| \leq \mathcal{O}\left(\sqrt{q} \left(L \Delta_k + \frac{\epsilon_f}{\Delta_k}\right)\right).$$

因此, 对于 $\mathcal{Y}_k$ 为 Λ -适定的迭代, 我们有 m_k 是 κ_{ef}, κ_{eg} -完全线性模型, 针对

$$\kappa_{eg}, \kappa_{ef} = \mathcal{O}\left(\sqrt{q} \left(L + \frac{\epsilon_f}{\Delta_{\min}^2}\right)\right).$$

由此, 结合定理 5.12 以及定理 5.13 证明中的论证, 我们可以将非几何校正迭代 (即 Q_k 被重新采样的迭代) 的数量以 $\mathcal{O}(\frac{n}{\epsilon^2})$ 为界。对于这些迭代, 我们仅使用一次预言机调用来评估 $f(x_k + \bar{s}_k)$; 然而, 对于紧跟在成功迭代或不成功迭代之后的迭代, 我们使用 $\mathcal{O}(q)$ 次预言机调用来评估 $f(x_{k+1} + Q_{k+1} y_i)$ 和 $f(x_{k+1} + Q_{k+1} z_i)$ 。对于所有其他迭代, 如算法 2 中那样, 我们最多使用 2 次预言机调用。最后, 根据定理 4.1, 连续几何校正迭代的最大数量为 $\mathcal{O}(q \log q)$ 。结果随之成立。 $\square$

6 数值实现与结果 本节中, 我们提出算法 2 的一种实现, 它包含其所有要素, 如自校正和几何校正步骤, 但此外还包括若干实用特性。其中一些特性借鉴自 Powell 算法, 另一些则是新的。正如我们将讨论的, 所有这些附加特性都提升了实际性能, 但使分析更加繁琐。然而, 最终算法的最坏情况复杂度的阶保持不变。

The practical implementation is given in Algorithm 5. It utilizes the two interpolation sets $\mathcal{Y}$ and $\mathcal{Z}$ to manage linear Lagrange polynomials while fitting quadratic models, as proposed in Algorithm 2. We compare our proposed algorithm to NEWUOA [21], which is arguably the most scalable of Powell’s algorithms and which maintains geometry of the full interpolation set by the use of quadratic Lagrange polynomials, and with DFOTR [2] which is a surprisingly efficient method that does not maintain any Lagrange polynomials and only updates the sample set based on the distance of the points to the TR center. In that respect DFOTR also maintains two separate sets, in that it does not reduce the trust region on steps that are not successful and when there are fewer than $n + 1$ points in the appropriate vicinity of the trust region center.

We also test a variation of Algorithm 2, to which current theory does not extend and which uses the quadratic Lagrange polynomials combined with a NEWUOA-like self-correcting rule in place of the rule described in Algorithm 2. As this method seems to provide improvement in high-dimensional setting, it gives motivation for the theory from Sections 3 and 4 to be extended to quadratic Lagrange polynomials in future work. All solvers will use a novel, adaptive fitting scheme. We test these algorithms on a collection of unconstrained problems from the CUTEst test set [17] and investigate results in low dimension, high dimension, and in randomized subspaces.

The following Section 6.1 describes the design choices of our proposed algorithms and how it still satisfies the theory. In Section 6.2 we discuss the testing methodology and numerical results.

6.1 Practical Implementations Algorithm 5 (GC-YZ-LIN) is a practical implementation of the geometry correction framework which maintains linear Lagrange polynomials for the set $\mathcal{Y}$ while interpolating a quadratic model using $\mathcal{Y} \cup \mathcal{Z} \cup \{x_k\}$. This method can be seen as the middle ground between DFOTR and NEWUOA, where the former makes only minimal effort to ensure good sample set geometry while the latter uses a significant amount of effort.

We now describe the changes implemented in Algorithm 5 as opposed to Algorithm 2. Algorithm 5 makes use of a resolution floor and small step gate, both of which are ideas borrowed from Powell’s methods.

The resolution floor σ_k is an adaptive lower bound on Δ_k which only gets decreased once no progress can be made for that resolution. That is, if $\Delta_k = \sigma_k$, the geometry is good, and the iteration is still unsuccessful we reset $\sigma_{k+1} = \theta \sigma_k$ for some $0 < \theta < 1$. We then allow Δ_k to shrink even on model improving iterations as long as Δ_k is larger than the floor σ_k . This is motivated by the observation that in higher dimensions it can be expensive to always ensure good geometry before shrinking, hence an alternative is to only ensure this good geometry at intervals throughout the trajectory of the algorithm.

A small step gate and small model gradient gate are used by checking the conditions $\|s_k\| \geq c_g \sigma_k$ and $\|g_k\| \geq \eta_2 \Delta_k$ respectively. The small step gate adds an additional condition under which an iteration is deemed successful, which we note improves the termination speed of the algorithm. In the small model gradient gate, before evaluating a proposed step, we check if the norm of the model gradient is small relative to the trust region radius. That is if we know the step is going to be rejected, we save a function evaluation by skipping the evaluation of the trial step and attempt a geometry correcting step and possibly shrink the trust region radius. The same principle applies for the small step gate.

With these two changes, the complexity analysis in Section 2 follows through with small modifications, provided that the following simple assumption holds.

ASSUMPTION 6.1. *On every iteration k , we have*

$$\|s_k\| \geq \kappa_{\text{step}} \min \left\{ \Delta_k, \frac{\|g_k\|}{\kappa_{bhm}} \right\}.$$

Note that if the trust region subproblem is solved exactly, the assumption is satisfied with $\kappa_{\text{step}} = 1$, while for the Cauchy step we have $\kappa_{\text{step}} = \frac{\kappa_{fcd}}{3}$. Then, for Lemma 2.3 to go through, we require that $c_g \leq \kappa_{\text{step}}$ and Assumption 6.1 to hold. It would then follow that

$$\kappa_{bhm} \Delta_k \leq \max\{\kappa_{bhm}, \eta_2\} \Delta_k \leq \|g_k\| \implies \min \left\{ \Delta_k, \frac{\|g_k\|}{\kappa_{bhm}} \right\} = \Delta_k,$$

hence $\|s_k\| \geq \kappa_{\text{step}} \Delta_k \geq c_g \sigma_k$, and a small Δ_k still implies a successful iteration, even with a small step gate. The rest of the complexity argument for the full-dimensional method follows very closely to the analysis in Section 2, by applying key results with σ_k instead of Δ_k and deriving a bound on the number of unsuccessful iterations until

实际实现见算法 5。它利用两个插值集 $\mathcal{Y}$ 和 $\mathcal{Z}$ 来管理线性拉格朗日多项式，同时拟合二次模型，如算法2所提出。我们将所提出的算法与NEWUOA [21], 进行比较，后者可以说是Powell算法中最具可扩展性的，并且通过使用二次拉格朗日多项式来维持完整插值集的几何；还将其与DFOTR[2] 进行比较，后者是一种出奇高效的方法，它不维护任何拉格朗日多项式，仅根据点到信赖域中心的距离来更新样本集。在这方面，DFOTR也维护两个独立的集合，即它在不成功的步骤上，以及在信赖域中心适当邻域内的点少于 $n + 1$ 个时，不缩小信赖域。

我们还测试了算法2的一个变体，当前理论并未将其涵盖，该变体使用二次拉格朗日多项式，并结合类似NEWUOA的自校正规则，以替代算法2中描述的规则。由于该方法似乎在高维设置中带来改进，这为未来工作将第3节和第4节的理论扩展到二次拉格朗日多项式提供了动机。所有求解器都将使用一种新颖的自适应拟合方案。我们在来自CUTEst测试集 [17] 的一组无约束问题上测试这些算法，并考察低维、高维以及随机子空间中的结果。

接下来的第6.1节描述了我们所提出算法的设计选择，以及它如何仍然满足该理论。在第6.2节中，我们讨论测试方法和数值结果。

6.1 实际实现 算法 5 (GC-YZ-LIN) 是几何校正框架的一种实际实现，它针对集合 $\mathcal{Y}$ 维护线性拉格朗日多项式，同时使用 $\mathcal{Y} \cup \mathcal{Z} \cup \{x_k\}$ 插值一个二次模型。该方法可被视为 DFOTR 与 NEWUOA 之间的折中，其中前者仅付出最小努力来确保良好的样本集几何，而后者则付出大量努力。

我们现在描述 算法 5 中相较于算法2所实现的改动。算法 5 使用分辨率下限以及 小步长门控，二者都是借鉴自 Powell方法的思想。

分辨率下限 σ_k 是 Δ_k 的自适应下界，仅当对于该分辨率无法再取得进展时，它才会被降低。也就是说，如果 $\Delta_k = \sigma_k$ ，几何良好，而迭代仍不成功，我们就针对某个 $0 < \theta < 1$ 重置 $\sigma_{k+1} = \theta \sigma_k$ 。然后我们允许 Δ_k 即使在模型改进迭代中也收缩，只要 Δ_k 大于下限 σ_k 。这源于以下观察：在更高维度中，总是在收缩前确保良好几何可能代价高昂，因此另一种做法是仅在算法轨迹中的某些间隔确保这种良好几何。

小步长门控和小模型梯度门控 分别通过检查条件 $\|s_k\| \geq c_g \sigma_k$ 和 $\|g_k\| \geq \eta_2 \Delta_k$ 来使用。小步长门控增加了一个额外条件，在该条件下迭代被视为成功；我们注意到这提高了算法的终止速度。在小模型梯度门控中，在评估建议步之前，我们会检查模型梯度的范数相对于信赖域半径是否较小。也就是说，如果我们知道该步将被拒绝，就通过跳过试探步的评估来节省一次函数求值，并尝试几何校正步骤，还可能收缩信赖域半径。同样的原则也适用于小步长门控。

通过这两处更改，第 2 节中的复杂度分析在经过少量修改后仍然成立，前提是以下简单假设成立。

假设 6.1. 在每次迭代 k 中，我们有

$$\|s_k\| \geq \kappa_{\text{step}} \min \left\{ \Delta_k, \frac{\|g_k\|}{\kappa_{bhm}} \right\}.$$

注意，如果信赖域子问题被精确求解，则该假设在 $\kappa_{\text{step}} = 1$ 下得到满足；而对于柯西步，我们有 $\kappa_{\text{step}} = \frac{\kappa_{fcd}}{3}$ 。然后，为使引理 2.3 成立，我们要求 $c_g \leq \kappa_{\text{step}}$ 以及假设 6.1 成立。由此便可推出

$$\kappa_{bhm} \Delta_k \leq \max\{\kappa_{bhm}, \eta_2\} \Delta_k \leq \|g_k\| \implies \min \left\{ \Delta_k, \frac{\|g_k\|}{\kappa_{bhm}} \right\} = \Delta_k,$$

因此 $\|s_k\| \geq \kappa_{\text{step}} \Delta_k \geq c_g \sigma_k$ ，并且较小的 Δ_k 仍意味着成功迭代，即使采用小步长门控。全维方法的复杂度论证的其余部分与第 2 节中的分析非常接近：通过使用 σ_k 代替 Δ_k 来应用关键结果，并推导出对不成功迭代次数的界，直到

Δ_k reaches the floor σ_k for each round of σ_k reductions. Hence the proposed method will obey the theory, at the cost of some logarithmic factors.

The model fitting procedure fits a quadratic model, which we call the “hedge” model, using the following somewhat elaborate sequence of steps. On each iteration we compute the minimum Frobenius norm (MFN) model and the minimum change Frobenius norm (MCFN) model. Both of these models are computed by solving

$$(6.1) \quad \min_{g,H} \|H_{prev} - H\|_F \quad \text{s.t.} \quad v^\top g + \frac{1}{2}v^\top H v = f(x_k + v) - f(x_k), \quad v \in \mathcal{Y} \cup \mathcal{Z}.$$

where the MFN model sets $H_{prev} = 0$ and the MCFN model sets H_{prev} to the previous iteration’s MCFN Hessian. For both models, an exponentially weighted moving average of a relative error score is maintained and is used to determine which model is to be used for the current iteration. Only one model is “active” at a time and if the error for the alternative model becomes lower than the active model, we switch which model is used to compute the trial step. This change comes from the observation that sometimes it may be beneficial to remember the current local curvature, while at other times it may be beneficial to fit a fresh model as previous model Hessians can become stale. The proposed scoring method is a way to adaptively decide which model is best suited for the current iteration. We remark that a similar idea was proposed in Powell’s original paper, where consecutive poor steps would trigger an MFN model to be fit instead of MCFN.

After this modeling procedure, to ensure a bounded model Hessian, we check if the fitted Hessian exceeds the bound, i.e. if $\|H\| > K$. If so, we fit g_k and H_k by solving the regression problem (4.1). We note that there are other ways to bound $\|H_k\|$, such as via clipping. Since several of the objective functions in our data set naturally have extreme curvature, to prevent the truncation of useful curvature information, we set K to be a very large number. As a result, the condition $\|H\| > K$ will rarely trigger, which will yield either one of the MFN or MCFN models throughout an overwhelming majority of the iterations. We note that setting K very large implies that for some problems κ_{bhm} becomes very large. However, it appears to happen only when L is similarly large, thus having large κ_{bhm} or reducing it has no bearing on the order of the theoretical complexity bound.

We also develop a version of Algorithm 5, which we call GC-YZ-V, where the self-correction step computes and updates quadratic Lagrange polynomials over the set $\mathcal{Y} \cup \mathcal{Z}$, similar to NEWUOA, while still maintaining the $\mathcal{Y}$ - $\mathcal{Z}$ separation. Specifically, for the self-correcting step, we compute the score of a point in the sample set to be:

$$\text{score}(y_i) = \max \left(1, \frac{\|y_i\|^2}{\max(0.1\Delta_k, \sigma_k)^2} \right)^3 \cdot \ell_i(s_k)^2,$$

and replace the point that achieved the maximum score by s_k . Intuitively, this attempts to simultaneously remove points that are deemed far away while attempting to improve geometry. After such self-correction, the geometry correction is carried out in the same manner as in GC-YZ-LIN. Since the pseudocode closely mirrors Algorithm 5 and is not currently accompanied by any theoretical guarantees, we omit it and present the results as a proof of concept. We see that this algorithm somewhat outperforms both GC-YZ-LIN and NEWUOA and believe that modifications to only the self-correction step can be ultimately covered by extending our theory.

The random subspace variation with geometry correction has a straightforward implementation, in that there is little deviation from the theoretical framework described in Algorithm 4. We call this algorithm GC-sub. The only deviation from the framework is the addition of a successful and unsuccessful iteration “patience” parameter. This value dictates the required number of successful or unsuccessful iterations before a subspace is redrawn. Intuitively, if we are making significant progress on a particular subspace, it may be beneficial to keep this subspace instead of redrawing. In our theoretical framework Algorithm 4 this parameter is set to 1. While one can extend the theory to any fixed value of this parameter, it would complicate the notation of Section 5. On the other hand our computational results suggest that 1 is the best value for this parameter at least in the current setting.

Δ_k 在每一轮 σ_k 缩减中都达到下限 σ_k 。因此，所提方法将遵循该理论，代价是一些对数因子。

模型拟合过程 拟合一个二次模型，我们称之为“Hedge”模型，采用以下略显繁琐的步骤序列。在每次迭代中，我们计算最小 Frobenius 范数（MFN）模型和最小变化 Frobenius 范数（MCFN）模型。这两个模型均通过求解得到

$$(6.1) \quad \min_{g,H} \|H_{prev} - H\|_F \quad \text{s.t.} \quad v^\top g + \frac{1}{2}v^\top H v = f(x_k + v) - f(x_k), \quad v \in \mathcal{Y} \cup \mathcal{Z}.$$

其中，MFN 模型设置 $H_{prev} = 0$ ，而 MCFN 模型将 H_{prev} 设为上一次迭代的 MCFN Hessian 矩阵。对于这两种模型，都会维护相对误差得分的指数加权移动平均，并用其确定当前迭代应使用哪个模型。每次只有一个模型处于“活跃”状态；如果另一个模型的误差低于活跃模型，我们就切换用于计算试探步的模型。这一变化源于以下观察：有时记住当前局部曲率可能是有益的，而在其他时候，拟合一个全新模型可能更有利，因为之前的模型海森矩阵可能会过时。所提出的评分方法是一种自适应决定哪个模型最适合当前迭代的方式。我们注意到，Powell 的原始论文中提出了类似思想：连续的不良步骤会触发拟合 MFN 模型而不是 MCFN 模型。

在此建模过程之后，为了确保有界模型 Hessian 矩阵，我们检查拟合的 Hessian 矩阵是否超出该界，即如果 $\|H\| > K$ 。若是，我们通过求解回归问题 (4.1) 来拟合 g_k 和 H_k 。我们注意到，还有其他为 $\|H_k\|$ 设定界的方法，例如通过裁剪。由于我们数据集中的若干目标函数天然具有极端曲率，为了防止有用的曲率信息被截断，我们将 K 设为非常大的数。因此，条件 $\|H\| > K$ 很少会触发，这将在绝大多数迭代中产生 MFN 或 MCFN 模型中的一种。我们注意到，将 K 设得非常大意味着对于某些问题， κ_{bhm} 会变得非常大。然而，这种情况似乎仅在 L 同样很大时才会发生，因此 κ_{bhm} 较大或将其减小，对理论复杂度界的阶都没有影响。

我们还开发了算法 5 的一个版本，我们称之为 GC-YZ-V；其中，自校正步骤在集合 $\mathcal{Y} \cup \mathcal{Z}$ 上计算并更新二次拉格朗日多项式，类似于 NEWUOA，同时仍保持 $\mathcal{Y}$ - $\mathcal{Z}$ 分离性。具体而言，对于自校正步，我们将样本集中某个点的得分计算为：

$$\text{score}(y_i) = \max \left(1, \frac{\|y_i\|^2}{\max(0.1\Delta_k, \sigma_k)^2} \right)^3 \cdot \ell_i(s_k)^2,$$

并用 s_k 替换取得最大得分的点。直观上，这试图在尝试改进几何结构的同时，移除被认为相距较远的点。经过这样的自校正后，几何校正按照与 GC-YZ-LIN 相同的方式进行。由于伪代码与算法 5 高度相似，且目前尚无任何理论保证，我们将其省略，并将结果作为概念验证给出。我们看到，该算法在某种程度上优于 GC-YZ-LIN 和 NEWUOA，并相信仅对自校正步骤进行修改最终可以通过扩展我们的理论来涵盖。

带几何校正的随机子空间变体具有直接的实现方式，因为它与算法 4 中描述的理论框架几乎没有偏差。我们将该算法称为 GC-sub。与框架的唯一偏差是增加了一个成功迭代和不成功迭代的“耐心”参数。该值决定了在重新抽取子空间之前所需的成功迭代或不成功迭代次数。直观地说，如果我们在某个特定子空间上取得显著进展，那么保留该子空间而不是重新抽取可能是有益的。在我们的理论框架算法 4 中，该参数被设为 1。尽管可以将理论推广到该参数的任意固定值，但这会使第 5 节的符号表示复杂化。另一方面，我们的计算结果表明，至少在当前设置下，1 是该参数的最佳取值。

Algorithm 5: Geometry-correcting algorithm with $\mathcal{Y}$ and $\mathcal{Z}$ separation (GC-YZ-LIN)

Inputs: A zeroth-order oracle $f(x) \approx \phi(x)$, $\Delta_0, x_0, \gamma \in (0, 1)$, $\eta_1 > 0, \eta_2 > 0, \Lambda > 1, \Lambda_{sc} \geq 1, \theta \in (0, \gamma)$, $\sigma_{\text{end}} \in (0, \Delta_0], c_g \in (0, 1]$, model Hessian bound K , averaging weight $\beta \in (0, 1]$.

Initialization: An initial set $\mathcal{Y}_0$ such that $|\mathcal{Y}_0| = n$, an initial set $\mathcal{Z}_0$ such that $|\mathcal{Z}_0| \leq n(n+1)/2$ and the function values $f(x_0), f(x_0 + y_i), y_i \in \mathcal{Y}_0, f(x_0 + z_i), z_i \in \mathcal{Z}_0$. Set the resolution floor $\sigma_0 = \Delta_0$. A set of Lagrange polynomials $\{\ell_i(x), i = 1, \dots, n\}$ in $\mathcal{P}$ for the set $\mathcal{Y}_0$, model errors $e_C, e_F \leftarrow \text{undefined}$, $H_{prev} \leftarrow \mathbf{0}$, set active model label $\alpha \leftarrow F$.

for $k = 0, 1, 2, \dots$ **do**

1 *Model building:* Construct MFN model g_F, H_F . Construct MCFN model g_C, H_C . Set $H_{prev} = H_C$.

2 If e_C or e_F is undefined, then $\alpha \leftarrow F$ and skip *model switching* step in Line 3.

3 *Model switching:* If $\alpha = F$ and $e_C < 0.8e_F$ then $\alpha \leftarrow C$, otherwise if $\alpha = C$ and $e_F < 0.8e_C$ then $\alpha \leftarrow F$.

4 If $\alpha = F$ then $g_k, H_k \leftarrow g_F, H_F$. If $\alpha = C$ then $g_k, H_k \leftarrow g_C, H_C$.

5 *Ensuring bounded model Hessians:* If $\|H_k\| > K$ set g_k, H_k by solving (4.1).

6 Construct the set of Lagrange polynomials $\{\ell_i(x), i = 1, \dots, n\}$ in $\mathcal{P}$ for the set $\mathcal{Y}_k$ and find max value

$$(j_k^*, s_k^*) = \arg \max_{j=1, \dots, n, s \in B(0, \Delta_k)} |\ell_j(s)|.$$

7 *Small model gradient gate:* If $\|g_k\| < \eta_2 \Delta_k$ perform the following checks:

 If $|\ell_{j_k^*}(s_k^*)| > \Lambda$ then perform “*Geometry correction of $\mathcal{Y}$ by replacing a “bad” point*”.

 If $|\ell_{j_k^*}(s_k^*)| \leq \Lambda$ and $\Delta_k \leq \sigma_k$ perform the *resolution update*:

$$\sigma_{k+1} = \max\{\theta\sigma_k, \sigma_{\text{end}}\}, \Delta_{k+1} = \max\{\frac{1}{2}\Delta_k, \sigma_{k+1}\}.$$

 If $|\ell_{j_k^*}(s_k^*)| \leq \Lambda$ and $\Delta_k > \sigma_k$, then $\Delta_{k+1} \leftarrow \max\{\gamma\Delta_k, \sigma_k\}$.

 Go to 1 and $k \leftarrow k + 1$.

8 Compute a trial step s_k as in Algorithm 1.

9 *Small step gate:* If $\|s_k\| < c_g\sigma_k$, do not evaluate $f(x_k + s_k)$. Perform the same checks as *small model gradient gate*.

10 Evaluate $f(x_k + s_k)$ and compute the ratio ρ_k as in Algorithm 1.

11 *Model scoring:* **for** $\mu \in \{F, C\}$ **do**

$$p_\mu \leftarrow (g_\mu)^\top s_k + \frac{1}{2} s_k^\top H_\mu s_k, \quad f_c \leftarrow f(x_k + s_k) - f(x_k)$$

$$\varepsilon_\mu \leftarrow \frac{|p_\mu - f_c|}{\max\{|f_c|, |p_\mu|\}}$$

 If e_μ is undefined, then $e_\mu = \varepsilon_\mu$, otherwise $e_\mu = \beta e_\mu + (1 - \beta)\varepsilon_\mu$.

12 *Successful iteration:* $\rho_k \geq \eta_1$. Perform the successful update as in Algorithm 2.

13 *Unsuccessful or Model Improving iteration:* $\rho_k < \eta_1$. Perform all applicable steps (i), ..., (iv) from Algorithm 2.

Model improving iteration: If $I_{imp} = 1$, set $\Delta_{k+1} = \max\{\gamma\Delta_k, \sigma_k\}$.

Unsuccessful iteration above the floor: If $I_{imp} = 0$ and $\Delta_k > \sigma_k$, set $\Delta_{k+1} = \max\{\gamma\Delta_k, \sigma_k\}$.

Unsuccessful iteration at floor: If $I_{imp} = 0$ and $\Delta_k \leq \sigma_k$, perform the *resolution update* from Line 7.

6.2 Testing and Results We now turn to the results. We state upfront that our numerical comparison yields the following observations.

1. The new adaptive Hessian fitting procedure improves performance for all solvers.
2. Methods with geometry correction, i.e. GC-YZ-LIN/V and NEWUOA, outperform DFOTR in higher dimensions.

算法 5: 具有 $\mathcal{Y}$ 和 $\mathcal{Z}$ 分离的几何校正算法 (GC-YZ-LIN)

输入: 一个零阶预言机 $f(x) \approx \phi(x)$, $\Delta_0, x_0, \gamma \in (0, 1)$, $\eta_1 > 0, \eta_2 > 0, \Lambda > 1, \Lambda_{sc} \geq 1, \theta \in (0, \gamma)$, $\sigma_{\text{end}} \in (0, \Delta_0], c_g \in (0, 1]$, 模型Hessian界 K , 平均权重 $\beta \in (0, 1]$ 。

初始化: 一个初始集合 $\mathcal{Y}_0$ 使得 $|\mathcal{Y}_0| = n$, 一个初始集合 $\mathcal{Z}_0$ 使得 $|\mathcal{Z}_0| \leq n(n+1)/2$ 以及函数值 $f(x_0), f(x_0 + y_i), y_i \in \mathcal{Y}_0, f(x_0 + z_i), z_i \in \mathcal{Z}_0$ 。设置分辨率下限 $\sigma_0 = \Delta_0$ 。一组拉格朗日多项式 $\{\ell_i(x) \mid i = 1, \dots, n\}$ 在 $\mathcal{P}$ 中, 对于集合 $\mathcal{Y}_0$, 模型误差 $e_C, e_F \leftarrow \text{未定义}$, $H_{prev} \leftarrow \mathbf{0}$, 设置活动模型标签 $\alpha \leftarrow F$ 。

对于 $k = 0, 1, 2, \dots$ **执行**

1 模型构建: 构建 MFN模型 g_F, H_F 。构建 MCFN模型 g_C, H_C 。设置 $H_{prev} = H_C$ 。

2 如果 e_C 或 e_F 未定义, 则 $\alpha \leftarrow F$ 并跳过第 3 行中的 模型切换 步骤。

3 模型切换: 如果 $\alpha = F$ 且 $e_C < 0.8e_F$, 则 $\alpha \leftarrow C$, 否则, 如果 $\alpha = C$ 且 $e_F < 0.8e_C$, 则 $\alpha \leftarrow F$ 。

4 如果 $\alpha = F$, 则 $g_k, H_k \leftarrow g_F, H_F$ 。如果 $\alpha = C$, 则 $g_k, H_k \leftarrow g_C, H_C$ 。

5 确保有界模型海森矩阵: 如果 $\|H_k\| > K$, 设置 g_k, H_k , 通过求解 (4.1)。

6 构造拉格朗日多项式集合 $\{\ell_i(x) \mid i = 1, \dots, n\}$ 在 $\mathcal{P}$ 中, 对于集合 $\mathcal{Y}_k$, 并求最大值

$$(j_k^*, s_k^*) = \arg \max_{j=1, \dots, n, s \in B(0, \Delta_k)} |\ell_j(s)|.$$

7 小模型梯度门控: 如果 $\|g_k\| < \eta_2 \Delta_k$ 执行以下检查:

 如果 $|\ell_{j_k^*}(s_k^*)| > \Lambda$ 则执行 “通过替换一个‘坏’点来对 $\mathcal{Y}$ 进行几何校正”。

 如果 $|\ell_{j_k^*}(s_k^*)| \leq \Lambda$ 且 $\Delta_k \leq \sigma_k$ 执行分辨率更新:

$$\sigma_{k+1} = \max\{\theta\sigma_k, \sigma_{\text{end}}\}, \Delta_{k+1} = \max\{\frac{1}{2}\Delta_k, \sigma_{k+1}\}.$$

 如果 $|\ell_{j_k^*}(s_k^*)| \leq \Lambda$ 且 $\Delta_k > \sigma_k$, 则 $\Delta_{k+1} \leftarrow \max\{\gamma\Delta_k, \sigma_k\}$ 。

 转到 1 并 $k \leftarrow k + 1$ 。

8 按照算法1计算试探步 s_k 。

9 小步长门控: 如果 $\|s_k\| < c_g\sigma_k$, 不要计算 $f(x_k + s_k)$ 。执行与小模型相同的检查梯度门控。

10 计算 $f(x_k + s_k)$, 并按照算法1计算比率 ρ_k 。

11 模型评分: **对于** $\mu \in \{F, C\}$ **执行**

$$p_\mu \leftarrow (g_\mu)^\top s_k + \frac{1}{2} s_k^\top H_\mu s_k, \quad f_c \leftarrow f(x_k + s_k) - f(x_k)$$

$$\varepsilon_\mu \leftarrow \frac{|p_\mu - f_c|}{\max\{|f_c|, |p_\mu|\}}$$

 如果 e_μ 未定义, 则 $e_\mu = \varepsilon_\mu$, 否则 $e_\mu = \beta e_\mu + (1 - \beta)\varepsilon_\mu$ 。

12 成功迭代: $\rho_k \geq \eta_1$ 。按照算法2执行成功更新。

13 不成功或模型改进迭代: $\rho_k < \eta_1$ 。执行所有适用步骤 (i)....., (iv) 来自算法2。

 模型改进迭代: 如果 $I_{imp} = 1$, 设置 $\Delta_{k+1} = \max\{\gamma\Delta_k, \sigma_k\}$ 。

 下限以上的不成功迭代: 如果 $I_{imp} = 0$ 和 $\Delta_k > \sigma_k$, 设置 $\Delta_{k+1} = \max\{\gamma\Delta_k, \sigma_k\}$ 。

 下限处的不成功迭代: 如果 $I_{imp} = 0$ 和 $\Delta_k \leq \sigma_k$, 执行第7行的分辨率更新。

6.2 测试与结果 现在我们来看结果。我们首先说明, 我们的数值比较得出以下观察结果。

1. 新的自适应Hessian拟合过程提升了所有求解器的性能。
2. 采用几何校正的方法, 即 GC-YZ-LIN/V 和 NEWUOA, 在更高维度上优于 DFOTR。

- Using the $\mathcal{Y}$ - $\mathcal{Z}$ separation, GC-YZ-LIN is able to largely match NEWUOA, with trade-offs between early-game and late-game performance. Transplanting NEWUOA’s self-correcting rule into the $\mathcal{Y}$ - $\mathcal{Z}$ framework, we obtain a variant that outperforms NEWUOA; no theoretical analysis has been developed for it yet.
- A practical implementation of a randomized subspace algorithm that is competitive with full space solvers remains difficult.

Here we give the details of our testing methodology. Table 6.1 outlines the parameters used for our proposed solver and DFOTR. We will use PRIMA’s [27] implementation of NEWUOA, which has fixed trust region parameters by default, so we omit restating it in the table. All problems are initialized in the same way, where the initial interpolation set $\mathcal{Y}_0$ is set to $\{\Delta_0 u_i\}$ where $u_i, i = 1, \dots, n$ are randomly rotated coordinate vectors, and radius $\Delta_0 = 0.5$. $\mathcal{Z}_0$ is then set to $\{-\Delta_0 u_i\}$. To allow the algorithm to run until an exhausted budget, we set the termination criteria to be $\Delta_{\text{end}} = 10^{-12}$ for DFOTR and $\rho_{\text{end}}, \sigma_{\text{end}} = 10^{-12}$ for NEWUOA and GC-YZ-LIN/V respectively. All trust region subproblems are solved exactly. We compare performances using data profiles at tolerances $\tau \in \{10^{-2}, 10^{-4}, 10^{-6}\}$ and report the area under curve score of each solver in the legend. We average the curves obtained from the different randomly rotated initial sets.

We remark on the choice of three parameters in GC-YZ-LIN that also show up in the complexity analysis. Firstly, the model Hessian bound K is set to be extremely large, namely 10^{100} and essentially inactive. This is largely a byproduct of testing solver performance on problems from CUTEst with naturally extreme curvature. For example, problems such as “SSBRYND” and “SCURLY10” have true Hessian norms greater than 10^{20} while “POWERSUM” can have a true Hessian norm even exceeding 10^{100} . Thus setting a universal, fixed, and small K will handicap the model’s ability to capture the function’s true curvature. Additionally, we set η_2 to be small, namely 5×10^{-9} . While ensuring the norm of the model gradient is not too small relative to the radius is necessary for the theory, a large η_2 tethering the trust region radius to the model gradient can be detrimental to problems with small local Lipschitz constants, for example, the Rosenbrock function outside of the “banana valley”. If there is still progress to be made, iterates in this region will be forced to have a small trust region radius and may stunt progress. Moreover, in practice it appears beneficial to have a relaxed choice for Λ . According to our theory, Λ -poisedness guarantees a certain amount of function decrease on successful iterations each time. However, ensuring a small Λ in practice comes at the cost of more function evaluations for geometry correction. We observe that the best performance comes when $\Lambda = 1000$, which suggests that although the geometry may not be perfect, the algorithm is still able to progress, and geometry correction should only take place when the geometry is exceedingly bad. We also make the distinction between Λ and Λ_{sc} , where the latter is a threshold on how large a Lagrange polynomial value at a step would need to be to initiate a self-correcting step. In practice we set is to 2, while the analysis applies as long as $\Lambda_{sc} \geq 1$.

These seemingly detrimental parameter choices for η_2 and Λ in terms of the complexity bounds can be explained by the fact that our analysis addresses the worst-case. Setting $\eta_2 = \sqrt{n}$ and $\Lambda = \Theta(1 + \frac{1}{n})$ essentially forces more model improving iterations before a successful step is allowed or trust region radius is reduced, to ensure best models and progress according to the worst case κ_{eg} constant. However, in practice the error between the model and the function at the trial step can be much smaller than the worst case κ_{eg} bound suggests, so too many model improvement steps may be wasteful. It would be interesting to explore adaptive choices for these constants in future work.

Low-dimensional tests. We begin our testing on 174 low-dimensional test problems with dimension between $2 \leq n \leq 5$. Since the dimension is low, we set the maximum allowable number of points in the sample set to be $(n+1)(n+2)/2$ points. Aside from the first few iterations, the sample set of GC-YZ-LIN/V and NEWUOA will eventually become full, resulting in an interpolation system that is fully determined, so the two models coincide and the switching is inactive. In contrast, DFOTR often removes several sample points at once, so the hedge model can be active throughout the trajectory. Figure 6.1 shows the performance of the solvers on this test set. All solvers seem to perform comparably well.

Medium-/High-dimensional tests. We also benchmark on test sets in dimension 30, 100, and 200, with 95, 101, and 92 problems respectively in each set. For these tests, we will use a maximum of $2n+1$ sample points in the total interpolation set. Firstly, Figure 6.2 compares the solvers using their default fitting routine against using the hedge model. We see that adopting this model strictly improves performance across the board. In what follows, to ensure fair testing, all solvers are benchmarked using the hedge model.

Figures 6.3, 6.4, and 6.5 show the data profiles for the 30-, 100- and 200- dimensional test sets respectively.

- 利用 $\mathcal{Y}$ - $\mathcal{Z}$ 分离, GC-YZ-LIN 能够在很大程度上与 NEWUOA 匹敌, 但在早期和后期表现之间存在权衡。将 NEWUOA 的自校正规则移植到 $\mathcal{Y}$ - $\mathcal{Z}$ 框架中, 我们得到了一个优于 NEWUOA 的变体; 但目前尚未对其开展理论分析。

- 要给出一种能与全空间求解器竞争的随机子空间算法的实用实现仍然很困难。

这里我们给出测试方法的细节。表 6.1 列出了我们提出的求解器和 DFOTR 所用的参数。我们将使用 PRIMA 的 [27] 版 NEWUOA 实现, 其默认具有固定的信赖域参数, 因此我们在表中不再重述。所有问题都以相同方式初始化, 其中初始插值集 $\mathcal{Y}_0$ 设为 $\{\Delta_0 u_i\}$, 而 $u_i, i = 1, \dots, n$ 是随机旋转后的坐标向量, 且半径 $\Delta_0 = 0.5$ 。 $\mathcal{Z}_0$ 随后设为 $\{-\Delta_0 u_i\}$ 。为使算法能运行到预算耗尽, 我们将终止准则设为: DFOTR 为 $\Delta_{\text{end}} = 10^{-12}$, NEWUOA 和 GC-YZ-LIN/V 分别设为 $\rho_{\text{end}}, \sigma_{\text{end}} = 10^{-12}$ 。所有信赖域子问题均精确求解。我们使用容差为 $\tau \in \{10^{-2}, 10^{-4}, 10^{-6}\}$ 的数据剖面比较性能, 并在图例中报告每个求解器的曲线下面积得分。我们对由不同随机旋转初始集合得到的曲线取平均。

我们评论 GC-YZ-LIN 中三个参数的选择, 这些参数也出现在复杂度分析中。首先, 模型 Hessian 界 K 被设置为极大, 即 10^{100} , 并且基本上不激活。这在很大程度上是对来自 CUTEst 且具有天然极端曲率的问题测试求解器性能的副产物。例如, “SSBRYND” 和 “SCURLY10” 等问题的真实 Hessian 范数大于 10^{20} , 而 “POWERSUM” 的真实 Hessian 范数甚至可能超过 10^{100} 。因此, 设置一个通用、固定且较小的 K 会削弱模型捕捉函数真实曲率的能力。此外, 我们将 η_2 设得很小, 即 5×10^{-9} 。尽管理论上需要确保模型梯度的范数相对于半径不太小, 但较大的 η_2 将信赖域半径与模型梯度绑定在一起, 可能对局部 Lipschitz 常数较小的问题不利, 例如 “香蕉谷” 之外的 Rosenbrock 函数。如果仍有进展可以取得, 该区域中的迭代点将被迫具有较小的信赖域半径, 并可能阻碍进展。此外, 在实践中, 对 Λ 采用较宽松的选择似乎是有益的。根据我们的理论, Λ -适定性保证每次成功迭代都有一定量的函数下降。然而, 在实践中确保较小的 Λ 的代价是更多用于几何校正的函数求值。我们观察到, 当 $\Lambda = 1000$ 时性能最佳, 这表明尽管几何可能并不完美, 算法仍能取得进展, 并且只有几何极差时才应进行几何校正。我们还区分 Λ 和 Λ_{sc} , 其中后者是一个阈值, 用于确定在某一步处拉格朗日多项式值需要多大才会启动自校正步。在实践中, 我们将其设为 2, 而只要 $\Lambda_{sc} \geq 1$, 分析就成立。

这些对于 η_2 和 Λ 而言在复杂度界方面看似不利的参数选择, 可以用我们的分析针对的是最坏情况这一事实来解释。设置 $\eta_2 = \sqrt{n}$ 和 $\Lambda = \Theta(1 + \frac{1}{n})$ 本质上强制在允许成功步或缩减信赖域半径之前进行更多的模型改进迭代, 以确保根据最坏情况 κ_{eg} 常数得到最佳模型和进展。然而, 在实践中, 试探步处模型与函数之间的误差可能远小于最坏情况 κ_{eg} 界所表明的大小, 因此过多的模型改进步骤可能是浪费的。在未来的工作中探索这些常数的自适应选择将会很有意义。

低维测试。 我们开始在 174 个低维测试问题上进行测试, 其维数在 $2 \leq n \leq 5$ 之间。由于维数较低, 我们将样本集中允许的最大点数设置为 $(n+1)(n+2)/2$ 个点。除了前几次迭代外, GC-YZ-LIN/V 和 NEWUOA 的样本集最终会变满, 从而产生一个完全确定的插值系统, 因此两个模型重合, 切换不活跃。相比之下, DFOTR 经常一次移除多个样本点, 因此对冲模型可以在整个轨迹中保持活跃。图 6.1 显示了求解器在此测试集上的性能。所有求解器似乎都表现相当好。

中/高维测试。 我们还对维数为 30、100 和 200 的测试集进行了基准测试, 每个测试集分别包含 95、101 和 92 个问题。对于这些测试, 我们将在总插值集中最多使用 $2n+1$ 个样本点。首先, 图 6.2 比较了使用默认拟合例程的求解器与使用对冲模型的求解器。我们发现, 采用该模型全面严格提升了性能。在下文中, 为保证测试公平, 所有求解器均使用对冲模型进行基准测试。

图 6.3、6.4 和 6.5 分别展示了维数为 30、100 和 200 的测试集的数据剖面。

Table 6.1: Solver settings used throughout testing.

	GC-YZ-LIN/V	GC-sub	DFOTR
Acceptance ratio η_1	0.01	0.1	0.05
Model gradient threshold η_2	5×10^{-9}	5×10^{-9}	—
“Very successful iteration” threshold	—	—	0.5
Expansion factor γ_{inc}	1.3	1.3	1.6
Contraction factor γ_{dec}	0.8	0.8	0.8
Resolution shrink θ	0.1	—	—
Poisedness threshold Λ	1000	1000	—
Self-correcting threshold Λ_{sc}	2	2	—
Hedge: EWMA decay β	0.8	—	—
Model Hessian bound K	10^{100}	—	—
Small step gate threshold c_g	0.5	—	—
Subspace dimension q	—	5	—
Interpolation set capacity	—	$2q$	—

表6.1: 整个测试过程中使用的求解器设置。

	GC-YZ-LIN/V	GC-sub	DFOTR
接受率 η_1	0.01	0.1	0.05
模型梯度阈值 η_2	5×10^{-9}	5×10^{-9}	—
“非常成功迭代” 阈值	—	—	0.5
扩张因子 γ_{inc}	1.3	1.3	1.6
收缩因子 γ_{dec}	0.8	0.8	0.8
分辨率收缩 θ	0.1	—	—
适定性阈值 Λ	1000	1000	—
自校正阈值 Λ_{sc}	2	2	—
Hedge: EWMA衰减 β	0.8	—	—
模型Hessian界 K	10^{100}	—	—
小步长门控阈值 c_g	0.5	—	—
子空间维数 q	—	5	—
插值集容量	—	$2q$	—

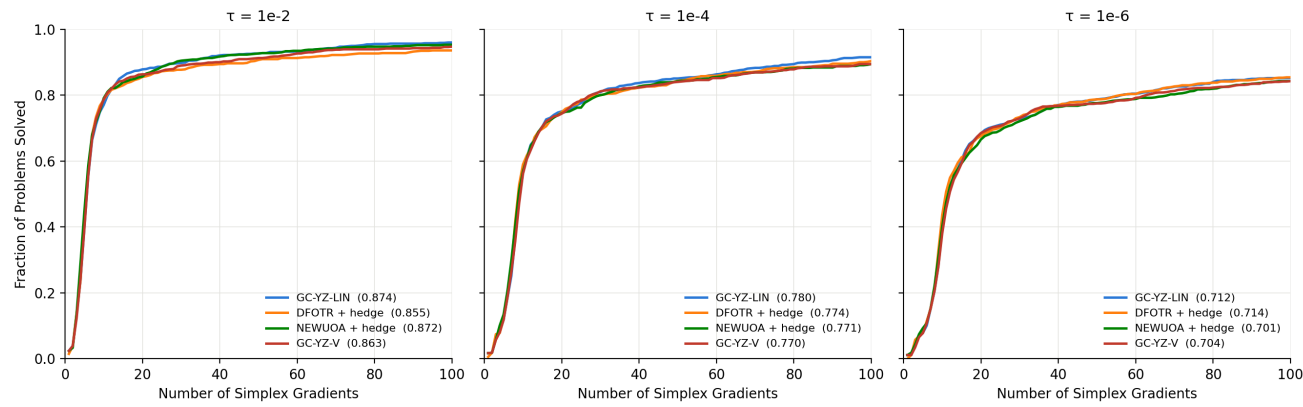
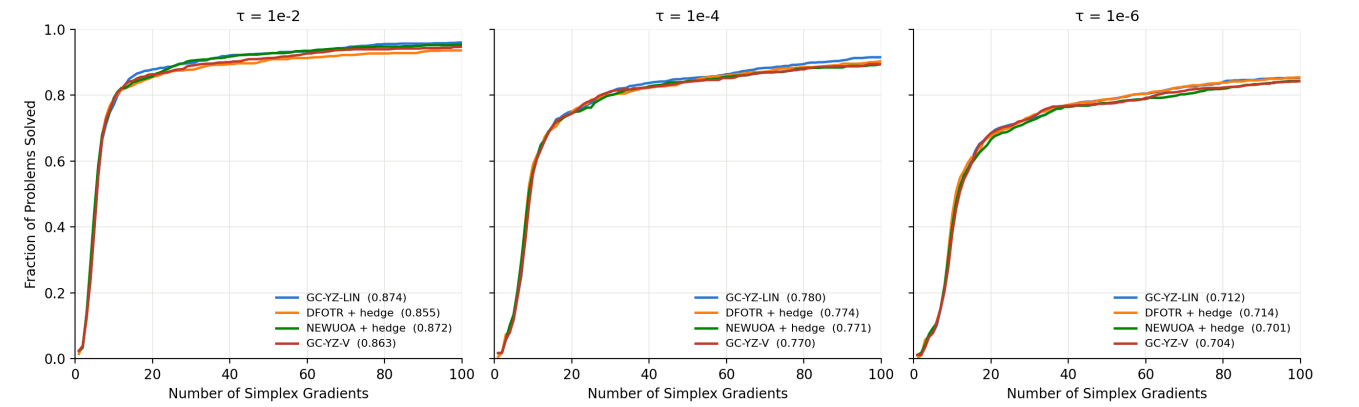
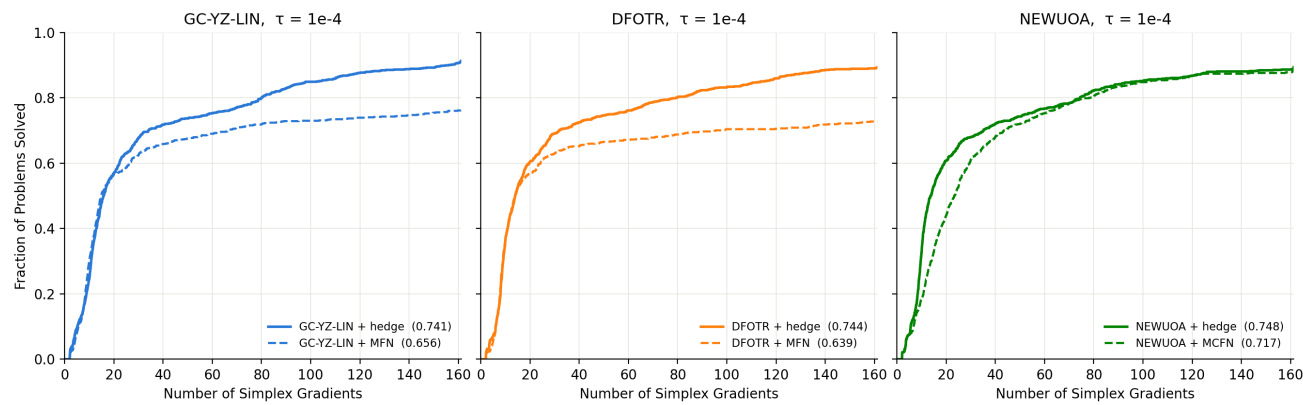
Figure 6.1: Data profiles for the low-dimensional test suite: $2 \leq n \leq 5$.图6.1: 低维测试集的数据剖面: $2 \leq n \leq 5$ 。

Figure 6.2: Data profiles comparing the hedge model with each solver's default fitting procedure.

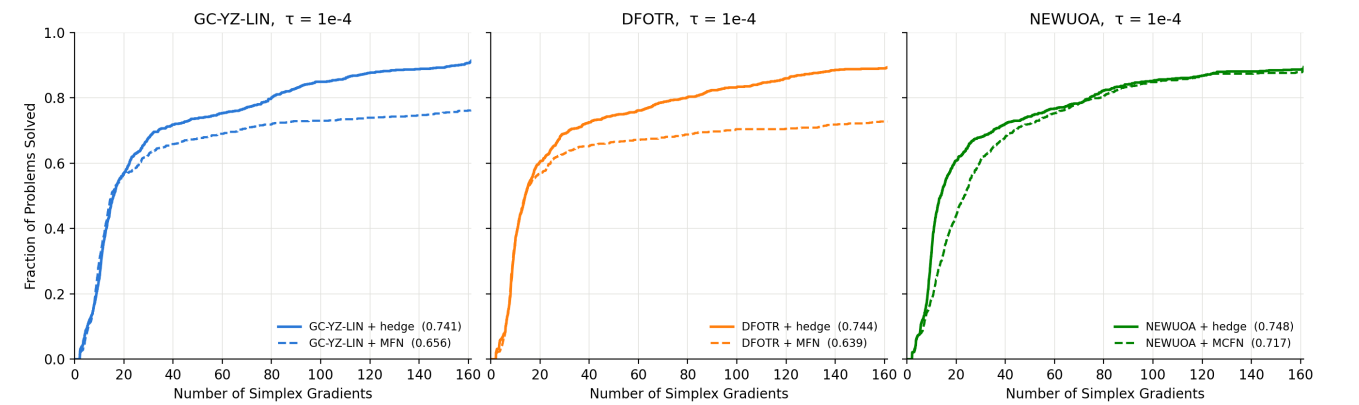


图6.2: 比较对冲模型与每个求解器默认拟合过程的数据剖面。

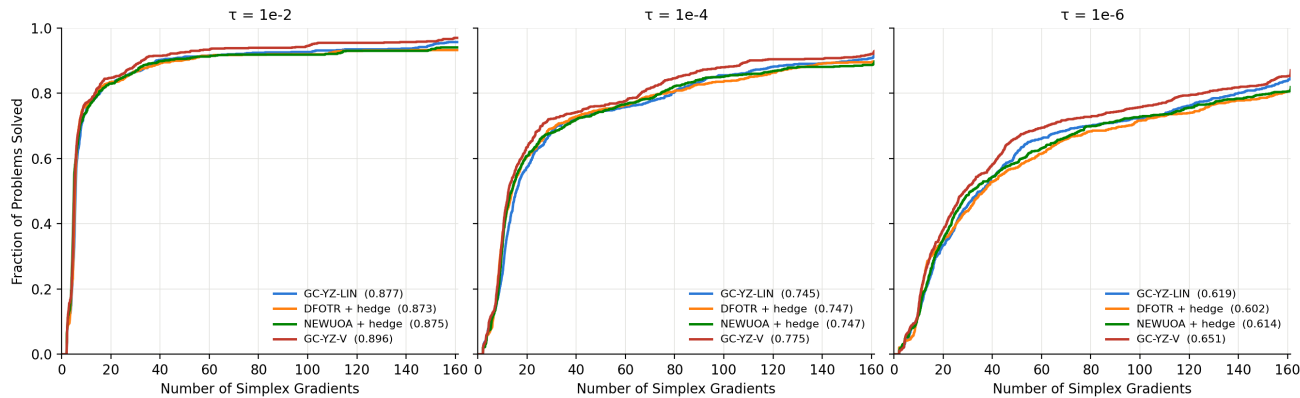


Figure 6.3: Data profiles for the test suite $n = 30$.

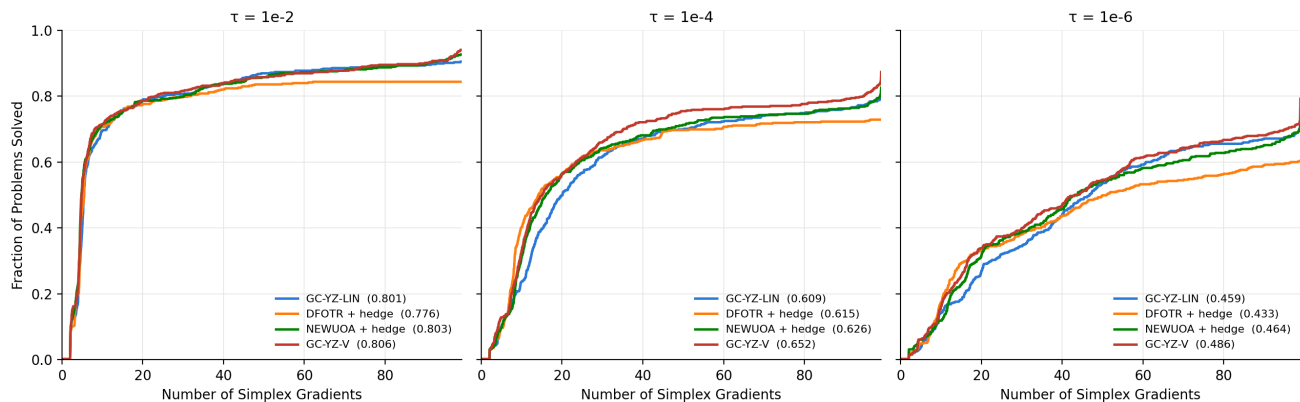


Figure 6.4: Data profiles for the test suite $n = 100$.

The first most apparent observation is the deterioration of DFOTR as dimension increases. This is largely due to the absence of a geometry management procedure. As dimension increases, the $2n + 1$ points become increasingly sparse and are more likely to be nearly degenerate. Thus, DFOTR performs increasingly aggressive shrinking to eventually force a geometry reset, which leads to radius collapse and premature termination, as seen in the significant flattening of the DFOTR data profile curves.

Another apparent observation is the almost strict improvement of GC-YZ-V over NEWUOA and GC-YZ-LIN. GC-YZ-V is a combination of several algorithmic features from both NEWUOA and GC-YZ-LIN. This result suggests that the use of NEWUOA-style self-correction may be more efficient than first ensuring all points are close, then checking the self-correction property. While the latter is more principled and easier to analyze theoretically, it may be slower in practice. This also suggests it may be worthwhile to maintain a separation of sets $\mathcal{Y}$ - $\mathcal{Z}$, even when quadratic Lagrange polynomials are maintained for the entire sample set $\mathcal{Y} \cup \mathcal{Z}$. Again, we note that this variant is not covered by current theory and is a line of future work.

Finally, we observe that GC-YZ-LIN is closely competitive with NEWUOA across dimensions and across tolerances. At $\tau = 10^{-2}$, the curves are nearly identical. At $\tau = 10^{-4}, 10^{-6}$, a pattern emerges where NEWUOA is quicker to solve more problems early in the run, however GC-YZ-LIN eventually catches up and overtakes NEWUOA later in the run. In general, the performance of the two methods is comparable, which is surprising given that GC-YZ-LIN only manages linear geometry for a subset of the sample set compared to NEWUOA's full quadratic geometry.

Results in subspaces. Figure 6.6 compares the performance of the random subspace method against GC-YZ-LIN. The subspace method sets $q = 5$, patience parameters equal to 1, and draws a fresh coordinate stencil on every subspace redraw. A maximum of $2q + 1$ points is used in the sample. We remark that varying choices of

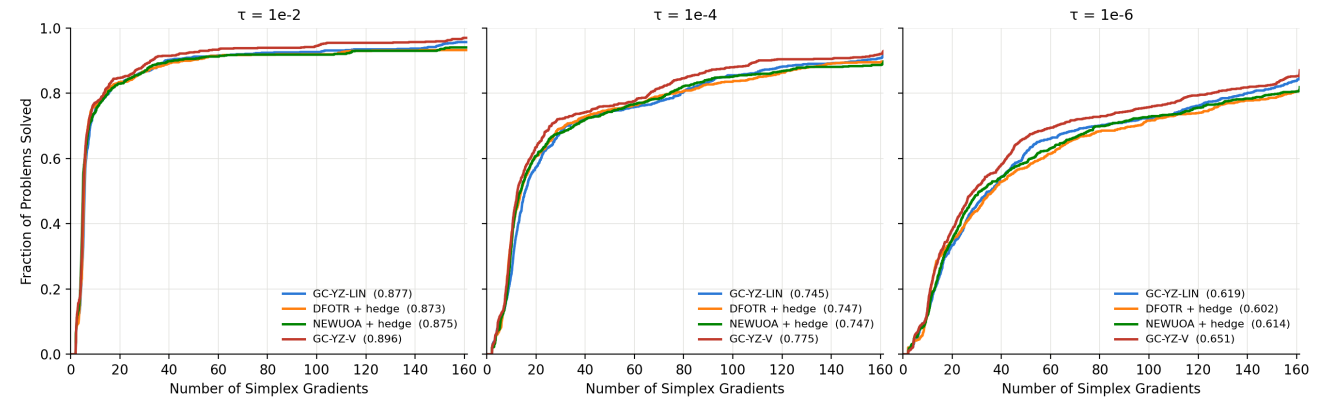


图 6.3: 测试套件 $n = 30$ 的数据剖面。

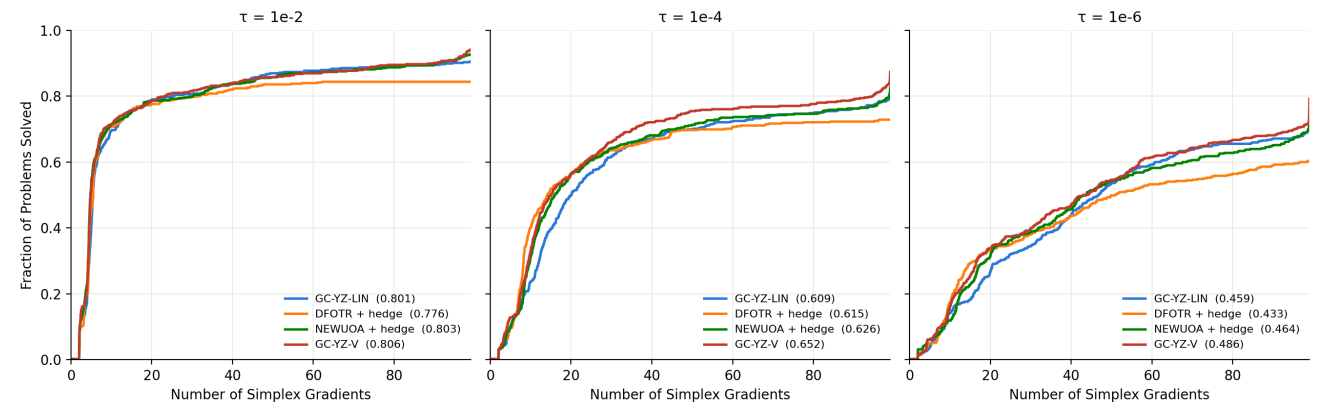


图 6.4: 测试套件 $n = 100$ 的数据剖面。

第一个最明显的观察结果是，随着维数增加，DFOTR 的性能会变差。这主要是由于缺少几何管理过程。随着维数增加， $2n + 1$ 点变得越来越稀疏，并且更可能接近退化。因此，DFOTR 会执行越来越激进的收缩，最终迫使其进行几何重置，这会导致半径坍缩和过早终止，正如 DFOTR 数据剖面曲线显著变平所显示的那样。

另一个明显的观察结果是，GC-YZ-V 相对于 NEWUOA 和 GC-YZ-LIN 几乎严格更优。GC-YZ-V 结合了来自 NEWUOA 和 GC-YZ-LIN 的若干算法特征。这一结果表明，使用 NEWUOA 式自校正可能比先确保所有点都彼此接近、再检查自校正性质更高效。尽管后者更有原则性，并且在理论上更容易分析，但在实践中可能更慢。这也表明，即使为整个样本集 $\mathcal{Y} \cup \mathcal{Z}$ 保留二次拉格朗日多项式，保持集合 $\mathcal{Y}$ 与 $\mathcal{Z}$ 的分离也可能是值得的。我们再次指出，该变体未被当前理论涵盖，并且是未来工作的一个方向。

最后，我们观察到，GC-YZ-LIN 在不同维数和不同容差下与 NEWUOA 都极具竞争力。在 $\tau = 10^{-2}$ 处，曲线几乎完全一致。在 $\tau = 10^{-4}, 10^{-6}$ 时，出现了一种模式：NEWUOA 在运行早期能更快地求解更多问题，然而 GC-YZ-LIN 最终会迎头赶上，并在运行后期超越 NEWUOA。总体而言，两种方法的性能相当，考虑到与 NEWUOA 的完整二次几何相比，GC-YZ-LIN 仅在样本集的一个子集上实现线性几何，这一点令人惊讶。

子空间中的结果。 图6.6比较了随机子空间方法与 GC-YZ-LIN 的性能。子空间方法设置 $q = 5$ ，将耐心参数设为 1，并在每次子空间重绘时抽取新的坐标模板。样本中最多使用 $2q + 1$ 个点。我们注意到，以下各项的不同选择：

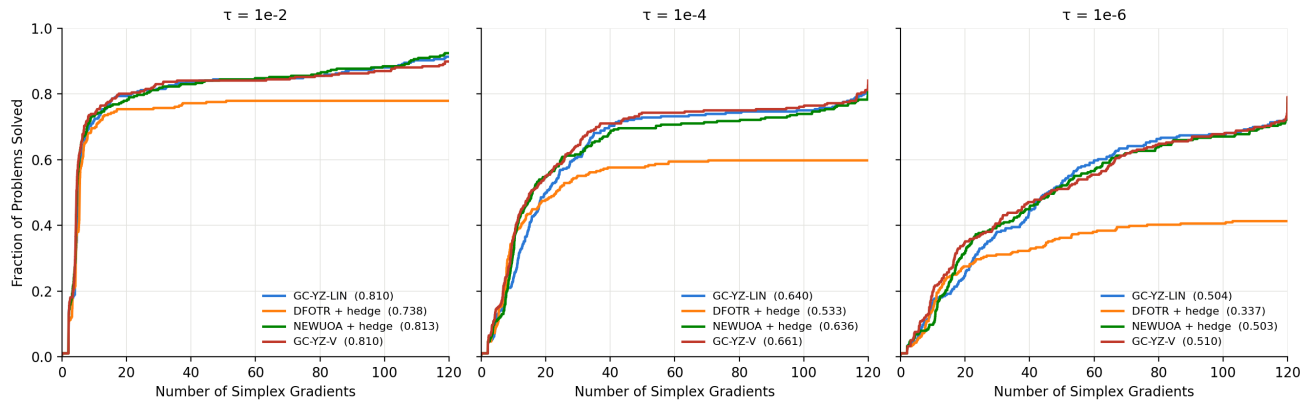


Figure 6.5: Data profiles for the test suite $n = 200$.

patience parameter, subspace dimension, re-draw strategy, and maximum number of allowed sample points were tested, and we opt to present the best performing combination found in dimension 30 and 100. It is evident that the subspace method does not perform as well as a full space solver and its performance deteriorates as tolerances get smaller. While theory suggests that as dimension increases, the competitiveness of subspace methods should as well, this is not something that is immediately evident. A plausible explanation for the discrepancy between theory and practice is as follows. The random subspace method needs at least $\mathcal{O}(q)$ points for each subspace redraw and the bound on the number of subspace redraws is not loose. Thus we arrive at the worst case complexity of order q without fail. However, in the full space solvers, we rarely need more than a few consecutive geometry correcting steps to achieve good geometry, hence in practice we are typically far from actually attaining the worst-case behavior described in the complexity bound. Another explanation is that subspace methods ultimately lose the advantage of quadratic approximation in the whole space, so may only be competitive where full space quadratic approximation is prohibitive or useless. A future implementation which initiates any sample set with as few as two sample points may improve performance of subspace methods. In addition reusing sample points from other subspaces can be beneficial. In conclusion, our contribution in terms of subspace TR method in this paper is mainly theoretical while practical approaches require further investigation. We would like to point the reader to the extensive empirical study carried out in [8].

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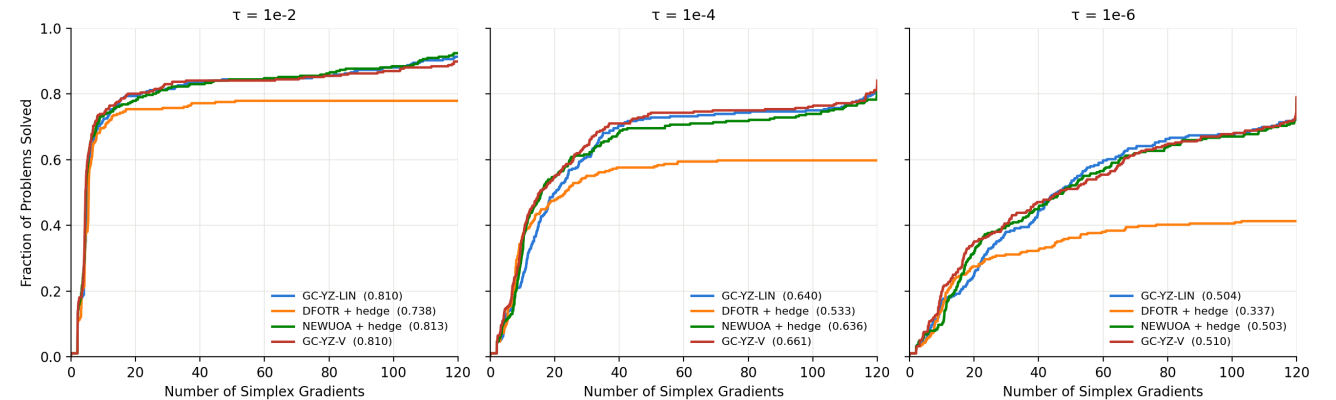


图 6.5: 测试套件 $n = 200$ 的数据剖面。

耐心参数、子空间维数、重抽取策略以及允许的最大样本点数都经过了测试，我们选择展示在维数 30 和 100 下找到的表现最佳的组合。显然，子空间方法的表现不如全空间求解器，并且随着容差变小，其性能会恶化。尽管理论表明，随着维数增加，子空间方法的竞争力也应增强，但这一点并不显而易见。对理论与实践之间差异的一种合理解释如下。随机子空间方法在每次子空间重绘时至少需要 $\mathcal{O}(q)$ 个点，而且对子空间重绘次数的界并不宽松。因此，我们必定会达到阶为 q 的最坏情况复杂度。然而，在全空间求解器中，我们很少需要超过少数几个连续的几何校正步骤就能获得良好的几何，因此在实践中，我们通常远未真正达到复杂度界所描述的最坏情况行为。另一种解释是，子空间方法最终失去了全空间二次逼近的优势，因此可能仅在全空间二次逼近不可行或无用的情况下才具有竞争力。未来若实现用少至两个样本点来初始化任意样本集，可能会改善子空间方法的性能。此外，重用来自其他子空间的样本点也可能有益。总之，本文在子空间信赖域方法方面的贡献主要是理论性的，而实用方法仍需进一步研究。我们想请读者参考在 [8]中开展的广泛实证研究。

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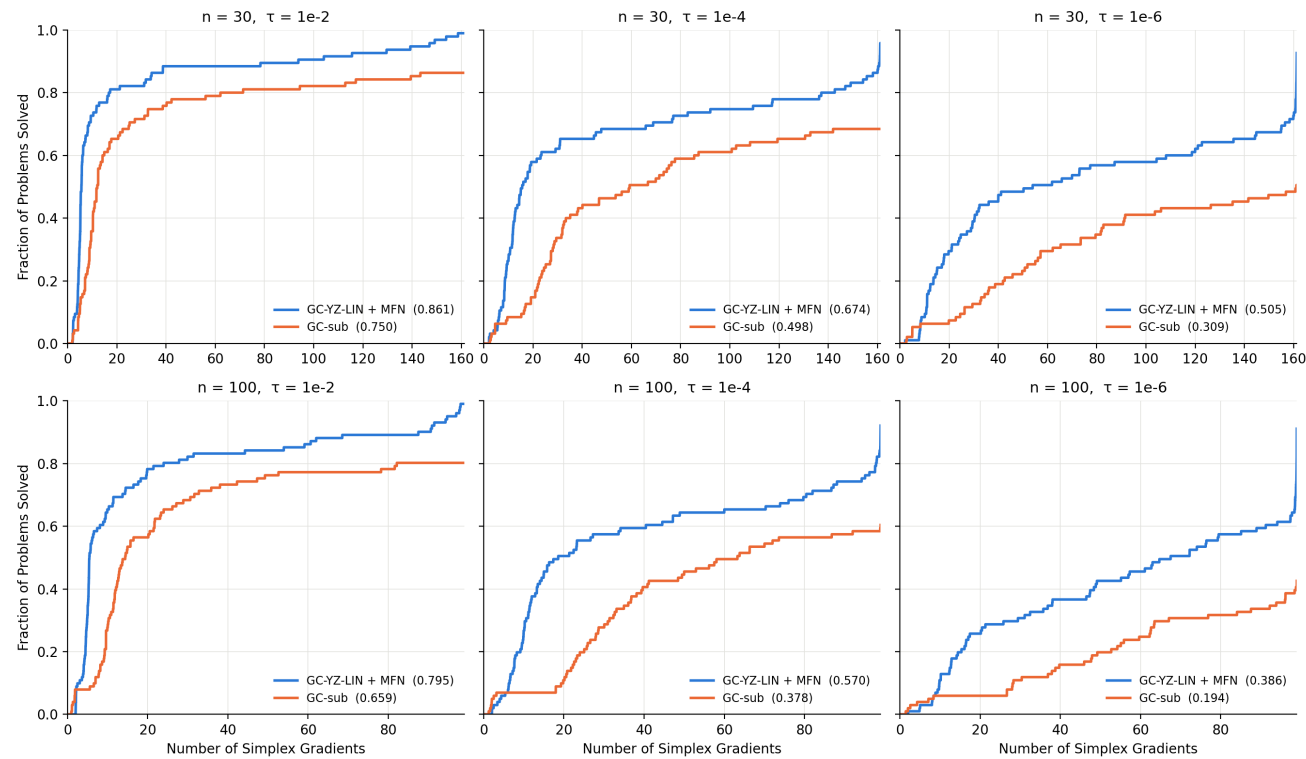


Figure 6.6: Data profiles for the random subspace implementation compared against GC-YZ-LIN for $n = 30, 100$.

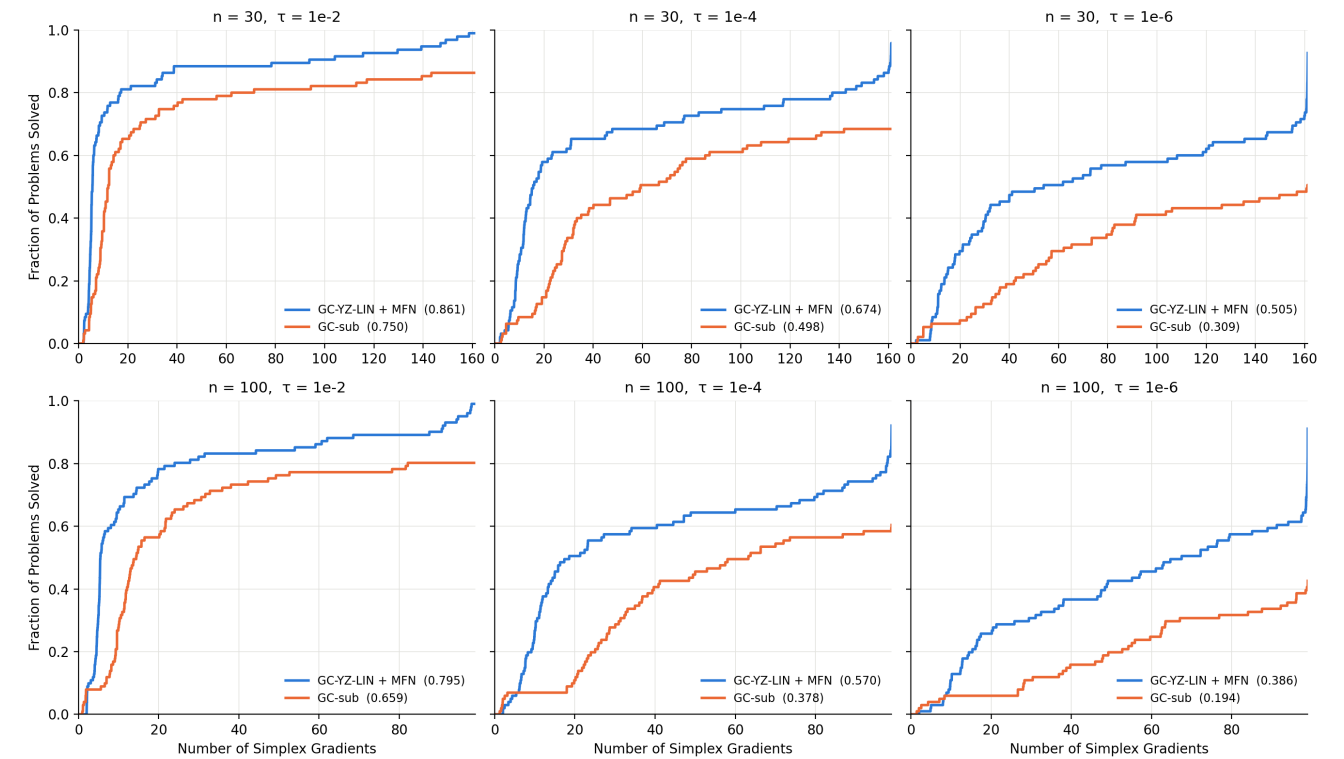


图 6.6: 在 $n = 30, 100$ 下, 随机子空间实现与 GC-YZ-LIN 的数据剖面对比。

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